

## Highlights

**A moment hierarchy for the voxel: closed-form single-site  $T$ -matrices, and the error of elastic volume-integral scattering isolated on a space-filling layer of cubes**

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- A moment hierarchy gives the cube's single-site T-matrix in closed form
- A voxel needs two bases: one for the medium and one for the wavefield
- The order is  $\min(2p+2, 2r+2)$  for field degree  $p$  and medium degree  $r$
- The single-site T-matrix's error is the medium's relative projection error
- The error is known in advance, from the medium and the grid alone

# A moment hierarchy for the voxel: closed-form single-site $T$ -matrices, and the error of elastic volume-integral scattering isolated on a space-filling layer of cubes

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## Abstract

To model seismic waves efficiently in a realistically complicated medium, volume-integral methods divide it into cubic voxels coupled through the background Green's tensor. Each voxel must represent the medium within it and the wavefield within it. We ask which basis each needs and what error a given choice leaves. We answer on a layer of voxels: because cubes tile space,  $n$  planes of voxels are the layer itself, and its exact solution isolates the discretisation error. With the wavefield expanded to Legendre degree  $p$  and the medium's contrast to degree  $r$  in each voxel, the scheme converges at order  $\min(2p + 2, 2r + 2)$ , with leading error  $c_p(kd)^{2p+2}$  in closed form; the orders hold for SV and SH waves and at oblique incidence. The single-site  $T$ -matrix's second-order error equals the relative projection error of the medium at degree  $\min(p, r)$ : the two bases must be matched, and the error is known from the medium before anything is solved. In a plane-stratified background the orders persist: the background changes the field the voxels must represent, not how well they represent it. The single site is the first member of a

hierarchy in the field's gradients, an elastodynamic extension of the equivalent inclusion method, whose coefficients are moments of the Green's tensor over the cube, in closed form on four constants. Its blocks gain accuracy in pairs: with third gradients it is 300 times more accurate than the uniform-strain closure against the exact sphere, and as a voxel scheme it is fourth order on a graded sphere.

*Keywords:* Elastic wave scattering, Volume integral equation, Foldy–Lax multiple scattering, Discrete dipole approximation, Galerkin method, Convergence analysis

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## 1. Introduction

*The problem this paper serves.* Seismic waves in the crust are scattered by heterogeneity on every scale, and the aim of the work this paper begins is an efficient model of the wavefield in a medium of realistic complication. The models that describe such a medium are parameterised in many ways (as layers or blocks, by smooth basis functions, or by interfaces), and a calculation that uses one must first sample it onto a grid. A volume-integral scattering calculation divides the medium into cells and treats every cell as a scatterer: a voxel carrying a single-site  $T$ -matrix, the operator that turns the field incident on the voxel into the sources it radiates, and coupled to every other voxel through the Green's tensor of the background, with the multiple-scattering series summed.

*The question: two bases.* A voxel is a small region in which two functions must be represented by a finite number of coefficients. One is the *medium*: the contrast of the elastic moduli and the density from the background,

16 which the model prescribes. The other is the *wavefield*: the displacement and  
 17 strain inside the voxel, which the calculation must find. Each is expanded  
 18 in a basis on the cell, and the two bases are separate choices. Here both  
 19 are Legendre polynomials, the wavefield to degree  $p$  and the contrast to  
 20 degree  $r$ ; the usual scheme, a uniform field in a cell of constant properties,  
 21 is  $p = r = 0$ . Neither choice is new in itself. In higher-order volume integral  
 22 methods for electromagnetics the material within an element is represented  
 23 by polynomials whose orders are independent of those of the field [1], as it  
 24 is in finite elements, and voxels that carry a linear field are in use [2]. What  
 25 has been missing is the law that relates the two bases: the order a given pair  
 26 attains, and the error it leaves. The answer decides the cost of everything  
 27 built on it, because a voxel that represents its contents more accurately can  
 28 be larger, and the number of voxels, not the size of each one's state, sets the  
 29 cost of the multiple-scattering solve.

30 *Why the question is hard to answer.* For the media such a scheme is built for,  
 31 there is no exact solution. Its error cannot be measured against the answer.  
 32 Comparison with another numerical method mixes the errors of both. And a  
 33 refinement study can mislead, because a voxel scheme can converge smoothly  
 34 to the wrong limit. §4 shows how that happens with the most natural choice  
 35 of coupling, the Green's tensor evaluated between voxel centres: it carries  
 36 a local term whose size per unit volume does not shrink as the voxels do,  
 37 and unless the single voxel's response is matched to cancel it, refinement  
 38 converges to the wrong answer without revealing it.

39 *The method: a layer of voxels.* This paper therefore answers the question  
 40 in the simplest configuration in which a voxel scheme runs in full (every

41 voxel, every coupling, the whole multiple-scattering series, at normal and  
 42 at oblique incidence) and the exact answer is nevertheless known: a layer  
 43 filled with voxels. Because cubes tile space, a layer of identical voxels does  
 44 not approximate a homogeneous layer; it *is* one. Any difference between  
 45 its reflection and transmission and those of the continuous layer is therefore  
 46 error of the discretisation alone, with no modelling error and no shape error  
 47 behind which to hide. A sphere, the usual test object for volume-integral and  
 48 discrete-dipole methods [3], cannot give this: its error mixes the discretisation  
 49 with the staircase approximation of a curved surface by cubes. The layer  
 50 plays the part of the patch test in finite elements: reproducing a uniform  
 51 medium correctly is necessary, though not sufficient, for a scheme to be  
 52 trusted in a heterogeneous one. The experiment is then made harder in  
 53 three steps, each with an exact solution of its own: a randomly stratified  
 54 layer, in which every cell is homogeneous but the cells differ; a layer whose  
 55 properties vary smoothly within every cell, which is where the basis for the  
 56 medium is tested; and a sphere whose contrast falls smoothly to zero at its  
 57 surface, which carries the layer's prediction into three dimensions without a  
 58 staircase. The coupling and the single site that the layer validates are the  
 59 same in any medium; a laterally heterogeneous medium remains to be tested.

60 *The answer.* (1) *The order is set by the weaker basis.* The scheme converges  
 61 at order  $\min(2p + 2, 2r + 2)$  in the cell size (§9). A contrast held constant in  
 62 each cell therefore caps every voxel at second order in a medium that varies  
 63 within the cells, however rich the basis of its field; and a contrast of higher  
 64 degree than the field is wasted. (2) *The leading error is in closed form, for*  
 65 *every grid.* For a voxel of side  $d$  and a wave of wavenumber  $k$  it is  $c_p(kd)^{2p+2}$

66 with  $c_0 = 1/12$ ,  $c_1 = 1/720$  and  $c_2 = 1/100800$  (§8); the orders 2, 4 and 6 of  
 67  $p = 0, 1$  and 2 hold for P, SV and SH waves at normal and oblique incidence.  
 68 For the collocation scheme, the uniform-field voxel in its simplest form, the  
 69 error is  $+(kd)^2/8$  in reflection and  $-(kd)^2/24$  in transmission (§7), and the  
 70 opposite signs explain why no correction of the single site can improve both  
 71 (§12). (3) *The error that remains is the single site's, and it is the projection*  
 72 *error of the medium.* The single-site  $T$ -matrix sums the rescattering within a  
 73 voxel to all orders and the coupling between voxels is exact, but the  $T$ -matrix  
 74 is that of the voxel as represented, its contrast projected to degree  $r$  and its  
 75 field held to degree  $p$ . The error of the term of second order in the contrast  
 76 equals the *relative projection error* of the medium, the fraction of the squared  
 77 norm of its profile that polynomials of degree  $\min(p, r)$  on the cells cannot  
 78 represent (Eq. (43)): exactly in the long-wave limit, and to within 1.3% for  
 79 all nine pairs with  $p, r \leq 2$  (§9). The wavefield basis must therefore carry  
 80 what the medium imprints on the field, not only the wave, and the two bases  
 81 must be matched. (4) *The error is known in advance.* The projection error  
 82 is a property of the medium and the grid alone, so the cell size and the  
 83 degree that a required accuracy needs are chosen before anything is solved.  
 84 (5) *Where the medium is piecewise constant on the cells the projection error*  
 85 *vanishes*, a constant suffices for the medium, and the degree of the field  
 86 alone sets the order: in a randomly stratified model one first-moment voxel  
 87 per model layer reaches a reflection accuracy of  $10^{-6}$ , where the uniform-field  
 88 voxel needs about forty (§9), and in the uniform layer it allows voxels 58 times  
 89 larger in each direction (Appendix Appendix I). (6) *In three dimensions* the  
 90 smoothly graded sphere converges at second order with the mean-only voxel,

91 as the layer predicts (§10), and at fourth order with the first-moment voxel  
 92 (§11).

93 *What makes the experiment exact.* The answer above can be trusted only if  
 94 nothing but the two bases is approximated. Three results, set against the  
 95 coupled-dipole method described below, secure that. (a) *An exact coupling.*  
 96 The integrated Green tensor and the filtered coupled dipoles approximate the  
 97 field of a cell; for cubes the source-cell average is exact, because the cube's  
 98 form factor vanishes on the reciprocal lattice (the tiling identity, §5), and  
 99 at normal incidence the lattice kernel is in closed form with no lattice sum.  
 100 (b) *No polarisability to choose.* The polarisability, a modelling choice in the  
 101 coupled-dipole method, is here a derived object (§6): the moments of the  
 102 Green's tensor over the cube, taken as distributions, yield Eshelby's  $\delta$  term  
 103 without adding it by hand, the cube's departure from the sphere as a pure  
 104 cubic anisotropy, and the closure's error against exact scattering by a sphere  
 105 as a real  $(ka)^2$  term in closed form. The self term the single site carries is the  
 106 one the averaged kernel omits, so the discrete system is a collocation of the  
 107 continuum equation. (c) *The elastic state.* A voxel carries a force (through  
 108 its density contrast) and a moment (through its stiffness contrast), so its  
 109 state has nine components, displacement and strain, the coupling is a  $9 \times 9$   
 110 tensor, and the single site's self-interaction is the elastic depolarisation of a  
 111 cube, the discrete counterpart of the inclusion tensor of Eshelby [4]. Cubes  
 112 serve all three: they tile space, so a uniform region cut into cubes is still that  
 113 region, and their regular lattice makes the coupling a convolution, solved by  
 114 fast transforms.

115 *The coupled-dipole method.* Replacing a heterogeneous body by a lattice of  
 116 small scatterers that interact through the background Green's function is the  
 117 multiple-scattering description of Foldy [5] and Lax [6]. In electromagnetics  
 118 it became a numerical method as the coupled-dipole method, or discrete  
 119 dipole approximation (DDA): each cell of a cubic grid is a polarisable point  
 120 dipole driven by the incident field and by the fields of every other dipole  
 121 [7, 8]. The method is a collocation of the volume integral equation for the  
 122 electric field, with the cell's self-interaction absorbed into its polarisability  
 123 [9], and Yurkin and Hoekstra [10] review it in that form. Its accuracy turns  
 124 on two ingredients, and this paper takes up both for elasticity. The first is the  
 125 polarisability. The Clausius–Mossotti value is the static response of a sphere;  
 126 the lattice dispersion relation of Draine and Goodman [11] adjusts it so that  
 127 an infinite lattice propagates waves as the continuum does to a given order  
 128 in  $kd$ ; Rahmani et al. [12] instead derive it from the exact static solution for  
 129 the whole body, so that the method is exact in the long-wavelength limit.  
 130 The second is the coupling. Integrating the Green tensor over the source  
 131 cell rather than sampling it at the centre improves the accuracy, most of  
 132 all for strong contrasts [13], and the filtered coupled dipoles of Piller and  
 133 Martin [14], revived by Yurkin et al. [15] for large refractive indices, replace  
 134 the point dipole by a source of finite extent whose spectrum is band-limited  
 135 to the grid. The coupling summed over a lattice extends the method to  
 136 periodic structures [16]. Its error converges at second order in the dipole  
 137 spacing when shape errors are absent [3], and for a cube, the one shape  
 138 a cubic grid represents without a staircase, Yurkin and Kahnert [17] reach  
 139 relative uncertainties of  $10^{-7}$  to  $10^{-3}$  by extrapolation in the grid. On the

140 mathematical side, Costabel et al. [18] show that the method is a finite section  
 141 of an infinite block Toeplitz matrix, stable for a large range of the contrast  
 142 but not for all of it, and Costabel [19] that in a one-dimensional model it  
 143 converges at second order in the interior, with a boundary layer in which  
 144 the consistency error does not vanish. The layer studied here is the elastic  
 145 counterpart of that cube: a body the grid represents exactly, so that the  
 146 discretisation error can be isolated. What this paper adds is the closed form:  
 147 for cells that tile space, the coupling that makes the lattice reproduce the  
 148 continuum exactly, and the order and the constant of the error that remains,  
 149 as functions of the two bases. In the terms used here the coupled-dipole  
 150 method is the pair  $p = r = 0$ , and its refinements improve the polarisability  
 151 and the coupling while leaving both bases constant in the cell.

152 *Its elastic counterparts.* The elastic problem has the same volume integral  
 153 equation [20], and every cell-based solver of it is a coupled-dipole method  
 154 whether or not it carries the name. Nestor [21] divided a two-and-a-half-  
 155 dimensional medium into cells that interact through the background Green's  
 156 tensor, each carrying a contrast renormalised by its own near field, with  
 157 the multiple-scattering series summed. Kanaun and Levin [22] discretise the  
 158 inclusion problem with Gaussian approximating functions whose matrix ele-  
 159 ments are analytic. Yang et al. [23] solve a stress-velocity form of the integral  
 160 equation on a grid by conjugate gradients with fast Fourier transforms, and  
 161 Touhei [24] does the same for a half space with a generalised Fourier trans-  
 162 form. Jakobsen [25] and Jakobsen et al. [26] build the seismic forward and  
 163 inverse problems on the transition operator of the discretised Lippmann-  
 164 Schwinger equation, the first in the acoustic approximation and the second

165 in arbitrary anisotropic elastic media with a nine-component state vector.

166 *This paper.* It extends the formulation of Nestor [21] to cubic voxels, and is  
167 ordered as the experiment is built. §§2–3 set out the discrete medium and  
168 the continuous layer it must reproduce; §4 shows why the point coupling  
169 fails, and §5 why the source-averaged coupling succeeds exactly. §6 derives  
170 the single-site  $T$ -matrix as the first member of a moment hierarchy in closed  
171 form, and sets out what that gives (§6.7). §§7–8 take up the wavefield basis,  
172 §9 the medium basis and its projection error, §10 the sphere, and §11 the  
173 hierarchy as a voxel scheme. §12 discusses what the results imply and §13  
174 concludes. Every result is derived symbolically in Mathematica and checked  
175 against an independent Python/Fortran implementation or an exact solution.

## 176 2. Setting

177 *The discrete medium.* Space is tiled by cubes of side  $d$  (half-side  $h = d/2$ ).  
178 Cube  $i$  carries a contrast  $\Delta\lambda, \Delta\mu, \Delta\rho$  from an isotropic background with  
179 Lamé parameters  $\lambda, \mu$ , density  $\rho$ , P and S speeds  $\alpha, \beta$  and P-wave modulus  
180  $M_P = \lambda + 2\mu$ . Time dependence is  $e^{-i\omega t}$ ;  $k_P = \omega/\alpha$ ,  $k_S = \omega/\beta$ . Coordinates  
181 are ordered  $(z, x, y)$  with  $z$  down. The field at a voxel is the nine-component  
182 state

$$\psi = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx}), \quad (1)$$

183 displacement followed by engineering strain. A voxel responds with a force  
184 and a moment (stress dipole); the  $9 \times 9$  single-site  $T$ -matrix maps the exciting  
185 state to them, and the  $9 \times 9$  lattice Green’s tensor  $\mathbf{K}(i, j)$  maps a source

186 voxel's force and moment to the state at voxel  $i$ . The Foldy–Lax system is

$$\psi_i = \psi_i^0 + \sum_{j \neq i} \mathbf{K}(i, j) \mathbf{T}_j \psi_j, \quad (2)$$

187 with  $\psi^0$  the incident state.

188 *Where the kernel comes from.* Write the medium as the background plus  
 189 the contrast. With time dependence  $e^{-i\omega t}$  the equation of motion is  $\partial_c \sigma_{bc} +$   
 190  $\rho \omega^2 u_b = 0$ , with  $\sigma_{bc} = c_{bcde} \varepsilon_{de}$  and summation over repeated indices. Split-  
 191 ting  $\rho = \rho_0 + \Delta \rho$  and  $c = c^0 + \delta c$ , with  $\delta c_{bcde} = \Delta \lambda \delta_{bc} \delta_{de} + \Delta \mu (\delta_{bd} \delta_{ce} + \delta_{be} \delta_{cd})$   
 192 the contrast stiffness, and moving the contrast terms to the right leaves the  
 193 background operator on the left and the contrast on the right as a body force,

$$f_b = \omega^2 \Delta \rho u_b + \partial_c m_{bc}, \quad m_{bc} = \delta c_{bcde} \varepsilon_{de}, \quad (3)$$

194 a force from the density contrast, and the divergence of the contrast stress  $m$ ,  
 195 the moment, from the stiffness contrast. With  $\mathbf{G}$  the Green's tensor of the  
 196 background, the displacement is  $u = u^0 + \int \mathbf{G} f \, d\mathbf{y}$ , and the moment term  
 197 integrates by parts,

$$\int G_{ab}(\mathbf{x}-\mathbf{y}) \frac{\partial m_{bc}}{\partial y_c} \, d\mathbf{y} = - \int \frac{\partial G_{ab}}{\partial y_c}(\mathbf{x}-\mathbf{y}) m_{bc} \, d\mathbf{y} = + \int \frac{\partial G_{ab}}{\partial x_c}(\mathbf{x}-\mathbf{y}) m_{bc} \, d\mathbf{y}, \quad (4)$$

198 the boundary term vanishing because  $m$  is confined to the scatterers, and the  
 199 last step holding because  $\mathbf{G}$  depends only on the separation. The moment  
 200 therefore radiates through the *receiver* derivative of  $\mathbf{G}$ , with a plus sign, when  
 201 the moment is the contrast stress itself. With the moment tensor defined  
 202 instead through its equivalent body force  $-\partial_c [M_{bc} \delta(\mathbf{x} - \mathbf{y})]$ , as in Aki and  
 203 Richards [27], the same field reads  $M_{bc} \partial G_{ab} / \partial y_c$ , the source derivative, with

204  $M = -m$ . The two are one convention written two ways; what matters is that  
 205 the derivative in the kernel and the sign in the contrast operator are chosen  
 206 together. The nine-component kernel follows (Fig. 1): its force columns are  
 207  $\mathbf{G}$  and its moment columns the receiver derivatives  $\partial_c G_{ab}$ , symmetrised over  
 208  $(b, c)$  with a factor  $\frac{1}{2}$  on the shear components; its displacement rows are  
 209 those entries and its strain rows their receiver derivatives, symmetrised in  
 210 the same way and doubled on the shear rows to give engineering strain. In  
 211 that convention the contrast operator is

$$\Delta = \text{blockdiag}(\omega^2 \Delta \rho \mathbf{I}_3, \Delta \mathcal{C}), \quad (5)$$

212 where  $\Delta \mathcal{C}$  is the  $6 \times 6$  Voigt form of  $\delta c$ , the contrast stiffness with  $\Delta \lambda + 2\Delta \mu$   
 213 and  $\Delta \lambda$  in the normal block and  $2\Delta \mu$  (not  $\Delta \mu$ ) on the shear diagonal.

214 *Averaging, and collocation.* Three operations in this paper are averages over  
 215 a cell, and they are distinct. The *source-cell average* integrates the kernel  
 216 over the cell that radiates:  $\langle \mathbf{K} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{K}(\mathbf{r} - \mathbf{y}) d\mathbf{y}$  is the field at  $\mathbf{r}$   
 217 of a uniform polarisation of the source cell. Every scheme in this paper uses  
 218 it, and §5 shows it to be exact. *Collocation* fixes where the discrete equation  
 219 is imposed: at the receiving cell's centre, with the internal field of each cell  
 220 uniform, so that the unknown is the field at the centre. The *receiver-cell*  
 221 *average* imposes the equation in the mean over the receiving cell instead;  
 222 with a uniform internal field that is the Galerkin scheme of degree zero,  
 223 called the mean-only voxel below. The two are both *uniform-field* voxels, a  
 224 term that refers to the internal field and not to the test, and they differ only  
 225 in the test function (a delta at the centre against the cell's indicator) and  
 226 in the constant of their  $O((kd)^2)$  error (Table 7). The *cell-mean contrast*,

227 finally, is an average of the material, not of the field: where the medium  
 228 varies within a cell, the contrast is projected onto the cell’s basis, its mean  
 229 for degree  $r = 0$ , rather than sampled at the centre (§§9, 10). In the language  
 230 of the coupled-dipole method, the point dipole is collocation with the point  
 231 kernel, the integrated Green tensor adds the source-cell average, and the  
 232 filtered coupled dipoles replace the point source by one of finite extent.

233 *The laterally uniform layer.* A layer of thickness  $D$  is filled with  $n$  planes  
 234 of voxels,  $d = D/n$ , and illuminated at normal incidence. The field is then  
 235 laterally uniform, only the zero-lateral-wavenumber (specular) component of  
 236 the lattice coupling acts, and Eq. (2) becomes exactly a chain of  $n$  planes  
 237 coupled by the specular lattice sums

$$\mathbf{S}(m) = \sum_{\mathbf{R}} \mathbf{G}(\mathbf{R} + md\hat{\mathbf{z}}), \quad (6)$$

238 where the sum runs over the lattice vectors  $\mathbf{R} = d(0, p, q)$ ,  $p, q$  integers, that  
 239 join the centres of the voxels in one plane (a square lattice of pitch  $d$ ), and  
 240  $m$  is the separation of the two planes in units of  $d$  (Fig. 2a,b). The question  
 241 is whether, and how fast, this chain converges to the continuous layer as  
 242  $n \rightarrow \infty$  at fixed  $D$ .

243 *Implementations.* Every result is derived symbolically in Mathematica and  
 244 compared with an independent numerical implementation or an exact so-  
 245 lution. “The package” below is the Python/Fortran implementation of the  
 246 voxel scheme (`cubic_scattering`): its cube  $T$ -matrix, its lattice kernels and  
 247 its stratified propagator, which serve both as the object under test and, where  
 248 validated, as a second implementation.

### 249 3. The continuous layer and its thin-layer expansion

250 At normal incidence a plane force drives a one-dimensional problem per  
 251 wave type, with modulus  $M$  ( $M_P$  for P,  $\mu$  for S), wavenumber  $k$  and plane  
 252 Green's function  $g(z) = \frac{i}{2Mk} e^{ik|z|}$ . The layer's reflection and transmission  
 253 follow from continuity of displacement and traction at its faces; they agree  
 254 with the stratified propagator used as the reference in this work to  $2.5 \times 10^{-8}$ ,  
 255 P and S (`ContinuumLimit_Reference.wl`).

256 For a layer of thickness  $D$  with modulus  $M_1$  and density  $\rho_1$  in a back-  
 257 ground  $M_0, \rho_0$ , write  $Z_j = \sqrt{M_j \rho_j}$  for the impedances,  $\zeta = Z_1/Z_0$  for their  
 258 ratio and  $k_1 = \omega \sqrt{\rho_1/M_1}$  for the layer's wavenumber. The reflection coeffi-  
 259 cient expands as

$$R = \frac{ik_1 D}{2} (\zeta - \zeta^{-1}) + \frac{(k_1 D)^2}{4} (\zeta^{-2} - \zeta^2) - \frac{i(k_1 D)^3}{24} (\zeta - \zeta^{-1}) (3\zeta^{-2} + 2 + 3\zeta^2) \\ + \frac{(k_1 D)^4}{48} (\zeta^2 - \zeta^{-2}) (3\zeta^2 - 2 + 3\zeta^{-2}) + O(D^5), \quad (7)$$

260 exact in the contrast (`ContinuumLimit_ThinLayer.wl`, which checks each  
 261 coefficient symbolically and the truncated series against the exact layer).  
 262 The first term is

$$\frac{i\omega D}{2} \sqrt{\frac{M_0}{\rho_0}} \left[ \frac{\rho_1 - \rho_0}{M_0} - \rho_0 \left( \frac{1}{M_1} - \frac{1}{M_0} \right) \right] : \quad (8)$$

263 to first order a thin layer is an interface across which the traction jumps by  
 264 the excess inertia and the displacement by the excess compliance. At oblique  
 265 incidence the same statement holds with the first-order system  $\partial_z \mathbf{b} = \mathbf{A} \mathbf{b}$   
 266 for the P-SV state  $\mathbf{b} = (u_x, u_z, t_{xz}, t_{zz})$ : the layer acts to  $O(D)$  as the jump  
 267  $\mathbf{I} + (\mathbf{A}_1 - \mathbf{A}_0)D$ , which reproduces the layer's  $O(D)$  reflection coefficients

268  $R_{PP}, R_{PS}$  to  $10^{-12}$ ; the series truncated at  $O(D^k)$  converges at exactly order  
 269  $k$ ; and a soft thin layer with shear modulus  $\mu_1 = D/\eta$  at fixed compliance  $\eta$   
 270 tends to the linear-slip interface as  $D \rightarrow 0$ . These expansions are the target  
 271 a single plane of voxels must reproduce order by order. A plane of voxels  
 272 that carry Legendre moments of their internal field to degree  $p$  reproduces  
 273 Eq. (7) term by term through  $O(D^{2p+2})$  and first differs at  $D^{2p+3}$  (§8): the  
 274 mean-only voxel reproduces the equivalent interface and the  $D^2$  term, the  
 275 first-moment voxel all four terms shown.

#### 276 4. The specular sums of the point Green's tensor

277 *Why transform the sum.* The specular sum (6) adds, at a receiver, the field  
 278 of every voxel of a plane  $m$  planes away. In that form it is a poor object to  
 279 study. The Green's tensor decays slowly with distance, so the sum converges  
 280 slowly, and it hides the one question that matters here: which part of it is  
 281 the continuum, and which part is the discretisation. The Poisson summation  
 282 formula [28] answers both at once, and is the standard tool for lattice sums  
 283 of wave fields [29]. For a function  $f$  on the plane and the square lattice of  
 284 pitch  $d$  it states

$$\sum_{\mathbf{R}} f(\mathbf{R}) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{f}(\mathbf{g}), \quad \hat{f}(\mathbf{g}) = \int f(\mathbf{x}) e^{-i\mathbf{g} \cdot \mathbf{x}} d^2x, \quad \mathbf{g} = \frac{2\pi}{d}(p, q), \quad (9)$$

285 with  $p, q$  integers: a sum of samples of  $f$  on the lattice equals a sum of samples  
 286 of its Fourier transform on the reciprocal lattice, whose pitch is  $2\pi/d$  (Fig. 3).  
 287 Applied to the specular sum,

$$\mathbf{S}(m) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{\mathbf{G}}(\mathbf{g}; md), \quad (10)$$

288 with  $\hat{\mathbf{G}}(\mathbf{g}; z)$  the two-dimensional (lateral) transform of the Green's tensor  
 289 between planes a distance  $z$  apart. Each term has a physical meaning. Trans-  
 290 formed laterally, the field of a plane of sources is a superposition of plane  
 291 waves, one for each lateral wavenumber, and the lattice selects the wavenum-  
 292 bers  $\mathbf{g}$ . The order  $\mathbf{g} = 0$  is the laterally uniform plane wave that a continuous  
 293 layer radiates: it is the continuum's term. Every other order has  $|\mathbf{g}| \geq 2\pi/d$ ,  
 294 far beyond the wavenumber of the field when the voxels are small compared  
 295 with the wavelength, so it is evanescent and decays away from the plane as  
 296  $e^{-|\mathbf{g}|md} = e^{-2\pi m\sqrt{p^2+q^2}}$ , independently of  $d$ . The transform therefore splits  
 297 the lattice sum exactly into the continuum and a discretisation correction,  
 298 and the correction converges after a few orders.

299 *The local term that survives.* In the strain-from-moment block the evanescent  
 300 orders carry two powers of  $|\mathbf{g}| \propto 1/d$ , so their contribution per voxel volume  
 301 does *not* vanish as  $d \rightarrow 0$ : at  $m = 1$  it is eleven times the continuum  
 302 term (`ContinuumLimit_SpecularSums.wl`, matched to the package's Ewald  
 303 lattice sum to  $5 \times 10^{-13}$ ). This surviving term is static. Summed over all  
 304 planes in slab order, the static lattice sum of the point kernel is exactly  
 305 the depolarisation of a uniform plate ( $-1/M_P$  on  $\varepsilon_{zz} \leftarrow M_{zz}$  and  $-1/(2\mu)$   
 306 on the normal shears, per unit volume) plus a cubic-symmetric remainder  
 307 (`ContinuumLimit_StaticLocal.wl`, to  $2 \times 10^{-10}$ ). The point kernel therefore  
 308 reproduces the continuum only if that remainder is cancelled by the voxel's  
 309 own self term.

## 310 5. The tiling identity

311 Replace the point coupling by the *source-cell average*  $\langle \mathbf{G} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{G}(\mathbf{r} -$   
 312  $\mathbf{y}) d\mathbf{y}$ , the field at a voxel centre of a uniform polarisation of the source voxel.  
 313 Each Poisson order in Eq. (10) is then multiplied by the cube’s form factor

$$F(\mathbf{g}) = \text{sinc}(g_x h) \text{sinc}(g_y h), \quad (11)$$

314 which vanishes at every  $\mathbf{g} \neq 0$  of the reciprocal lattice since  $\text{sinc}(p\pi) = 0$  for  
 315 integer  $p \neq 0$ . This is the Fourier form of the fact that cubes tile the plane,  
 316  $\sum_{\mathbf{R}} \chi(\mathbf{x} - \mathbf{R}) = 1$  for the cube’s indicator  $\chi$ : a partition of unity (Fig. 4).  
 317 Two exact statements follow.

318 *Between planes.* For  $m \neq 0$  every evanescent order is removed and

$$\mathbf{S}_{\text{avg}}(m) = \frac{1}{d^2} \sum_{W=P,S} \hat{\mathbf{G}}_W(0; md) \text{sinc}(k_W h), \quad (12)$$

319 the continuum’s plane-wave term times the vertical form factor. The pack-  
 320 age’s cell-averaged kernel agrees in all 81 entries to  $2.8 \times 10^{-16}$  (`ContinuumLimit_`  
 321 `AveragedSums.wl`).

322 *Within a plane.* The sum over *all* cells of the plane, the voxel’s own included,  
 323 is the field at the mid-plane of a uniform plate of thickness  $d$  (Fig. 2c):

$$d^3 \mathbf{S}_{\text{avg}}(0) + \mathbf{S}_{\text{self}} = d^3 \mathbf{P}_0, \quad \mathbf{S}_{\text{self}} = \int_{\text{cube}} \mathbf{G}(-\mathbf{y}) d\mathbf{y}, \quad (13)$$

324 where  $\mathbf{P}_0 = d^{-3} \int_{-h}^h \mathbf{G}_{\text{plate}}(-z') dz'$  is elementary: in the one-dimensional  
 325 transform each entry is  $c_0 + A/(Mk_z^2 - \rho\omega^2)$ , so it averages to  $c_0/d + A(e^{ikh} -$   
 326  $1)/(dMk^2)$ . Its only nonzero entries are  $u \leftarrow f$  on the diagonal and the  
 327 three normal strain-from-moment entries, whose local weights  $-1/M_P$  and

328  $-1/(2\mu)$  are the plate depolarisation of §4. Statically the identity holds for  
 329 all 21 independent components of the strain-from-moment tensor to  $4 \times 10^{-9}$ ,  
 330 with every cell integral evaluated analytically (below).

331 *Consequence: an exact kernel with no lattice sum.* At normal incidence the  
 332 same-plane block of the cell-averaged Foldy–Lax kernel is

$$\mathbf{S}_{\text{avg}}(0) = \mathbf{P}_0 - \mathbf{S}_{\text{self}}/d^3, \quad (14)$$

333 with no Ewald splitting, near shell or asymptotic tail. The route it replaces  
 334 (Ewald summation with a near shell averaged by quadrature and an  $O(d^2)$   
 335 Taylor tail beyond it) carries a static strain-from-moment error of  $9.4 \times 10^{-4}$   
 336 at its default settings, falling to  $2.6 \times 10^{-5}$  only when both the shell radius  
 337 and the quadrature order are raised; Eq. (14) is now the package’s kernel at  
 338 normal incidence.

339 *Cell integrals without quadrature.* Write the static (Kelvin) tensor as  $G_{bc} =$   
 340  $(c_A + c_B) \delta_{bc}/r - c_B \partial_b \partial_c r$  with  $c_B = (1/\mu - 1/M_P)/8\pi$  and  $c_A = 1/(4\pi\mu) - c_B$ .  
 341 Every cell integral of its derivatives is a cell integral of derivatives of  $1/r$  and  
 342  $r$ . One application of the divergence theorem moves it onto faces  $x_p = c_p \pm h$ ,  
 343 which never contain the origin, and there the two-dimensional antiderivative  
 344 in the face coordinates is continuous, so the face integral is a sum over the  
 345 face’s corners. The distributional term at  $r = 0$  is carried by the divergence  
 346 theorem itself. (A triple antiderivative evaluated at the cube’s corners is  
 347 *not* usable for repeated derivatives: its arctangent terms jump across the  
 348 coordinate planes through the origin, and differentiating twice there discards  
 349 exactly that term.) The self term so obtained reproduces the classical closed  
 350 form of the cube’s depolarisation [30, 31] to machine precision.

## 351 6. The single site: the cube $T$ -matrix from moment integrals

352 Every error of a voxel scheme is made at the single site (§12), so the single  
353 site is derived here rather than imported. The classical route imports it. For  
354 an ellipsoid, Eshelby’s uniformity theorem makes the internal strain under a  
355 uniform load uniform, with a depolarisation tensor in closed form [4, 32, 33];  
356 its dynamic generalisation is known for spheres and ellipsoids [34, 35, 36].  
357 For a cube or a general polyhedron the static field of a uniform eigenstrain  
358 is also in closed form [30, 37, 31], and it has been computed for a cuboid in  
359 an anisotropic matrix [38], but the strain it produces is not uniform, so there  
360 is no exact uniform-strain inclusion to import. Closed-form Eshelby tensors  
361 for polyhedral inclusions with polynomial eigenstrains have since been given  
362 [39], the dynamic inclusion problem has been formulated for shapes other  
363 than the ellipsoid, cubes among them [40], and the Green’s tensor has been  
364 integrated analytically over a cuboid for the coupled-dipole method, to an  
365 error of  $O((kd)^4)$  [41]; these are the static and electromagnetic counterparts of  
366 individual moments below. In scattering theory the volume integral equation  
367 of an elastic inclusion [20] is closed in the same spirit by a self-consistent  
368 or effective-field approximation [42, 43]: a first-order (Born) scatterer [44]  
369 renormalised by the geometric series of its own near field, the step that turns  
370 a contrast into an effective contrast. Nestor [21] introduced the single site  
371 in this form, summing the series of the cell’s self integral, with the singular  
372 near field evaluated over an infinitesimal cell. The moments below evaluate  
373 the same self integral over the finite cube, as a distribution. The elastic  $T$ -  
374 matrix of Waterman [45] represents a scatterer in spherical multipoles, the  
375 natural basis for a sphere; a voxel in a lattice needs one compatible with the

376 lattice, and the moments over the cube provide it. The construction below  
 377 makes the approximation of a voxel explicit rather than implicit: it derives  
 378 the cube’s  $T$ -matrix from the moments of the elastodynamic Green’s tensor  
 379 over the cube, with the uniform-strain closure as the first truncation of a  
 380 hierarchy.

381 *Its origin in the equivalent inclusion method.* In its static form the con-  
 382 struction is the equivalent inclusion method with polynomial eigenstrains  
 383 of Moschovidis and Mura [46]: the eigenstrain is expanded about the centre  
 384 of the inhomogeneity, the consistency equation is expanded there in a Taylor  
 385 series, and the coefficient of each monomial is equated, which amounts to  
 386 imposing the Lippmann–Schwinger equation and its derivatives at the cen-  
 387 tre [47, §2.2]. Johnson et al. [48, 49] applied an expansion about the origin  
 388 to a cuboidal inhomogeneity, and Wu et al. [39] expanded the eigenstrain  
 389 of a polyhedron to second degree about its centroid; both report that the  
 390 expansion about the centre does not follow the field near the edges, corners  
 391 or vertices. Fu and Mura [34] extended the method to time-harmonic waves  
 392 for an ellipsoid, with a second unknown for the density mismatch, and in the  
 393 cases they report evaluated it with uniform eigenstrains. Brisard et al. [47]  
 394 replaced the Taylor form by a Galerkin form of the same equation, because  
 395 the Taylor form need not give an invertible system and a higher degree need  
 396 not improve it. What is added here, to our knowledge, is the elastodynamic  
 397 hierarchy, with density and stiffness contrasts together, to any degree and for  
 398 a body that is not an ellipsoid; its moments as distributions in closed form;  
 399 and the structure and the error of its truncations, tested degree by degree  
 400 against the exact solution for a sphere. The moments here are integrals of

the Green's tensor against monomials over the body. They are not moments in the sense of the method of moments [50], of which the hierarchy is the instance with monomials about the centre as expansion functions and the  $\delta$  function and its derivatives at the centre as testing functions; and the exact static response of an inclusion of any shape to polynomial loads is described by the polarisation and elastic moment tensors of Ammari and Kang [51].

### 6.1. The closed hierarchy

Let the voxel be  $V = [-h, h]^3$  with contrast  $\delta c_{ijkl} = \Delta\lambda \delta_{ij}\delta_{kl} + \Delta\mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$  and  $\Delta\rho$ . The hierarchy is stated below for any degree; its third degree is written out first, being the lowest at which the stiffness channel converges at fourth order (§6.4). Expand the internal displacement about the centre to third degree and project the Lippmann–Schwinger equation onto 1,  $x_p$ ,  $x_px_q$  and  $x_px_qx_v$ . The unknowns are  $u_i(0)$ ,  $\partial_p u_i(0)$ ,  $\partial_p \partial_q u_i(0)$  and  $\partial_p \partial_q \partial_v u_i(0)$ , and the result is the closed system

$$(\delta_{ij} - \omega^2 \Delta\rho \mathbf{G}_{ij})u_j + (\mathbf{N}_{in,k}^r \delta c_{nksj} - \frac{1}{2}\omega^2 \Delta\rho \mathbf{K}_{ij}^{rs})\partial_r \partial_s u_j = u_i^0, \quad (15)$$

$$\begin{aligned} &(\delta_{pr}\delta_{ij} + \omega^2 \Delta\rho \mathbf{N}_{ij,p}^r - \mathbf{M}_{in,pk} \delta c_{nkrj})\partial_r u_j \\ &+ (\frac{1}{6}\omega^2 \Delta\rho \mathbf{B}_{ij,p}^{rst} - \frac{1}{2}\mathbf{P}_{in,pk}^{st} \delta c_{nkrj})\partial_r \partial_s \partial_t u_j = \partial_p u_i^0, \end{aligned} \quad (16)$$

$$-\omega^2 \Delta\rho \mathbf{M}_{ij,pq} u_j + (\delta_{qr}\delta_{ps}\delta_{ij} + \mathbf{Q}_{in,pqk}^r \delta c_{nksj} - \frac{1}{2}\omega^2 \Delta\rho \mathbf{P}_{ij,pq}^{rs})\partial_r \partial_s u_j = \partial_p \partial_q u_i^0, \quad (17)$$

$$\begin{aligned} &(\omega^2 \Delta\rho \mathbf{Q}_{ij,pqv}^r - \mathbf{X}_{in,pqvk} \delta c_{nkrj})\partial_r u_j \\ &+ (\delta_{pr}\delta_{qs}\delta_{vt}\delta_{ij} + \frac{1}{6}\omega^2 \Delta\rho \mathbf{J}_{ij,pqv}^{rst} - \frac{1}{2}\mathbf{Z}_{in,pqvk}^{st} \delta c_{nkrj})\partial_r \partial_s \partial_t u_j = \partial_p \partial_q \partial_v u_i^0, \end{aligned} \quad (18)$$

all fields evaluated at the centre. Dropping the third gradients, that is Eq. (18) and the second line of Eq. (16), leaves the system of second degree; dropping the second gradients as well, that is Eq. (17) and the  $\partial\partial u$  terms of Eq. (15), leaves the uniform-strain closure. Equations (15) and (17) are the same at second and third degree, because every term the third degree would add to them vanishes by parity (§6.3). The coefficients are ten moments of the background Green's tensor  $\mathbf{G}$  over the voxel

$$\begin{aligned}
\mathbf{G}_{ij} &= \int_V G_{ij} \, dV, & \mathbf{K}_{ij}^{rs} &= \int_V G_{ij} x_r x_s \, dV, & \mathbf{N}_{in,k}^r &= \int_V \partial'_k G_{in} x_r \, dV, \\
\mathbf{M}_{in,pk} &= \int_V \partial'_p \partial'_k G_{in} \, dV, & \mathbf{P}_{ij,pq}^{rs} &= \int_V \partial'_q \partial'_p G_{ij} x_r x_s \, dV, & \mathbf{Q}_{in,pqk}^r &= \int_V \partial'_q \partial'_p \partial'_k G_{in} x_r \, dV, \\
\mathbf{B}_{ij,p}^{rst} &= \int_V \partial'_p G_{ij} x_r x_s x_t \, dV, & \mathbf{J}_{ij,pqv}^{rst} &= \int_V \partial'_v \partial'_q \partial'_p G_{ij} x_r x_s x_t \, dV, & \mathbf{X}_{in,pqvk} &= \int_V \partial'_v \partial'_q \partial'_p \partial'_k G_{in} \, dV, \\
\mathbf{Z}_{in,pqvk}^{rs} &= \int_V \partial'_v \partial'_q \partial'_p \partial'_k G_{in} x_r x_s \, dV,
\end{aligned} \tag{19}$$

the first six being those of the second degree and the last four those the third gradients add. The derivatives are taken with respect to the source point,  $\partial' = -\partial$ . That sign reaches the moments with an odd number of derivatives,  $\mathbf{N}$ ,  $\mathbf{Q}$ ,  $\mathbf{B}$  and  $\mathbf{J}$ , and no symmetry, parity or scaling check can detect it, so it is fixed by convention. The moments are graded by  $(D, W)$ , the number of derivatives and the degree of the weight:  $\mathbf{G}(0, 0)$ ,  $\mathbf{K}(0, 2)$ ,  $\mathbf{N}(1, 1)$ ,  $\mathbf{M}(2, 0)$ ,  $\mathbf{P}(2, 2)$ ,  $\mathbf{Q}(3, 1)$ , and  $\mathbf{B}(1, 3)$ ,  $\mathbf{J}(3, 3)$ ,  $\mathbf{X}(4, 0)$ ,  $\mathbf{Z}(4, 2)$ .

*The hierarchy at any degree.* Nothing in the construction is special to the third degree. Let the internal displacement be expanded to degree  $\ell$ ,

$$u_j(\mathbf{x}) = \sum_{|W| \leq \ell} \frac{x_W}{|W|!} U_{j;W}, \quad U_{j;W} = \partial_W u_j(0), \quad x_W = x_{w_1} \cdots x_{w_m}, \tag{20}$$

the sum running over ordered sets of indices  $W = (w_1, \dots, w_m)$ , and let the strain be its derivative. Write the moments with any number of derivatives and any weight as

$$\mathbf{g}_{in}[D; W] = \langle \partial_D G_{in}, x_W \mathbf{1}_V \rangle, \quad (21)$$

the derivatives being those of  $G_{in}$  as a function of its argument and the pairing distributional (§6.2). The ten moments of Eq. (19) are these  $\mathbf{g}$  themselves: they are the integrals of  $\partial'_D G_{in}(\mathbf{0} - \mathbf{x})$ , and the factor  $(-1)^{|D|}$  of the source derivative cancels the parity  $(-1)^{|D|}$  of  $\partial_D G_{in}$ , so that, for example,  $\mathbf{N}_{in,k}^r = \mathbf{g}_{in}[k; r]$  and  $\mathbf{Z}_{in,pqvk}^{rs} = \mathbf{g}_{in}[pqvk; rs]$ . Beyond the third degree the moments are written as  $\mathbf{g}$  only. Imposing the Lippmann–Schwinger equation and its derivatives to order  $\ell$  at the centre gives, for every set of derivative indices  $P$  with  $|P| \leq \ell$ ,

$$U_{i;P} - (-1)^{|P|} \omega^2 \Delta \rho \sum_{|W| \leq \ell} \frac{\mathbf{g}_{ij}[P; W]}{|W|!} U_{j;W} + (-1)^{|P|} \sum_{|W| \leq \ell-1} \frac{\mathbf{g}_{in}[Pk; W]}{|W|!} \delta c_{nkrj} U_{j;rW} = \partial_P u_i^0(0). \quad (22)$$

Equations (15)–(18) are the case  $\ell = 3$ , with the moments of Eq. (19) put for the  $\mathbf{g}$ ; their truncations above are  $\ell = 2$  and the uniform-strain closure  $\ell = 1$ . With the field rather than the eigenstrain as the unknown, the moments  $\mathbf{g}[D; W]$  play the part of the polynomial Eshelby tensors of the equivalent inclusion method [33, Ch. 4], which Brisard et al. [47] call influence pseudotensors, here for the elastodynamic kernel and for the density channel as well. For an ellipsoid the static self term of a polynomial eigenstrain is itself a polynomial of the same degree, so the expansion about the centre is exact in the self term and approximate only in the interaction with other bodies [47]; for a cube it is approximate in both. The cube is centrosymmetric, so a

452 moment vanishes unless  $|D| + |W|$  is even, and the system needs the grades

$$\begin{aligned}
(D, W) &= (m, W), \quad W \leq \ell && \text{(density),} \\
(D, W) &= (m + 1, W), \quad W \leq \ell - 1 && \text{(stiffness),}
\end{aligned}
\quad m = 0, \dots, \ell, \quad D+W \text{ even.}
\tag{23}$$

453 Table 1 lists what each degree adds. The unknowns number  $3\binom{\ell+3}{3}$ : 12, 30, 60 for  $\ell = 1, 2, 3$ .

Table 1: The grades  $(D, W)$  of moments,  $D$  derivatives on the Green's tensor and a weight of degree  $W$ , that the hierarchy of degree  $\ell$  needs beyond those of degree  $\ell - 1$ .

| $\ell$ | new grades                                     | moments in all      |
|--------|------------------------------------------------|---------------------|
| 1      | (0, 0), (1, 1), (2, 0)                         | 3                   |
| 2      | (0, 2), (2, 2), (3, 1)                         | 6: G, K, N, M, P, Q |
| 3      | (1, 3), (3, 3), (4, 0), (4, 2)                 | 10: with B, J, X, Z |
| 4      | (0, 4), (2, 4), (4, 4), (5, 1), (5, 3)         | 15                  |
| 5      | (1, 5), (3, 5), (5, 5), (6, 0), (6, 2), (6, 4) | 21                  |

454

455 *Two independent systems.* In either sum of Eq. (22) the unknown  $U_{i;P}$  is  
456 coupled to  $U_{j;W}$ , or to  $U_{j;rW}$ , through a moment of grade  $(|P|, |W|)$  or  $(|P| +$   
457  $1, |W|)$ , and on a centrosymmetric body that moment vanishes unless  $|P|$  and  
458 the degree of the unknown it multiplies have the same parity. The hierarchy  
459 of any degree therefore separates exactly, at every frequency and contrast,  
460 into two systems: one for the even gradients  $u, \partial\partial u, \dots$  and one for the odd  
461 gradients  $\partial u, \partial\partial\partial u, \dots$ . Raising the degree from odd to even adds a block  
462 to the even system alone, and from even to odd a block to the odd system  
463 alone. At  $\ell = 3$  the even system is Eqs. (15) and (17), unchanged from  $\ell = 2$ ,  
464 and the odd system is Eqs. (16) and (18). Of the sixty unknowns, twenty-one

465 belong to the even system and thirty-nine to the odd. The leading radiation  
 466 of a density contrast,  $\int_V \Delta \rho u \, dV$ , is drawn from the even system, and that of  
 467 a stiffness contrast,  $\int_V \delta c : \nabla u \, dV$ , from the odd one. The separation is thus  
 468 the structure behind the result of §6.4, that the blocks improve the density  
 469 and the stiffness channels alternately. At first degree it is implicit in the  
 470 dynamic equivalent inclusion method for an ellipsoid, where the force and  
 471 the strain systems uncouple at the centre [34]; here it holds at every degree,  
 472 for any centrosymmetric body.

## 473 6.2. The moments as distributions

474 Write the elastodynamic tensor as

$$G_{in}(\mathbf{r}) = \frac{1}{4\pi\mu} \left[ \delta_{in} g_S(r) + \frac{1}{k_S^2} \partial_i \partial_n (g_S(r) - g_P(r)) \right], \quad g_c(r) = \frac{e^{ik_c r}}{r} \quad (24)$$

475 [52, 27]. Expanding  $g_c$  in powers of  $r$  reduces every moment in Eq. (19) to a  
 476 finite sum of scalar moments

$$E[m; d_1 \dots d_D; w_1 \dots w_W] = \langle \partial_{d_1} \dots \partial_{d_D} r^m, x_{w_1} \dots x_{w_W} \mathbf{1}_V \rangle. \quad (25)$$

477 For  $m = -1$  and  $D \geq 2$  the classical integrand is not absolutely integrable.  
 478  $\partial^2(1/r)$  decays as  $r^{-3}$ , its angular average vanishes, and the value of the inte-  
 479 gral depends on the shape of the region excluded around the origin. This is  
 480 the source of the  $\delta$ -function term of the inclusion problem, which a numerical  
 481 quadrature misses. Without it the depolarisation is wrong; for a sphere it  
 482 vanishes altogether (§6.3). It is not added afterwards here. The derivatives  
 483 are distributional from the start, and each is moved onto the voxel's faces by  
 484 pairing with the test function  $w \mathbf{1}_V$ , using  $\partial_q \mathbf{1}_V = -n_q \delta_{\partial V}$ :

$$\langle \partial_q T, w \mathbf{1}_V \rangle = \oint_{\partial V} n_q w T \, dA - \langle T, (\partial_q w) \mathbf{1}_V \rangle. \quad (26)$$

485 The surface term is evaluated on the faces, a distance  $h$  from the origin, where  
 486 every derivative of  $r^m$  is smooth. The recursion lowers  $D$  and  $W$  together  
 487 and ends on convergent integrals. The face integrals are done in closed form  
 488 by the corner sums of §5. The sharp check is the sum rule

$$\sum_p E[-1; p, p; ] = \nabla^2(1/r) \text{ paired with } \mathbf{1}_V = -4\pi, \quad (27)$$

489 which the face route satisfies exactly and a spherical excision returns as  
 490 0. The closed forms contain logarithms of the units  $(2 \pm \sqrt{3})^n$  of the Pell  
 491 equation  $x^2 - 3y^2 = 1$ . Left unreduced, these cancel catastrophically: for  
 492 example,  $\log(72010600134783751 - 41575339372323900\sqrt{3})$  is a cancellation  
 493 of 35 digits. They are therefore reduced to  $n \log(2 - \sqrt{3})$ , which is exact.  
 494 Because every power of  $r$  is paired this way, the dynamic moments keep the  
 495 even-power terms of the expansion  $(k^2r, r^3, \dots)$  as well as the odd ones.  
 496 A route that integrates only the polynomial (odd-power) terms of  $\mathbf{G}$  misses  
 497 those terms, at a real  $O((kh)^2)$  cost (§6.6).

498 *Every grade, in closed form on four constants.* Closed forms for monomial-  
 499 weighted integrals of  $1/r$  over bricks are known [53], as are closed forms for  
 500 the potential and its first two derivatives over polyhedra with polynomial  
 501 density, from the gravity literature [54]. The engine used here differs in  
 502 taking derivatives of any order as distributions, in treating the kernel  $r$  as  
 503 well as  $1/r$ , and in the even-power terms of the dynamic series. The recursion  
 504 (26) does not depend on the grade, so the same engine gives every scalar  
 505 moment of Table 1. Each is evaluated once and stored. The scalar moments  
 506 of every grade through  $\ell = 5$ , with up to six derivatives and weights to  
 507 the fifth degree, have been evaluated for the kernels  $r^{-1}$  and  $r$ : 4224 closed

508 forms. (The second term of Eq. (24) places two further derivatives on  $r$ ;  
 509 those moments are stored through  $\ell = 3$  and are evaluated numerically, by  
 510 the second route below, beyond it.) Every one of them is a power of the  
 511 voxel's width times

$$q_0 + q_1 \sqrt{3} + q_2 \pi + q_3 \log(2 - \sqrt{3}), \quad q_i \text{ rational.} \quad (28)$$

512 The integration returns many of them in other forms, with  $\log(\sqrt{3} - 1)$ ,  
 513  $\log(1 + \sqrt{3})$ , inverse hyperbolic tangents and logarithms of powers of two,  
 514 which are equal in value but not canonical: two equal moments can then differ  
 515 in appearance, and sums of large logarithms that cancel lose digits. Each  
 516 stored form is therefore reduced to Eq. (28) by an integer-relation search at  
 517 400 digits, with  $\log 2$  and  $\log 3$  admitted to the search, and the reduction  
 518 is accepted only if it agrees with the stored form to 1200 digits. All 4224  
 519 reduce, and the coefficients of  $\log 2$  and  $\log 3$  are zero in every one.

520 Every grade is gated by two checks whose answers do not depend on  
 521 the engine, the Laplacian sum rules of Eq. (A.1), which test the  $\delta$  function  
 522 of the inclusion problem and its derivatives where a quadrature fails, and  
 523 confirmed by a second route that shares nothing with the faces of the cube,  
 524 to  $1.1 \times 10^{-12}$  or better (Appendix Appendix A).

### 525 *6.3. Parity, and the nine-component block*

526 The cube is centrosymmetric. A moment that couples a field quantity of  
 527 degree  $a$  to one of degree  $b$  therefore vanishes unless  $a + b$  is even.  $u$ ,  $\partial u$  and  
 528  $\partial \partial u$  have degrees 0, 1 and 2, so at second degree Eq. (16) contains the first  
 529 gradient alone:  $\partial u$  is determined by its own block and couples to neither  $u$   
 530 nor  $\partial \partial u$ . At any degree it couples to nothing of even degree; from  $\ell = 3$

531 it couples to  $\partial\partial\partial u$  (the second line of Eq. (16)). The leading scattering by  
532 a stiffness contrast is the stress dipole  $\int_V \delta c : \nabla u \, dV$ , which needs only  $\partial u$ .  
533 The leading far field of the stiffness contrast is therefore fixed by Eq. (16),  
534 and *no treatment of the second-gradient block (17) can change it*. What  
535 Eq. (17) contributes reaches the far field only through the  $O((kh)^2)$  coupling  
536 in Eq. (15). That is the order, and the origin (the variation of the field across  
537 the voxel), of the collocation error of §7, which the first-moment voxel of §8  
538 removes, as does the hierarchy itself once its third-gradient block is kept  
539 (§6.4).

540 With  $\Delta\rho = 0$ , Eq. (16) at second degree is the  $9 \times 9$  operator  $\delta_{pr}\delta_{ij} -$   
541  $\mathbf{M}_{in,pk} \delta c_{nkrj}$ , built from  $\mathbf{M}$  alone. Under  $O_h$  the nine gradient components  
542 decompose as  $A_{1g} \oplus E_g \oplus T_{1g} \oplus T_{2g}$ , and the operator is diagonal in that basis.  
543 Its eigenvalues are the inverse strain concentrations:

$$A_{1g} \text{ (dilatation)} : \quad 1 + \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{\lambda + 2\mu}, \quad (29)$$

$$T_{2g} \text{ (off-diagonal shear)} : \quad 1 + 2\Delta\mu S_{\text{off}}, \quad S_{\text{off}} = \frac{\pi(\lambda + 2\mu) - \sqrt{3}(\lambda + \mu)}{3\pi\mu(\lambda + 2\mu)}, \quad (30)$$

$$E_g \text{ (diagonal shear)} : \quad 1 + 2\Delta\mu S_{\text{diag}}, \quad S_{\text{diag}} = \frac{3\sqrt{3}(\lambda + \mu) + 2\pi\mu}{6\pi\mu(\lambda + 2\mu)}. \quad (31)$$

544 The density channel is the displacement block of Eq. (15) with the second  
545 gradients dropped, an amplification  $(1 - \omega^2\Delta\rho \mathbf{G})^{-1}$  with  $\mathbf{G}$  the cube integral  
546 of  $\mathbf{G}$ .

547 *The cube against the sphere.* For a sphere, Eshelby's tensor gives a dilata-  
548 tional channel  $1 + \Delta K / (\lambda + 2\mu)$ , with  $\Delta K = \Delta\lambda + \frac{2}{3}\Delta\mu$ , and a single deviatoric

channel  $1 + 2\Delta\mu S_{\text{sph}}$ , with  $2\mu S_{\text{sph}} = 2(4 - 5\nu)/[15(1 - \nu)]$ . The cube's dilatational channel, Eq. (29), is the sphere's exactly: its eigenvalue involves  $\mathbf{M}$  only through the contraction  $\int_V \partial_i \partial_n G_{in} dV$ , and the static Navier equation makes  $\partial_i \partial_n G_{in} = -\delta(\mathbf{r})/(\lambda + 2\mu)$ , whose integral does not depend on the shape. The cube's shear channels differ from each other. Their isotropic average, however, is the sphere's, also exactly:

$$\frac{1}{5}(3S_{\text{off}} + 2S_{\text{diag}}) = S_{\text{sph}}, \quad S_{\text{diag}} - S_{\text{off}} = \frac{(\lambda + \mu)(5\sqrt{3} - 2\pi)}{6\pi\mu(\lambda + 2\mu)} > 0. \quad (32)$$

The  $\sqrt{3}$  terms cancel in the average. What a cube adds to a sphere at this order is therefore a pure cubic anisotropy. It splits the deviatoric channel about the sphere's value, weighted 3 : 2 by the dimensions of  $T_{2g}$  and  $E_g$ , and the diagonal shear is the more depolarised. At the background of §7 ( $\nu = 7/32$ ),  $2\mu S_{\text{off}} = 0.4314$ ,  $2\mu S_{\text{diag}} = 0.5929$  and the sphere's value is 0.4960.

Where Eshelby enters. Equations (29)–(31) follow from the moment integrals and the group theory alone. No ellipsoidal result is used, and no  $\delta$  term is added by hand. What corresponds to Eshelby is the *truncation*. Keeping  $\partial u$  and discarding  $\partial \partial u$  is exactly the uniform-internal-strain ansatz, and by the uniformity theorem that ansatz is exact if and only if the inclusion is an ellipsoid. For a sphere the construction is therefore exact in the static limit, and for a cube it is an approximation. The hierarchy (22) states that approximation as the omitted blocks, rather than as an unquantified shape correction. A polynomial expansion about the centre cannot represent the edge and corner concentrations of an isolated cube's internal field at any finite degree; Table 4 measures what each degree leaves. A voxel in a uniform layer

572 has no such concentrations: its neighbours have the same properties, so its  
 573 internal field is the continuum's, which is smooth. That is why a low-degree  
 574 internal field suffices in a lattice (§§7–8) although it is only an approximation  
 575 for an isolated cube.

576 *Against the exact sphere.* These statements are tested against the exact  
 577 elastic Mie solution, with no Eshelby formula in between (Appendix Ap-  
 578 pendix F). Applied to a ball, the construction's dilatational and shear chan-  
 579 nels equal Mie's static concentrations to  $2 \times 10^{-10}$ , and the cube's dilata-  
 580 tional channel and isotropic shear average meet the same numbers; taken  
 581 as a principal value the moment has no depolarisation at all, so the whole  
 582 static response is the  $\delta$  term. At finite frequency the closure departs from  
 583 Mie only by a real term of order  $(ka)^2$  at first order in the contrast, given  
 584 in closed form in Eq. (F.1); the radiation damping is exact. That term is  
 585 the uniform-field ansatz's own error, the order the hierarchy assigns to the  
 586 omitted second gradients.

#### 587 6.4. What each block of the hierarchy buys

588 A hierarchy invites the question of what its next block is worth. The  
 589 answer has a simple form, and it is not the one a count of unknowns suggests:  
 590 the blocks help in pairs.

591 *The experiment.* The layer of §7 at normal incidence is cut into  $n$  cells, and  
 592 in each the hierarchy of degree  $\ell$  is imposed: the displacement is its Taylor  
 593 polynomial of degree  $\ell$  about the centre, the strain the derivative of that, and  
 594 the equation and its first  $\ell$  derivatives hold at the centre, with the  $\delta$  terms  
 595 kept. In one dimension the moments are elementary and the exact layer is

known, so the order of each truncation can be measured for a density contrast,  
a modulus contrast and both. The schemes are computed symbolically to 60  
digits for  $\ell = 1$  to 6 and, to  $\ell = 3$ , independently in double precision.

*The prediction.* A cell radiates the integral of its source against the kernel.  
If the source is held to a polynomial of degree  $d$  about the centre, the first  
power it omits is  $d + 1$ . When  $d + 1$  is odd, that term integrates to zero  
against the even part of the kernel and the error is two orders smaller; when  
 $d + 1$  is even it does not. The density source is the displacement, of degree  $\ell$ ;  
the stiffness source is the strain, of degree  $\ell - 1$ . So a density contrast gains  
an order pair at every even  $\ell$ , and a stiffness contrast at every odd  $\ell$ . This is  
the parity of §6.3, read as an order of convergence.

Table 2: The order of convergence of the hierarchy of degree  $\ell$  on the layer, measured from 16 to 32 cells in reflection and in transmission at  $\omega = 300 \text{ rad s}^{-1}$ , and its reflection error on 16 cells with both contrasts. Every measured order equals the predicted one to the digits shown.

|                       | $\ell = 1$           | 2                    | 3                     | 4                     | 5                     | 6                     |
|-----------------------|----------------------|----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| density contrast only | 2                    | 4                    | 4                     | 6                     | 6                     | 8                     |
| modulus contrast only | 2                    | 2                    | 4                     | 4                     | 6                     | 6                     |
| both                  | 2                    | 2                    | 4                     | 4                     | 6                     | 6                     |
| error, both, $n = 16$ | $5.4 \times 10^{-6}$ | $1.4 \times 10^{-6}$ | $5.8 \times 10^{-12}$ | $9.5 \times 10^{-13}$ | $2.6 \times 10^{-18}$ | $3.1 \times 10^{-19}$ |

*The result.* Table 2: all thirty-six orders are as predicted. The second gradi-  
ents raise the density channel to fourth order and leave the stiffness channel  
at second, with a constant three to four times smaller in reflection; the third  
gradients bring the stiffness channel to fourth order; and so on in pairs.

611 With both contrasts, fourth order therefore needs  $\ell = 3$  and sixth order  
612  $\ell = 5$ . The truncation of Eqs. (15)–(18) at second gradients is one block  
613 short of the next order for a stiffness contrast, which is why it improves the  
614 uniform-strain closure by a constant and not by an order.

615 *The error in closed form.* At first order in the contrast a cell whose source is  
616 held to its Taylor polynomial of degree  $d$  returns the exact term times  $1 + E_d$ ,  
617 where, with  $h$  the half-width,

$$E_d = \begin{cases} (-1)^{(d-1)/2} \frac{(kh)^{d+1}}{(d+2)!} & d \text{ odd, } R \text{ and } T, \\ (-1)^{d/2} \frac{(kh)^{d+2}}{(d+2)!} & d \text{ even, } R, \\ (-1)^{d/2+1} \frac{(d+1)(kh)^{d+2}}{(d+3)!} & d \text{ even, } T, \end{cases} \quad (33)$$

618  $R$  and  $T$  standing for reflection and transmission, with  $d = \ell$  for the den-  
619 sity term and  $d = \ell - 1$  for the stiffness term. It follows from the series  
620 of  $\int_{-1}^1 P_d(ikhx) e^{\mp ikhx} dx$  with  $P_d$  the Taylor polynomial of  $e^{ikhx}$ , and it is  
621 confirmed symbolically for  $d = 0$  to 6 and against the scheme on one cell,  
622 at a contrast of  $10^{-20}$  and  $kh = 0.03$ , to 0.5% in all twelve cases. For  
623  $d = 0$  it is  $(kd)^2/8$  in reflection and  $-(kd)^2/24$  in transmission: the con-  
624 stants of the collocation scheme, Eq. (36), which are thus the first rung  
625 of a ladder (`ContinuumLimit_GradientHierarchy.wl`, `scripts/measure_`  
626 `layer_taylor_hierarchy.py`).

### 627 6.5. The single site against a sphere and a cube

628 The hierarchy is a statement about one scatterer, so it is tested first on  
629 one scatterer. The assembly of Eq. (22), with the full tensor moments in

630 their dynamic form (the series of  $e^{ikr}/r$  taken to  $r^{11}$ ), is the same for any  
631 centrosymmetric body; only the scalar moments change.

632 *A sphere.* For a ball the scalar moments are elementary, by the surface route  
633 of Eq. (26) on  $|\mathbf{r}| = a$ , and the exact answer is known. A homogeneous  
634 sphere of radius  $a = 10$  m with the contrast of §7 is illuminated by a P plane  
635 wave; the hierarchy’s polynomial source is radiated to nine scattering angles  
636 between  $11^\circ$  and  $169^\circ$ , and the error is the largest component of the difference  
637 from the exact displacement [55, 56] over the largest exact component. The  
638 prediction, from the pairs of §6.4 with  $kh$  replaced by  $ka$ , is an error of  
639  $O((ka)^2)$  for  $\ell = 1$  and 2 and of  $O((ka)^4)$  for  $\ell = 3$ , and  $O((ka)^4)$  already  
at  $\ell = 2$  for a density contrast alone. Table 3 bears the prediction out. At

Table 3: One homogeneous sphere by the hierarchy of degree  $\ell$ : relative far-field error against exact scattering, and the order between the two lowest frequencies. All nine orders are as predicted.

| contrast     | $\ell$ (unknowns) | $k_S a = 0.4$        | 0.2                  | 0.1                  | order | predicted |
|--------------|-------------------|----------------------|----------------------|----------------------|-------|-----------|
| density only | 1 (12)            | $6.4 \times 10^{-3}$ | $1.6 \times 10^{-3}$ | $3.9 \times 10^{-4}$ | 2.0   | 2         |
|              | 2 (30)            | $4.8 \times 10^{-5}$ | $2.6 \times 10^{-6}$ | $1.8 \times 10^{-7}$ | 3.9   | 4         |
|              | 3 (60)            | $1.5 \times 10^{-5}$ | $9.5 \times 10^{-7}$ | $6.9 \times 10^{-8}$ | 3.8   | 4         |
| moduli only  | 1                 | $1.9 \times 10^{-2}$ | $5.0 \times 10^{-3}$ | $1.1 \times 10^{-3}$ | 2.3   | 2         |
|              | 2                 | $5.4 \times 10^{-3}$ | $1.4 \times 10^{-3}$ | $3.4 \times 10^{-4}$ | 2.0   | 2         |
|              | 3                 | $6.2 \times 10^{-5}$ | $4.1 \times 10^{-6}$ | $2.0 \times 10^{-7}$ | 4.3   | 4         |
| both         | 1                 | $1.4 \times 10^{-2}$ | $3.1 \times 10^{-3}$ | $7.2 \times 10^{-4}$ | 2.1   | 2         |
|              | 2                 | $3.2 \times 10^{-3}$ | $6.9 \times 10^{-4}$ | $1.8 \times 10^{-4}$ | 1.9   | 2         |
|              | 3                 | $4.6 \times 10^{-5}$ | $2.5 \times 10^{-6}$ | $1.4 \times 10^{-7}$ | 4.2   | 4         |

640

641  $k_{Sa} = 0.4$  the error of the uniform-strain closure is 1.4%, that of the second-  
642 gradient system 0.32%, and that of the third-gradient system 0.0046%: three  
643 hundred times below the closure. The systems are well conditioned, the sixty  
644 equations of  $\ell = 3$  having a condition number of 5.9 (`scripts/measure_`  
645 `ball_gradient_hierarchy.py`).

646 *A cube.* The same assembly with the cube's moments gives the single site of  
647 a voxel. No exact solution exists for a cube, so the reference is the same cube,  
648 of half-width  $h = 10$  m, solved by the Legendre voxels of §8 on refined grids,  
649 a cube being the one body such a grid represents exactly. Solutions with first  
650 and second moments on  $2^3$  and  $3^3$  cells differ among themselves by at most  
 $2 \times 10^{-5}$ , which is the accuracy of the reference. Table 4 shows what distin-

Table 4: One homogeneous cube of half-width  $h$  by the hierarchy of degree  $\ell$ : relative far-field error against the converged reference.

| $\ell$ (unknowns) | $k_{Sh} = 0.4$       | 0.2                  | 0.1                  | 0.05                 |
|-------------------|----------------------|----------------------|----------------------|----------------------|
| 1 (12)            | $2.1 \times 10^{-2}$ | $5.4 \times 10^{-3}$ | $2.8 \times 10^{-3}$ | $2.7 \times 10^{-3}$ |
| 2 (30)            | $3.2 \times 10^{-3}$ | $1.6 \times 10^{-3}$ | $1.9 \times 10^{-3}$ | $2.4 \times 10^{-3}$ |
| 3 (60)            | $3.5 \times 10^{-4}$ | $3.4 \times 10^{-4}$ | $2.9 \times 10^{-4}$ | $3.1 \times 10^{-4}$ |

651  
652 guishes a cube from a sphere. Each truncation has a floor that does not fall  
653 with the frequency, because the internal field of an isolated cube is not a poly-  
654 nomial even statically: it concentrates at the edges and corners, as the static  
655 expansions about the centre of Johnson et al. [49] and Wu et al. [39] found for  
656 the cuboid and for polyhedra. The floor is  $2.7 \times 10^{-3}$  for the uniform-strain  
657 closure and nearly the same with second gradients, as §6.3 requires, the static  
658 stress dipole being fixed by the first-gradient block and coupled next to the

third. The third-gradient block lowers it ninefold, to  $3 \times 10^{-4}$ . Above the floor the error falls with the frequency as on the sphere: at  $k_S h = 0.4$  the third-gradient system is sixty times more accurate than the closure. The dip of the second-gradient row at  $k_S h = 0.2$  is a cancellation between its static and dynamic terms, which have opposite signs. The condition numbers lie between 11 and 13 (`scripts/measure_cube_gradient_hierarchy.py`). These results come from one implementation, checked against the exact sphere and against the converged cube; the moments it uses are those confirmed by two routes in §6.2.

#### 6.6. The collocation closure

The collocation closure takes the internal field of a voxel as uniform, the truncation of §6.3 for the density and stiffness channels together. The voxel's own field at its centre is then  $\mathbf{S}_{\text{self}} \Delta e$  for internal state  $e$ , where  $\mathbf{S}_{\text{self}}$  is the nine-component form of the moments  $\mathbf{G}$  and  $\mathbf{M}$ . By parity it is block-diagonal. So  $e = \psi + \mathbf{S}_{\text{self}} \Delta e$  and

$$\mathbf{T} = d^3 \Delta (\mathbf{I} - \mathbf{S}_{\text{self}} \Delta)^{-1}. \quad (34)$$

With the kernel (14) the self term cancels. Substituting  $\psi = (\mathbf{I} - \mathbf{S}_{\text{self}} \Delta)e$  into Eq. (2),

$$e_i = \psi_i^0 + d^3 \sum_j \mathbf{K}_{\text{all}}(i, j) \Delta e_j, \quad \mathbf{K}_{\text{all}}(i, i) = \mathbf{P}_0, \quad \mathbf{K}_{\text{all}}(i, j) = \mathbf{S}_{\text{avg}}(i - j), \quad (35)$$

an identity checked on the matrices to  $4 \times 10^{-16}$ . Every block of  $\mathbf{K}_{\text{all}}$  is an *exact* cell integral of the one-dimensional continuum kernel (Eqs. (12), (13)), so Eq. (35) is precisely the collocation of the continuum Lippmann–Schwinger

equation with a strain piecewise constant over voxels. It does not contain  $\mathbf{S}_{\text{self}}$  at all.

The package’s cube  $T$ -matrix is this closure statically: both blocks agree with Eq. (34) to  $2 \times 10^{-5}$  at  $k_S h = 0.01$ , the difference falling as  $(kh)^2$  (`ContinuumLimit_CubeT.w1`). It departs dynamically by about  $0.25(kh)^2$  through its far-field form-factor corrections. Its self integrals are exact in the static and radiation parts but omit the real even-power terms of the Green’s tensor’s expansion ( $k^2 r, r^3, \dots$ ), an error of 0.4–0.6  $(kh)^2$  against the analytic self term, whose series agrees with the closed-form elastodynamic tensor to  $10^{-14}$ . By Eq. (35) that error cannot enter the closure scheme.

#### 6.7. *What a closed-form single-site $T$ -matrix gives*

The moment hierarchy costs more to set up than the alternatives: moments of the Green’s tensor to fourth derivatives, evaluated as distributions and reduced to closed form. This subsection collects what that buys. Each point is a property of the  $T$ -matrix that follows from the physics, the Lippmann–Schwinger equation of one body, rather than from a model fitted to it, and each rests on a result of this section.

*One derivation, from first principles.* The whole single site comes from one construction: the Lippmann–Schwinger equation of the body, the Green’s tensor of the background, and the truncation of the internal field at a stated degree. The static depolarisation, its dynamic corrections, the radiation damping and every higher block are consequences of that one equation, evaluated by one engine (§6.2); none is taken from a separate theory and joined to the others. Because each piece follows from the same equation, the pieces

cannot be inconsistent with one another, and the limits below (the static, the spherical and the low-frequency) are results of the construction rather than conditions imposed on it. The equivalent inclusion method rests on the same foundation [46, 34]; what the hierarchy adds is that, for a body that is not an ellipsoid and at finite frequency, every coefficient is in closed form and the error of each truncation is known.

*Nothing is fitted.* Every coefficient of Eq. (22) is a moment of the background Green’s tensor over the body. There is no polarisability to choose and no depolarisation factor to calibrate: as in the equivalent inclusion method, the single site is derived, with the uniform-strain closure as its first member (§6.3), where other routes import it from the ellipsoid [4, 33] or close the volume integral equation with an effective field [42, 43].

*The singular self-interaction is exact, and is not added by hand.* The  $\delta$ -function content of the inclusion problem, which a quadrature misses, is part of the moments from the start (§6.2): the sum rule of Eq. (27) holds exactly, and the traces of Eq. (A.1) hold at every grade. The same moments with a spherical exclusion give no depolarisation at all; the whole static response of a sphere is the  $\delta$  term (§6.3).

*The exact limits are built in.* The dilatational channel of the cube equals the sphere’s exactly, because it depends on the moments only through a contraction that does not depend on the shape, and the cube’s isotropic shear average equals the sphere’s exactly; both are confirmed against the exact Mie solution, to  $2 \times 10^{-10}$  and  $10^{-8}$  (§6.3). At finite frequency the closure departs from Mie only through a real term of order  $(ka)^2$ , and the

727 imaginary  $(ka)^3$  terms, the radiation damping, agree with Mie exactly in all  
728 three channels (Eq. (F.1)).

729 *The error is stated before anything is solved.* A truncation of the hierarchy  
730 is a statement of which blocks are omitted, so its error is a property of  
731 the omitted blocks and can be predicted. The predictions hold: the orders  
732 come in parity pairs, all thirty-six on the layer and all nine on the sphere as  
733 predicted (Tables 2 and 3); the leading error factor is in closed form, Eq. (33);  
734 and the separation into an even and an odd system (§6.1) says which channel  
735 each block improves.

736 *It improves block by block, on fixed constants.* Raising the degree adds mo-  
737 ments; it adds no numerical integration. The scalar moments are a power of  
738 the body's width times Eq. (28), evaluated once and stored (through  $\ell = 5$   
739 for the kernels  $r^{-1}$  and  $r$ , and through  $\ell = 3$  for the extra derivatives of the  
740 second term of Eq. (24)), and are confirmed by a second route that shares  
741 nothing with the first (§6.2). The accuracy of the single site is therefore set  
742 by its degree, not by a quadrature: against the exact sphere at  $k_{Sa} = 0.4$   
743 the error falls from 1.4% for the uniform-strain closure to 0.0046% with third  
744 gradients, three hundred times smaller, from a system of sixty equations with  
745 a condition number of 5.9 (Table 3). That each degree improves on the one  
746 below is shown here for one sphere and one cube, not in general: in the Tay-  
747 lor form of the equivalent inclusion method, for interacting inhomogeneities,  
748 a higher degree need not improve the solution and the system need not be  
749 invertible [57, 47].

750 *It says what a body's shape does.* The construction resolves the response into  
 751 the irreducible channels of the body's symmetry. For the cube it shows that,  
 752 at the order of the uniform-strain closure, a cube is a sphere plus a pure cubic  
 753 anisotropy, which splits the deviatoric channel about the sphere's value in  
 754 the ratio 3 : 2 (Eq. (32)), and the higher blocks measure what the corners  
 755 cost (Table 4).

756 *It describes a scatterer, not only a cell.* The assembly of Eq. (22) is the same  
 757 for every centrosymmetric body; only the scalar moments change, as the ball  
 758 and the cube of §6.5 show. The result is the  $T$ -matrix of one heterogeneity,  
 759 which enters a multiple-scattering solve as it stands, and the same moments  
 760 are the self block of a voxel scheme, whose singular work they then do once  
 761 and in closed form (§11).

762 *What it does not give.* A polynomial about the centre cannot follow the  
 763 edge and corner concentrations of an isolated cube's internal field, so each  
 764 truncation there has a floor that does not fall with the frequency ( $3 \times 10^{-4}$   
 765 with third gradients; Table 4). And in a lattice the Legendre voxel of §8 is the  
 766 more economical in unknowns: both are fourth order on a smoothly graded  
 767 sphere, the hierarchy with sixty unknowns a voxel and the first-moment voxel  
 768 with thirty-six (§11). The hierarchy's case is not the number of unknowns;  
 769 it is that it yields the single site itself, in closed form, with its error known.

## 770 **7. Convergence, and the closed-form error**

771 *The experiment.* Every convergence test of this paper is the same experiment,  
 772 varied in one respect at a time, and it is set out here once.

- 773 1. *The medium.* The background has  $\alpha = 5000$  and  $\beta = 3000 \text{ m s}^{-1}$  and  
774  $\rho = 2500 \text{ kg m}^{-3}$ . The layer occupies  $0 < z < D$  with  $D = 2 \text{ m}$ ,  
775 and differs from the background by  $\Delta\lambda = 2 \text{ GPa}$ ,  $\Delta\mu = 1 \text{ GPa}$  and  
776  $\Delta\rho = 100 \text{ kg m}^{-3}$ : 6.4% in the P modulus, 4.4% in the shear modulus  
777 and 4% in the density.
- 778 2. *The source.* A plane force at  $z_s = -12 \text{ m}$ , above the layer, radiates a  
779 plane wave towards it. The incident field in the layer is then known in  
780 closed form, and it is smooth there because the source is outside.
- 781 3. *What is measured.* The scattered displacement, total less incident, at  
782 two observers outside the layer: at  $z = -1 \text{ m}$  above it, where it is the  
783 reflected wave, and at  $z = 4 \text{ m}$  below it, where it is the change the layer  
784 makes to the transmitted wave. Outside the layer the scattered field is  
785 a single outgoing wave, so one observer on each side holds everything;  
786 the two are reported separately, as reflection / transmission.
- 787 4. *The exact answer.* The reflection and transmission of the layer from  
788 continuity of displacement and traction at its two faces (§3); for a  
789 layer whose properties vary with depth, from the differential equation  
790 integrated to 30 digits.
- 791 5. *The error.*  $e = |u_{\text{scheme}} - u_{\text{exact}}| / |u_{\text{exact}}|$  at each observer: relative to the  
792 scattered field, not to the total, so that an error of  $10^{-6}$  is a millionth  
793 of what the layer scatters.
- 794 6. *The refinement.* The layer is cut into  $n$  planes of voxels of side  $d =$   
795  $D/n$ , with  $n$  from 1 to 32, at  $\omega = 60, 300$  and  $600 \text{ rad s}^{-1}$ , for which  
796  $k_P D = 0.024, 0.12$  and  $0.24$  and  $k_S D = 0.04, 0.2$  and  $0.4$ . Three  
797 frequencies show that an order is not an accident of one.

798     7. *The arithmetic.* The errors of the higher-degree voxels fall below the  
799     round-off of double precision, so the symbolic implementation computes  
800     them to 40 digits; the independent numerical implementation, in double  
801     precision, agrees with it down to its own floor of about  $10^{-13}$ .

802     The later sections change one item each: the incident wave to SV or SH and  
803     its angle (§8), the medium to a stratified or a graded one and the background  
804     to a stratified one (§9), and the body to a sphere (§10).

805     *Second order.* For a layer  $D = 2$  m with the contrast  $\Delta\lambda = 2$  GPa,  $\Delta\mu =$   
806     1 GPa,  $\Delta\rho = 100$  kg m $^{-3}$  in a background  $\alpha = 5000$ ,  $\beta = 3000$  m s $^{-1}$ ,  $\rho =$   
807     2500 kg m $^{-3}$ , illuminated by a P plane force, the reflected displacement of  
808     the chain converges to the continuous layer's at order 2.000 for  $n = 4 \rightarrow$   
809     16 at  $\omega = 60, 300$  and  $600$  rad s $^{-1}$ , with the collocation (35) and with the  
810     package *T*-matrix alike (`ContinuumLimit_Chain.wl`). The error depends on  
811      $k_S h$  alone. Throughout, an order quoted for a pair of grids is the apparent  
812     order  $\ln(e_1/e_2)/\ln(n_2/n_1)$  between them, with  $e$  the relative error against  
813     the exact solution and  $n$  the refinement; the orders in the legend of Fig. 6  
814     are least-squares slopes of  $\ln e$  against  $\ln n$  over the grids marked as fitted,  
815     with their standard errors. The two agree where a sequence is asymptotic  
816     and differ where it is not, which the apparent orders show.

817     *The leading error in closed form.* In Eq. (35) the kernel is integrated exactly;  
818     the only approximation is the internal field's variation across a voxel. At first  
819     order in the contrast the discrete scattered field is therefore a midpoint rule  
820     applied to the continuum's Born integral, with the kernel's cell integral exact.  
821     For reflection the integrand carries the two-way phase  $e^{2ikz}$ ; for transmission

822 it is flat. With  $k$  the background wavenumber of the wave concerned ( $k_P$  for  
 823 an incident P wave,  $k_S$  for an incident S wave),  $R$  the reflection coefficient  
 824 and  $T^{\text{sc}}$  the scattered part of the transmitted field, for *every*  $n$

$$\frac{R_{\text{disc}}}{R} = \frac{\text{sinc}(kh)}{\text{sinc}(2kh)} = 1 + \frac{(kd)^2}{8} + O(d^4), \quad \frac{T_{\text{disc}}^{\text{sc}}}{T^{\text{sc}}} = \text{sinc}(kh) = 1 - \frac{(kd)^2}{24} + O(d^4), \quad (36)$$

825 confirmed for  $n = 1$  to 16 to  $10^{-4}$  of the error itself (`ContinuumLimit_`  
 826 `ErrorConstant.wl`); for an incident S wave, with  $k = k_S$ , the same forms  
 827 and those of Eq. (39) hold to  $4 \times 10^{-4}$  (`ContinuumLimit_IncidentS.wl`, §8).  
 828 At the full contrast the reflection constant is unchanged (0.124 against 1/8);  
 829 the transmission constant grows to  $-0.092$ .

830 *Why a uniform-field single site stops at second order.* The two constants have  
 831 opposite signs, so no scalar correction of  $\mathbf{T}$ , no single  $O((kh)^2)$  factor, can  
 832 cancel both. The package  $T$ -matrix illustrates it: its form-factor corrections  
 833 reduce the reflection error by a factor 4.45 and *increase* the transmission  
 834 error by 1.7, at every  $n$ . The missing term in Eq. (36) is the first moment  
 835 of the internal field over each voxel, which radiates through the derivative of  
 836 the kernel. A fourth-order scheme must carry it: a single site with a gradient  
 837 channel, and a kernel whose cell integrals include the kernel's derivative.

## 838 8. Fourth order and beyond: Legendre moments of the voxel field

839 Let each voxel carry the mean *and* the first moment of its internal field.  
 840 On voxel  $j$ , with local coordinate  $s \in [-h, h]$  and Legendre functions  $\phi_b(s) =$   
 841  $P_b(s/h)$  ( $\phi_0 = 1$ ,  $\phi_1 = s/h$ ,  $\phi_2 = (3s^2/h^2 - 1)/2$ , ...), the field is  $w =$   
 842  $\sum_b c_{b,j} \phi_b$ ,  $w = (u, \varepsilon)$  the displacement and strain of the P channel, and the

843 polarisation is  $q = (\omega^2 \Delta \rho u, \Delta M \varepsilon)$  with  $\Delta M = \Delta \lambda + 2\Delta \mu$ . Testing the  
844 continuum equation

$$w(z) = w^0(z) + \int \mathbf{K}(z-z') q(z') dz', \quad \mathbf{K} = \begin{pmatrix} g & g' \\ g' & g'' \end{pmatrix}, \quad g'' = -k^2 g - \delta(z)/M_P, \quad (37)$$

845 with the same functions on every voxel gives

$$\left( \int_i \phi_a^2 dz \right) c_{a,i} = \int_i \phi_a w^0 dz + \sum_{j,b} \left[ \iint_j \phi_a(z) \mathbf{K}(z-z') \phi_b(z') dz dz' \right] \Delta_1 c_{b,j}, \quad (38)$$

846  $\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$ . Between voxels the double integral is separable, a  
847 product of Legendre moments of  $e^{\pm i k z}$ ; within a voxel it is split at  $z' = z$   
848 and the local term contributes  $-\int_i \phi_a \phi_b dz / M_P$ . Every entry is closed-form.  
849 Keeping only  $\phi_0$  recovers a second-order scheme; keeping  $\phi_0$  and  $\phi_1$  con-  
850 verges at fourth order in both reflection and transmission (`ContinuumLimit_`  
`FourthOrder.wl`; Table 5). A single plane of first-moment voxels represents

Table 5: Relative error of the scattered displacement of the layer, reflection / transmission, at  $\omega = 300 \text{ rad s}^{-1}$ , against the number of voxel planes  $n$ : collocation, the mean-only voxel and the first-moment voxel of Eq. (38). The order is 4.00 at  $\omega = 60$  and  $600 \text{ rad s}^{-1}$  too.

| $n$   | collocation                               | mean only                                 | mean and first moment                       |
|-------|-------------------------------------------|-------------------------------------------|---------------------------------------------|
| 1     | $1.8 \times 10^{-3} / 1.3 \times 10^{-3}$ | $1.2 \times 10^{-3} / 1.8 \times 10^{-3}$ | $2.8 \times 10^{-7} / 2.8 \times 10^{-7}$   |
| 4     | $1.1 \times 10^{-4} / 8.3 \times 10^{-5}$ | $7.5 \times 10^{-5} / 1.1 \times 10^{-4}$ | $1.1 \times 10^{-9} / 1.1 \times 10^{-9}$   |
| 16    | $7.0 \times 10^{-6} / 5.2 \times 10^{-6}$ | $4.7 \times 10^{-6} / 6.9 \times 10^{-6}$ | $4.2 \times 10^{-12} / 4.2 \times 10^{-12}$ |
| order | 2 / 2                                     | 2 / 2                                     | 4 / 4                                       |

851

852 the 2 m layer 6500 times more accurately than a single plane of uniform ones.

853 The collocation column is the scheme of §6 and reproduces the chain of the  
854 full  $9 \times 9$  system to all printed digits.

855 *Higher moments.* Carrying the Legendre moments to degree  $p$  changes noth-  
856 ing in the construction: the moments and the same-cell double integrals of  
857  $P_b(s/h)$  against exponentials are again closed-form, and for  $p \leq 1$  the general-  
858 degree solver reproduces the chain above exactly. The second-moment voxel  
859 ( $p = 2$ ) converges at sixth order and the third-moment voxel ( $p = 3$ ) at  
860 eighth, in reflection and transmission (Table 6; `ContinuumLimit_SecondMoment`.  
861 `w1`); the second-moment voxel’s order is 6.00 at  $\omega = 60, 300$  and  $600 \text{ rad s}^{-1}$ .  
The errors fall to  $10^{-22}$ , so they are computed in 40-digit arithmetic.

Table 6: Relative error of the scattered displacement of the layer, reflection / transmission,  
at  $\omega = 300 \text{ rad s}^{-1}$ , for voxels carrying Legendre moments to degree  $p$ . The order is the  
mean over the halvings  $n = 2 \rightarrow 4 \rightarrow 8$ .

| $n$   | first moment ( $p = 1$ )                    | second moment ( $p = 2$ )                   | third moment ( $p = 3$ )                    |
|-------|---------------------------------------------|---------------------------------------------|---------------------------------------------|
| 1     | $2.8 \times 10^{-7} / 2.8 \times 10^{-7}$   | $2.8 \times 10^{-11} / 4.2 \times 10^{-11}$ | $1.5 \times 10^{-15} / 1.5 \times 10^{-15}$ |
| 2     | $1.7 \times 10^{-8} / 1.7 \times 10^{-8}$   | $4.4 \times 10^{-13} / 6.5 \times 10^{-13}$ | $6.0 \times 10^{-18} / 6.0 \times 10^{-18}$ |
| 4     | $1.1 \times 10^{-9} / 1.1 \times 10^{-9}$   | $6.9 \times 10^{-15} / 1.0 \times 10^{-14}$ | $2.4 \times 10^{-20} / 2.4 \times 10^{-20}$ |
| 8     | $6.7 \times 10^{-11} / 6.7 \times 10^{-11}$ | $1.1 \times 10^{-16} / 1.6 \times 10^{-16}$ | $9.2 \times 10^{-23} / 9.2 \times 10^{-23}$ |
| order | 4 / 4                                       | 6 / 6                                       | 8 / 8                                       |

862

863 *The leading error in closed form.* Without scattering, the Galerkin solution  
864 is the  $L^2$  projection  $\Pi$  of the incident field onto each voxel’s basis, and the  
865 kernel is integrated over each voxel exactly; at first order in the contrast the  
866 error is therefore the kernel’s cell integral applied to  $\Pi w^0 - w^0$ . The field  
867 varies as  $e^{iks}$  across a voxel and the kernel toward an outside observer as  $e^{iqs}$ ,

868  $q = +k$  for reflection and  $q = -k$  for transmission, so for every scheme and  
869 every  $n$

$$\frac{\text{discrete}}{\text{exact}} = \frac{\int_{-h}^h e^{iqs} \Pi[e^{iks}] ds}{\int_{-h}^h e^{i(q+k)s} ds}, \quad (39)$$

870 with  $\Pi$  the centre value for collocation, the cell mean for the mean-only voxel  
871 and the projection onto  $\{P_0, \dots, P_p\}$  for the voxel of degree  $p$  (`ContinuumLimit_`  
872 `FourthOrderError.wl`, `ContinuumLimit_SecondMoment.wl`). The results  
are in Table 7. The chain of Eq. (38) reproduces the forms for collocation

Table 7: The leading error of the discrete layer in closed form, Eq. (39): discrete over exact scattered field at first order in the contrast, for every number of voxel planes  $n$ .

|               | reflection                        | transmission                      |
|---------------|-----------------------------------|-----------------------------------|
| collocation   | $1 + (kd)^2/8 + O(d^4)$           | $1 - (kd)^2/24 + O(d^4)$          |
| mean only     | $1 + (kd)^2/12 + O(d^4)$          | $1 - (kd)^2/12 + O(d^4)$          |
| first moment  | $1 - (kd)^4/720 + O(d^6)$         | $1 - (kd)^4/720 + O(d^6)$         |
| second moment | $1 + (kd)^6/100800 + O(d^8)$      | $1 - (kd)^6/100800 + O(d^8)$      |
| third moment  | $1 - (kd)^8/25401600 + O(d^{10})$ | $1 - (kd)^8/25401600 + O(d^{10})$ |
| degree $p$    | $1 + (-1)^p c_p (kd)^{2p+2}$      | $1 - c_p (kd)^{2p+2}$             |

873  
874 and degrees 0 and 1 at  $n = 1$  to 8 to  $10^{-4}$  of the error, the remainder being  
875 the second order in the contrast, and those for degrees 2 and 3 to  $4 \times 10^{-7}$   
876 at a contrast reduced by  $10^{-6}$ . The constant for degree  $p$  follows from the  
877 Legendre expansion  $e^{iks} = \sum_j i^j (2j+1) j_j(kh) P_j(s/h)$ . The projection drops  
878 the terms  $j > p$ , led by  $j = p+1$ , whose coefficient is  $O((kh)^{p+1})$ ; its moment  
879 against  $e^{iqs}$  is  $2h i^{p+1} j_{p+1}(qh) = O((qh)^{p+1})$ . With  $j_n(x) \approx x^n/(2n+1)!!$  and

880  $h = d/2,$

$$c_p = \frac{2p+3}{4^{p+1}[(2p+3)!!]^2} = \frac{[(p+1)!]^2}{(2p+2)!(2p+3)!}, \quad c_0 = \frac{1}{12}, \quad c_1 = \frac{1}{720}, \quad c_2 = \frac{1}{100800},$$

(40)

881 and the leading error carries  $(q/k)^{p+1}$ : the same sign in reflection and trans-  
 882 mission for odd  $p$ , opposite signs for even  $p$ . The first-moment voxel's leading  
 883 error therefore does not depend on the direction in which the field is observed.  
 884 Eq. (40) is checked symbolically for  $p = 0$  to 5 (every lower power of  $kd$  van-  
 885 ishes) in Mathematica and, independently, in sympy.

886 *One plane of voxels and the thin-layer series.* Refining at fixed thickness is  
 887 one limit; thinning a single plane is the other. For one plane of voxels ( $n = 1$ ,  
 888  $d = D$ ) the series in  $D$  of the reflection coefficient, computed exactly at the  
 889 full contrast, coincides with that of the exact layer, Eq. (7), term by term  
 890 through  $D^{2p+2}$  and first differs at  $D^{2p+3}$ , for  $p = 0$  to 3 and for a layer both  
 891 stiffer and softer than its surroundings (`ContinuumLimit_SecondMoment.`  
 892 `w1`). Each further Legendre moment adds two terms of the thin-layer series.

893 *In three dimensions.* The three-dimensional first-moment voxel carries the  
 894 nine-component state with Legendre weights  $\{1, x/h, y/h, z/h\}$ , 36 unknowns  
 895 in all, tested with the same functions and coupled by cell-to-cell double in-  
 896 tegrals of the Green's tensor. For a laterally uniform layer at normal in-  
 897 cidence it reduces *exactly* to Eq. (38) (`ContinuumLimit_Reduction3D.w1`).  
 898 The lateral first moments are never excited: the lattice and the incident  
 899 field are symmetric under reflection about every voxel's centre planes, and  
 900  $x/h, y/h$  are odd there. The mean and the  $z$ -moment carry weights that  
 901 are constant laterally, so their lateral form factor  $\text{sinc}(g_x h) \text{sinc}(g_y h)$  van-

902 ishes on the reciprocal lattice and the plane sum of the double integrals  
 903 is the plate-to-plate integral in  $z$  alone. Two facts mark where genuinely  
 904 three-dimensional structure begins. The lateral first moment's form fac-  
 905 tor,  $F_1(g) = i(gh \cos gh - \sin gh)/(gh)^2$ , does not vanish on the recipro-  
 906 cal lattice ( $F_1 = i(-1)^p/p\pi$ ), so the gradient channels couple through the  
 907 evanescent orders. And at oblique incidence the mean's own zeros shift,  
 908  $\text{sinc}((2\pi p/d + k_x)h) = (-1)^p k_x h/p\pi + O(k_x^2 h^2)$ , so the tiling identity holds  
 909 only to  $O(k_{\parallel}h)$ .

910 *Other angles, other waves, and degree two.* The orders and their constants  
 911 hold beyond normal incidence and incident P. At oblique incidence, where  
 912 the tiling identity holds only to  $O(k_{\parallel}h)$  and the gradient channels couple  
 913 through the evanescent orders, the mean-only and first-moment voxels remain  
 914 second and fourth order in the reflected P and the converted S wave; for  
 915 incident SV and SH waves the orders, and with  $k = k_S$  the constants, are  
 916 those of incident P; and the voxel of degree two converges at sixth order  
 917 for P, SV and SH waves, its error on four planes at normal incidence being  
 918 the  $c_2(kd)^6$  of Eq. (40) to 4%. Appendix Appendix B gives the spectral  
 919 construction, Eq. (B.1), the tests, and the second implementations that check  
 920 them; Figure 6(a,b) shows the convergence.

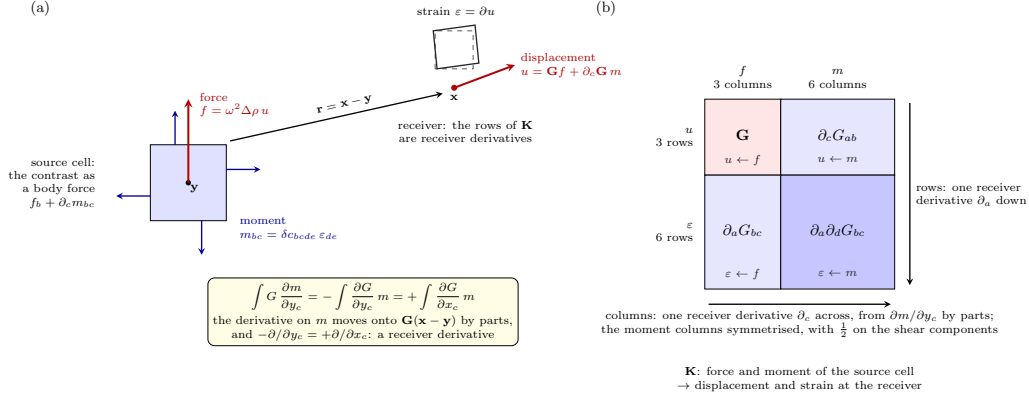


Figure 1: Where the kernel comes from. (a) A source cell at  $\mathbf{y}$  acts on the background through the body force of Eq. (3): a force  $f$  from its density contrast, and the divergence of its contrast stress  $m$ , the moment, from its stiffness contrast (arrows on the faces). At the receiver  $\mathbf{x}$  the state is the displacement  $u$  and the strain  $\varepsilon$ . The moment's derivative moves onto  $\mathbf{G}(\mathbf{x} - \mathbf{y})$  by parts, Eq. (4), and becomes a receiver derivative with a plus sign. (b) The block structure of the  $9 \times 9$  kernel  $\mathbf{K}$  that results: the force columns are  $\mathbf{G}$ , the moment columns carry one receiver derivative across, the strain rows one receiver derivative down, and the strain-from-moment block two.

*Alt text:* Two panels. On the left, a shaded square cell with a dot at its centre marked  $\mathbf{y}$ ; a thick red arrow labelled force rises from the centre, and four blue arrows labelled moment point outward from the cell's faces. A long arrow labelled  $\mathbf{r}$  equals  $\mathbf{x}$  minus  $\mathbf{y}$  runs from the cell to a red dot marked  $\mathbf{x}$  at the upper right, from which a red arrow labelled displacement leaves and above which a small dashed square deformed into a parallelogram is labelled strain. A boxed line below reads: the integral of  $G$  times the  $y$ -derivative of  $m$  equals minus the integral of the  $y$ -derivative of  $G$  times  $m$ , equals plus the integral of the  $x$ -derivative of  $G$  times  $m$ . On the right, a two-by-two grid of shaded blocks with rows labelled  $u$ , three rows, and strain, six rows, and columns labelled  $f$ , three columns, and  $m$ , six columns; the blocks read  $\mathbf{G}$ , the first derivative of  $G$ , the first derivative of  $G$ , and the second derivative of  $G$ ; an arrow down the right side reads rows: one receiver derivative down, and an arrow along the bottom reads columns: one receiver derivative across.

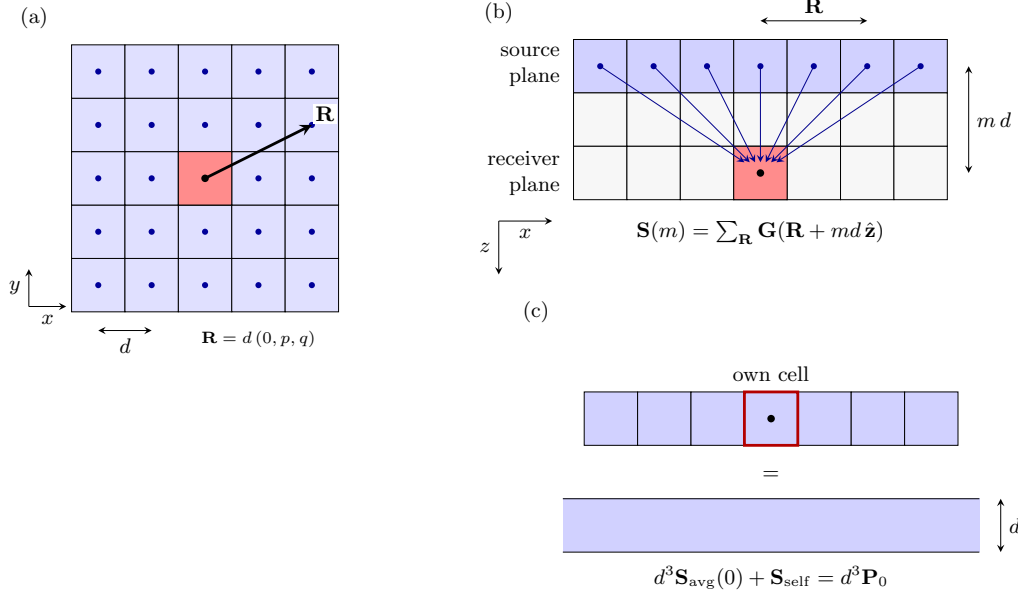


Figure 2: The lattice and its sums. (a) Plan view of one plane of voxels: their centres form a square lattice of pitch  $d$ , and a lattice vector  $\mathbf{R} = d(0, p, q)$  joins the centre of one voxel (shaded darker) to another. (b) Cross-section ( $z$  down), between planes: the voxel of a receiver plane (shaded darker) is coupled to every voxel of a source plane  $m$  planes above it, one for each lattice vector  $\mathbf{R}$ ; the sum over  $\mathbf{R}$  is the specular lattice sum of Eq. (6). (c) Within a plane: the sum over every cell of the plane, the receiver’s own cell (outlined) included, is the field of a uniform plate of thickness  $d$ , the tiling identity of Eq. (13).

*Alt text:* Three schematic panels. On the left, a plan view of a five-by-five grid of square voxels with a dot at each centre; the centre voxel is highlighted, an arrow labelled  $\mathbf{R}$  runs from its centre to another voxel centre, and the spacing between neighbouring centres is marked  $d$ . At the top right, a cross-section through three rows of voxels: arrows from the centre of every voxel in the top row converge on a highlighted voxel two rows below, with the plane separation  $m d$  marked. Below it, a single row of voxels with one outlined as the receiver’s own cell is shown equal to a continuous shaded band of the same thickness  $d$ , a uniform plate.

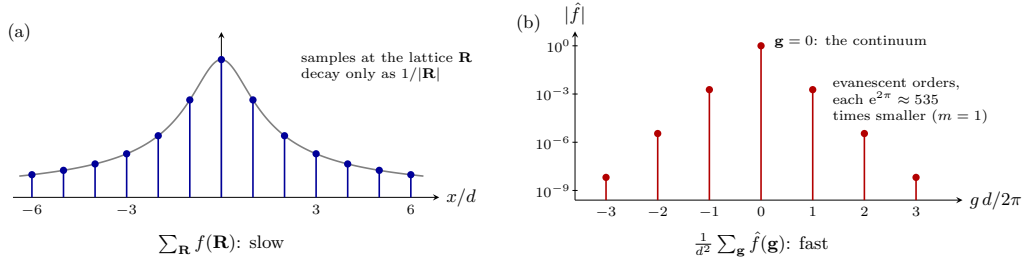


Figure 3: The Poisson summation formula (9) applied to a plane of voxels, along one lateral axis. (a) Real space: the field of each voxel of a source plane at a receiver one plane away, sampled at the lattice points; the samples decay only as  $1/|\mathbf{R}|$ , so their sum converges slowly. (b) The same sum after Poisson summation, on a logarithmic scale: samples of the lateral Fourier transform at the reciprocal-lattice orders  $\mathbf{g} = 2\pi p/d$ . The order  $\mathbf{g} = 0$  is the continuum's plane wave; every other order is evanescent and, at  $m = 1$ , smaller than its neighbour by  $e^{2\pi} \approx 535$ . Magnitudes are schematic: the static decay  $e^{-|\mathbf{g}|md}$  is shown.

*Alt text:* Two panels. On the left, a smooth bell-shaped curve that decays slowly on both sides is sampled by vertical stems at integer positions from minus six to six, the stems remaining clearly visible far from the centre. On the right, on a logarithmic vertical axis from one down to ten to the minus nine, a stem of height one at zero is labelled as the continuum, and stems at plus and minus one, two and three fall by about three powers of ten per step and are labelled as evanescent orders.

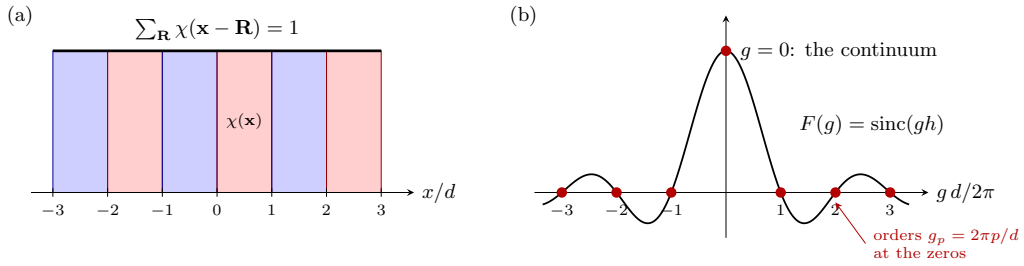


Figure 4: Why cubes make the coupling exact. (a) Cubes tile the plane: the indicator functions  $\chi$  of the translated cells sum to exactly one everywhere, a partition of unity. (b) The same fact in Fourier space: the cube's lateral form factor  $F(g) = \text{sinc}(gh)$  (shown along one axis) vanishes at every reciprocal-lattice order  $g_p = 2\pi p/d$  with  $p \neq 0$ , so of all the orders in the lattice sum (10) only  $g = 0$ , the continuum's plane wave, survives the source-cell average.

*Alt text:* Two panels. On the left, adjacent box functions of unit height and unit width, alternately shaded, lie side by side along an axis marked in voxel widths, and a thick horizontal line at height one marks their sum. On the right, the sinc curve, equal to one at the origin and oscillating with decaying amplitude, is plotted against wavenumber in units of  $2\pi/d$ ; dots mark the reciprocal-lattice orders, one at height one at the origin and all the others on the axis at the zeros of the curve.

## 921 9. A heterogeneous layer

922 A heterogeneous medium is not a layer of identical cells. The simplest  
923 heterogeneous case with an exact answer is a stratified model built from  
924 layers of constant properties, with real interfaces between them: the form of  
925 many layered crustal velocity models. The last test takes such a model. A  
926 layer  $D = 4$  m thick is divided into eight model layers of 0.5 m, and in each  
927 one  $\Delta\lambda$ ,  $\Delta\mu$  and  $\Delta\rho$  are drawn uniformly within  $\pm 15\%$  of the background  
928 (the values are listed in `ContinuumLimit_Heterogeneous.w1`). The result is  
929 a random stratification with physical contrasts. The voxels refine the model's  
930 layering, with  $m$  planes of voxels in each model layer, so no voxel straddles  
931 a model interface: the natural discretisation when the model's interfaces are  
932 known. At  $\omega = 1500 \text{ rad s}^{-1}$  the layer is  $k_P D = 1.2$  and  $k_S D = 2.0$  thick, so  
933 the field reverberates between the model interfaces. The schemes are those  
934 of §§7–8, with the contrast now changing from plane to plane; nothing else  
935 in them changes. In the terms of the experiment of §7, the test is this.

- 936 1. *The medium.* The eight sets of contrasts are drawn once, from a fixed  
937 seed, and the same eight are used in every run, so that every grid and  
938 every scheme sees the same model.
- 939 2. *Normal incidence.* The source is the plane force at  $z_s = -12$  m, and the  
940 scattered displacement is taken at  $z = -1$  m, where it is the reflected  
941 wave, and at  $z = D + 2 = 6$  m, where it is the change in the transmitted  
942 wave. The P problem has the modulus contrast  $\Delta\lambda + 2\Delta\mu$  and the wave  
943 speed  $\alpha$ ; the S problem is the same one-dimensional problem with  $\Delta\mu$   
944 and  $\beta$ .
- 945 3. *Oblique incidence.* A P plane wave is incident at  $20^\circ$  to the vertical.

946 What is measured is the pair of displacement amplitude vectors of  
 947 the reflected P and the reflected S plane waves,  $R_{PP}$  and  $R_{PS}$ , in the  
 948 specular order. The lateral sums of the coupling (§8) run over the  
 949 reciprocal-lattice orders  $|p|, |q| \leq 8$ .

950 4. *The exact answer.* The state of displacement and traction, two compo-  
 951 nents at normal incidence and four for P–SV at  $20^\circ$ , is carried across  
 952 each model layer by the matrix exponential of that layer’s system ma-  
 953 trix times its thickness, the eight matrices are multiplied in order,  
 954 and the radiation conditions of the two half-spaces are imposed on  
 955 the product. The arithmetic carries 30 to 40 digits. The reference  
 956 shares nothing with the voxel scheme but the model. It is checked  
 957 against a third algorithm, the reflection–transmission recursion of Ken-  
 958 nett [58] over the nine interfaces: the same-type coefficients,  $R_{PP}$  and  
 959  $R_{SS}$  at normal incidence and  $R_{PP}$  at  $20^\circ$ , agree to  $10^{-14}$  (`scripts/`  
 960 `gate_heterogeneous_reference_vs_kennett.py`). Only same-type  
 961 coefficients are compared, because they do not depend on how the up-  
 962 and downgoing waves are normalised; a converted coefficient does. One  
 963 sign differs, by convention: at normal incidence the one-dimensional  
 964 model’s  $R_{PP}$  is a ratio of vertical displacements, the recursion’s a ra-  
 965 tio of amplitudes along each wave’s own direction, and the reflected P  
 966 wave travels upwards.

967 5. *The error.* As in §7, the difference from the exact scattered quan-  
 968 tity relative to its size: for the amplitude vectors at  $20^\circ$ ,  $|\mathbf{R}_{\text{scheme}} -$   
 969  $\mathbf{R}_{\text{exact}}|/|\mathbf{R}_{\text{exact}}|$ .

970 6. *The refinement.* Each model layer is cut into  $m$  planes of voxels, so

971 that the layer holds  $n = 8m$ :  $m = 1, 2, 4, 8, 16$  at normal incidence  
972 and  $m = 1, 2, 4$  at  $20^\circ$ , where the lateral sums make each run dearer.  
973 The orders are apparent orders, from  $m = 4$  to 16 and from  $m = 2$  to  
974 4.

The errors are in Table 8 and Fig. 6(c).

Table 8: Relative error against the exact stratified layer, for  $m$  planes of voxels per model layer (§9). At normal incidence the order is measured from  $m = 4$  to 16, where it is also 2.00 for collocation and 4.00 in transmission; at  $20^\circ$  it is measured from  $m = 2$  to 4.

|                          |               | $m = 1$              | $m = 2$              | $m = 4$              | order |
|--------------------------|---------------|----------------------|----------------------|----------------------|-------|
| P, normal, $R$           | mean only     | $1.9 \times 10^{-3}$ | $4.7 \times 10^{-4}$ | $1.2 \times 10^{-4}$ | 2.00  |
|                          | first moment  | $8.5 \times 10^{-7}$ | $5.3 \times 10^{-8}$ | $3.3 \times 10^{-9}$ | 4.00  |
| S, normal, $R$           | mean only     | $5.3 \times 10^{-3}$ | $1.3 \times 10^{-3}$ | $3.3 \times 10^{-4}$ | 2.00  |
|                          | first moment  | $6.2 \times 10^{-6}$ | $3.9 \times 10^{-7}$ | $2.4 \times 10^{-8}$ | 4.00  |
| P, $20^\circ$ , $R_{PP}$ | mean only     | $1.4 \times 10^{-3}$ | $3.6 \times 10^{-4}$ | $9.0 \times 10^{-5}$ | 2.00  |
|                          | first moments | $3.3 \times 10^{-7}$ | $2.1 \times 10^{-8}$ | $1.3 \times 10^{-9}$ | 4.00  |
| P, $20^\circ$ , $R_{PS}$ | mean only     | $2.8 \times 10^{-3}$ | $6.9 \times 10^{-4}$ | $1.7 \times 10^{-4}$ | 2.00  |
|                          | first moments | $1.4 \times 10^{-6}$ | $8.8 \times 10^{-8}$ | $5.5 \times 10^{-9}$ | 4.00  |

975

976 *The interfaces cost nothing.* Every order is the one measured for the uniform  
977 layer. At a model interface the strain of the true field jumps. A voxel basis  
978 that is discontinuous from voxel to voxel represents that jump exactly when  
979 the interface lies on a voxel boundary, so the interface adds no error of its  
980 own. The only requirement is that voxel boundaries coincide with the model's  
981 interfaces. A voxel that straddled an interface would carry a contrast that is  
982 not uniform within it, a case not examined here.

983 *What fourth order buys.* For a reflection error of  $10^{-6}$  at normal incidence  
984 (P), the mean-only voxel needs 347 planes of voxels, about 43 per model  
985 layer. That figure is not a run: it is the second-order law carried from the  
986 finest grid computed, 128 planes, whose error is  $7.3 \times 10^{-6}$ , to the target,  
987  $n = 128 (7.3 \times 10^{-6} / 10^{-6})^{1/2}$ . The first-moment voxel needs 8, one per model  
988 layer, and that figure is a run: the model's own layering already gives  $8.5 \times$   
989  $10^{-7}$ . Counting unknowns (9 per voxel against 36), that is 3123 against 288  
990 per vertical column of voxels, a factor of eleven. In three dimensions, with  
991 the lateral grid refined as the vertical one is, the number of voxels falls by  
992  $(347/8)^3 \approx 8 \times 10^4$  and the number of unknowns by about  $2 \times 10^4$ . The same  
993 model has been solved independently in Python, with a contrast for each  
994 plane: the raw reflected displacements agree with the Mathematica scheme  
995 to  $1.1 \times 10^{-15}$  at normal incidence and at  $20^\circ$ , with and without the first  
996 moments. Lateral heterogeneity needs the full three-dimensional solver; it is  
997 the first extension discussed below.

998 *A medium that varies within a cell.* The stratified model is exact on its own  
999 grid: every cell is homogeneous. A smoothly varying medium is not, and  
1000 sampling it as cells of constant properties is an approximation of its own.  
1001 Let the contrast of cell  $j$  be its Legendre projection to degree  $r$ ,  $\Delta(z) =$   
1002  $\Delta_0 \sum_{c \leq r} f_c^{(j)} P_c(s/h)$ , with the voxel's field of degree  $p$  as in §8. The product  
1003 of contrast and field is again a Legendre sum,  $P_c P_b = \sum_e A_{cbe} P_e$ , with  $A_{cbe} =$   
1004  $(2e+1) \begin{pmatrix} c & b & e \\ 0 & 0 & 0 \end{pmatrix}^2$  the square of a Wigner  $3j$  symbol, so the coupling of Eq. (38)

1005 becomes

$$\iint_j \phi_a(z) \mathbf{K}(z-z') \Delta(z') \phi_b(z') dz dz' = \sum_{c \leq r} \sum_{e \leq p+r} f_c^{(j)} A_{cbe} \left[ \iint_j \phi_a(z) \mathbf{K}(z-z') \phi_e(z') dz dz' \right] \Delta_0 : \quad (41)$$

1006 the same kernel moments, now to degree  $p+r$ , and nothing else new. Against  
 1007 the exact graded layer, a high-precision solution of  $(M(z)u')' + \omega^2 \rho(z)u = 0$ ,  
 1008 the order of convergence is

$$\min(2p+2, 2r+2), \quad (42)$$

1009 or  $2p+2$  when degree  $r$  represents the profile exactly (Table 9; `ContinuumLimit_`  
 1010 `GradedContrast.wl`). A contrast that is constant in each cell therefore  
 1011 caps every voxel at second order, however rich its field basis, and for the  
 1012 smooth profile the voxel of degree 2 with a linear contrast has the error  
 1013 of the voxel of degree 1 to within 0.06%. At first order in the contrast  
 1014 this is what one expects: the contrast's projection error is orthogonal to  
 1015 the polynomials of degree  $r$ , so against the smooth product of the inci-  
 1016 dent and adjoint fields it contributes at  $O(h^{2r+2})$ . To keep order  $2p+2$   
 1017 the contrast must be represented to the degree of the field. The closed forms  
 1018 of the construction are tested one by one by an independent implementa-  
 1019 tion (`scripts/crosscheck_graded_contrast.py`): the Legendre moments  
 1020 of the kernel against  $2h i^n j_n(ch)$  and 30-digit quadrature (to  $2 \times 10^{-16}$ ), the  
 1021 linearisation coefficients against the  $3j$  symbols (exactly), and the same-cell  
 1022 double integrals against 30-digit quadrature ( $1.4 \times 10^{-16}$ ). It then assem-  
 1023 bles the scheme without linearisation, evaluating the projected contrast at  
 1024 the quadrature nodes, against its own 30-digit solution of the graded layer;  
 1025 its errors agree with those of Table 9 to  $10^{-3}$ , and at  $\omega = 1200 \text{ rad s}^{-1}$  it

Table 9: A contrast that varies within the cells: the field of each voxel to Legendre degree  $p$ , its contrast to degree  $r$ . Relative reflection error at  $n = 16$  planes and the order ( $n = 4 \rightarrow 8 \rightarrow 16$ ),  $\omega = 300 \text{ rad s}^{-1}$ , for the profiles  $1 + (2z/D - 1)/2$  (linear) and  $1 + \frac{1}{2} \sin(2\pi z/D)$  (smooth) times the contrast of §7. Predicted: Eq. (42), and  $2p + 2$  for the linear profile once  $r \geq 1$ .

| $p$ | $r$ | linear profile        |       | smooth profile        |       |
|-----|-----|-----------------------|-------|-----------------------|-------|
|     |     | error                 | order | error                 | order |
| 0   | 0   | $7.7 \times 10^{-5}$  | 2.00  |                       |       |
| 1   | 0   | $7.6 \times 10^{-5}$  | 2.00  | $5.5 \times 10^{-5}$  | 1.95  |
| 1   | 1   | $6.4 \times 10^{-11}$ | 4.00  | $1.4 \times 10^{-7}$  | 3.95  |
| 2   | 1   | $2.3 \times 10^{-17}$ | 6.00  | $1.4 \times 10^{-7}$  | 3.95  |
| 2   | 2   |                       |       | $1.6 \times 10^{-10}$ | 5.98  |

1026 measures the orders of Eq. (42) again.

1027 *Which basis sets the error.* Equation (42) gives the order. The size of the  
1028 error, and which of the two bases it belongs to, appear when the scattered  
1029 field is separated by its order in the contrast: the Born term  $T_1$ , and the  
1030 second-order term  $T_2$  in which the field scattered once is scattered again.

1031 Appendix Appendix G gives both terms for the exact layer in closed  
1032 form (Eqs. (G.2)–(G.7)) and for the scheme, where the first needs no solve  
1033 (Eq. (G.8)), and shows how both are read from solutions at four scaled  
1034 contrasts (Eq. (G.9)). The terms must be separated: at the full contrast the  
1035 terms of third and higher order are 7% of the second and carry errors of their  
1036 own.

1037 With  $f(z)$  the profile and  $\Pi_q$  the  $L^2$  projection onto the Legendre poly-  
1038 nomials of degree at most  $q$  in each cell, define the *relative projection error*

1039 of the medium at degree  $q$ ,

$$\mathcal{E}_q = \frac{\sum_{\text{cells}} \int (f - \Pi_q f)^2 dz}{\int f^2 dz} = 1 - \frac{\|\Pi_q f\|^2}{\|f\|^2}, \quad \|f\|^2 = \int f^2 dz : \quad (43)$$

1040 the fraction of the squared norm of the profile that polynomials of degree  $q$   
 1041 on the cells cannot represent. It is a property of the medium and the grid  
 1042 alone; it involves neither the wave nor the solution. For the smooth profile  
 1043 and every one of the nine pairs with  $p, r \leq 2$ , the relative error of  $T_2$  is

$$\frac{|T_2^{\text{scheme}} - T_2|}{|T_2|} = \mathcal{E}_{\min(p,r)} \quad (44)$$

1044 to within 1.3% at  $\omega = 300 \text{ rad s}^{-1}$  on 4 to 16 cells: the ratio of the two sides  
 1045 is 0.998 to 1.005 when  $\min(p, r)$  is 0 or 1, and 1.010 to 1.013 when it is 2.  
 1046 The error of the Born term is smaller by two to six orders of magnitude. The  
 1047 error of the whole is therefore that of  $T_2$ , the projection error times  $|T_2|/|T_1|$   
 1048 (0.044 here): for  $p = r = 1$  on 16 cells that predicts  $1.6 \times 10^{-7}$ , and Table 9  
 1049 has  $1.4 \times 10^{-7}$ .

1050 Equation (44) is exact in the long-wave limit (Appendix Appendix G).  
 1051 There the strain scattered once is the profile itself, a cell holds it as  $\Pi_{\min(p,r)} f$ ,  
 1052 and the second scattering takes place in the cell of the first: the law is  
 1053 a statement about the single site, whose  $T$ -matrix is exact for the cell as  
 1054 represented and whose error is that the cell is not the medium. Measured,  
 1055 the part of  $T_2$  between different cells carries at most 0.002 of the error, and  
 1056 the departure from the law falls as  $(kD)^2$ .

1057 Three consequences follow. The two bases must be matched: a contrast  
 1058 of higher degree than the field is wasted (the pairs  $(p, r) = (0, 1)$  and  $(0, 2)$   
 1059 have the errors of  $(0, 0)$  to three digits), and a field of higher degree than  
 1060 the contrast improves only the Born term. The wavefield basis is there to

1061 carry what the medium imprints on the field, not only the wave: at these  
 1062 frequencies the wave itself varies little across a cell. And the error is known  
 1063 from the medium, before anything is solved:  $\mathcal{E}_q$  is a property of the profile  
 1064 and the cells. It vanishes when the medium is piecewise constant on the cells,  
 1065 which is the stratified model above; there the contrast needs no more than a  
 1066 constant in each cell, and the degree of the field alone sets the order.

1067 *A stratified background.* Everything above takes the background to be a  
 1068 whole space. The method is meant for a background that is plane strati-  
 1069 fied, with the voxels carrying the departure from it, and the results must  
 1070 hold there. They do, in the one case tested: normal incidence, P, with the  
 1071 background two half-spaces joined at  $z = z_I$  below the layer, the lower one  
 1072 faster and denser (displacement reflection coefficient  $-0.19$ ). The voxels then  
 1073 interact through the Green’s function of that background,

$$g(z, z') = \frac{i}{2M_P k} \left[ e^{ik|z-z'|} + \rho_I e^{ik(2z_I - z - z')} \right], \quad z, z' < z_I, \quad (45)$$

1074 with  $\rho_I$  the reflection coefficient of the interface: the whole-space term and  
 1075 a reverberation that is smooth and carries no local term. The incident field  
 1076 is the field of the source in the background, and the exact answer is the  
 1077 reflection of the whole stack less that of the background. With the interface  
 1078 3 m below the layer, and again with the lowest voxel against it, the scheme  
 1079 converges at order 2.00, 4.00 and 6.0 for  $p = 0, 1$  and 2 on the uniform  
 1080 layer, and at 1.98, 3.98 and 5.98 on the smooth profile with  $r = p$ . The  
 1081 first-order term is predicted to round-off ( $5 \times 10^{-16}$ ): it is the sum over  
 1082 four paths, the incident wave travelling down or up and the scattered wave  
 1083 leaving up or down, of the exact first-order term of each path times the

1084 factor  $\sum_{a \leq p} (2a + 1) j_a(k_{\text{in}} h) j_a(k_{\text{out}} h) / j_0((k_{\text{in}} - k_{\text{out}}) h)$  for that path's pair  
 1085 of wavenumbers, whose leading terms are the constants of Eq. (40). And the  
 1086 relative error of  $T_2$  over  $\mathcal{E}_p$  is 1.00 to 1.02 at  $kD = 0.12$  on 4 to 16 cells for all  
 1087 three degrees, departing from one by  $1.1 \times 10^{-4}$  and  $3 \times 10^{-5}$  at  $kD = 0.012$   
 1088 for the two positions of the interface: Eq. (44) holds as in the whole space  
 1089 (`scripts/measure_layer_stratified_background.py`; the exact stack is  
 1090 computed in 30-digit arithmetic). The reason is the one given for Eq. (G.10):  
 1091 the reverberation adds no local term, so the second scattering in the cell of  
 1092 the first is unchanged. This check is by one implementation against an exact  
 1093 solution.

1094 *Oblique incidence, and a three-layer background.* The same holds at oblique  
 1095 incidence, for P, SV and SH waves, in the two half-spaces and in a three-layer  
 1096 background in which the layer of voxels reverberates between two interfaces  
 1097 (Appendix Appendix C). At  $20^\circ$  the orders are 2, 4 and 6 for  $p = 0, 1$  and  
 1098 2 in all thirty-six cases (Table C.15), and the errors are close to those of the  
 1099 whole space: the background changes the field the voxels must represent,  
 1100 not how well they represent it. Both terms of the error remain known in  
 1101 advance. The scheme's first-order term equals a closed-form prediction, built  
 1102 from the thirty-six pairs of incident and radiated plane waves (Eq. (C.3)), to  
 1103  $2 \times 10^{-15}$ ; and the law of Eq. (44) holds to 3% for  $p \leq 1$  and 6% for  $p = 2$  at  
 1104  $k_S D = 0.2$ , and to 0.6% at  $k_S D = 0.067$ , in every background and for every  
 1105 wave type (Table C.16).

## 1106 10. A sphere: what the layer predicts in three dimensions

1107 The layer and the stratified layer are laterally uniform. A sphere is not,  
 1108 and elastic scattering by a sphere, homogeneous [55, 56] or radially varying  
 1109 [60], has an exact solution: it is the usual benchmark for volume-integral and  
 1110 discrete-dipole methods. This section solves two spheres with the full three-  
 1111 dimensional voxel scheme: the sphere is voxelised on an  $n_{\text{sub}}^3$  grid, the Foldy-  
 1112 Lax system is solved with the cell-averaged coupling by an FFT-accelerated  
 1113 iterative solver, and a P wave is incident. For the graded sphere the voxel is  
 1114 the mean-only voxel of §8 in three dimensions: each cell carries the mean of its  
 1115 nine-component state, tested with the same constant, and the  $L^2$  projection  
 1116 of its contrast, which is its cell mean; every cell that overlaps the sphere is  
 1117 kept, and the cells are coupled by cell-to-cell double integrals of the Green’s  
 1118 tensor, one block per orbit of the cube group (Appendix Appendix E). For  
 1119 the sharp sphere the voxel is the package’s cube  $T$ -matrix, which differs from  
 1120 the collocation closure only through its far-field corrections (§6.6) and has  
 1121 the same order, each cell keeping the contrast of its centre. The far field is  
 1122 computed from the solution for the incident plane wave itself, each voxel’s  
 1123 displacement and strain carrying the plane-wave phase at its centre; adding  
 1124 to that phase a linear variation about the scatterer’s centre, as a Taylor  
 1125 expansion of the incident field would, counts the wave’s variation twice and  
 1126 leaves an error of order  $(ka)^2$  that no refinement removes. In the terms of  
 1127 the experiment of §7, the test is this.

- 1128 1. *The body.* A sphere of radius  $a = 10$  m (Fig. 5) in the background of  
 1129 §7, carrying the contrast of that section, at  $\omega = 150$  and  $300 \text{ rad s}^{-1}$ ,  
 1130 for which  $k_S a = 0.5$  and  $1$  and  $k_P a = 0.3$  and  $0.6$ .

- 1131 2. *The incident wave.* A P plane wave travelling along  $+z$ .
- 1132 3. *The grid.* The cube of side  $2a$  that circumscribes the sphere is cut into  
1133  $n_{\text{sub}}^3$  cells of side  $2a/n_{\text{sub}}$ ; which of them are kept is stated with each  
1134 voxel above.
- 1135 4. *The solve.* The iteration is carried to a relative residual of  $10^{-11}$ , far  
1136 below every discretisation error reported, so that the errors are the  
1137 scheme's and not the solver's.
- 1138 5. *The exact answer.* The partial-wave series of the sphere, summed to  
1139 order 10 at  $k_S a = 0.5$  and 11 at  $k_S a = 1$ , and evaluated as the exact  
1140 scattered displacement at the observers, not as its asymptotic form.
- 1141 6. *What is measured.* The scattered displacement vector at nine scattering  
1142 angles, equally spaced from  $11^\circ$  to  $169^\circ$  in the  $z$ - $x$  plane: forward and  
1143 backward scattering and everything between, P and S together.
- 1144 7. *The error.* The largest component of the difference between the scheme's  
1145 far field and the exact displacement over the nine angles, divided by the  
1146 largest component of the exact one. It is a maximum, not an average,  
1147 so that no angle is hidden.
- 1148 8. *The distance.* The solution of the scheme is the field inside the voxels,  
1149 and the scattered field outside follows from it at any distance. For the  
1150 far-field test it is read with the asymptotic formula, which keeps the  
1151 term in  $1/r$  alone: a choice of this test, not a property of the scheme.  
1152 The reference is the exact field, so the two differ by the near-field terms,  
1153 of relative size  $1/(kr)$ , however good the scheme. The graded sphere  
1154 is therefore observed at  $5 \times 10^8$  radii, where those terms are  $10^{-8}$  of  
1155 the peak: at  $5 \times 10^4$  radii they are  $1.4 \times 10^{-4}$ , which is 4% of the

1156 discretisation error at 12 voxels across and most of it at 24, and would  
 1157 be read as a failure to converge. The sharp sphere, whose errors are  
 1158 a hundred times larger, is observed at  $5 \times 10^4$  radii. The near field is  
 1159 tested separately, with a readout that drops nothing (Table D.17).

1160 *A sphere without a staircase.* A voxelised sphere with a sharp surface changes  
 1161 its shape with every grid. A sphere whose contrast falls smoothly to zero at  
 1162 its surface does not: the voxels that cross the surface carry a vanishing con-  
 1163 trast. The body here has a homogeneous core  $r < a/2$  carrying the contrast  
 1164 of §7, and across the shell  $a/2 < r < a$  the contrast falls to zero as the  
 1165 smoothstep  $s(x) = 10x^3 - 15x^4 + 6x^5$ ,  $x = (a - r)/(a - a/2)$ , which vanishes  
 1166 like  $(a - r)^3$  at the surface. Its exact scattering follows, order by order, from  
 1167 the radial equations of Takeuchi and Saito [60], integrated across the shell  
 1168 from the core's regular solutions and matched to outgoing waves at  $r = a$ .  
 1169 The equations are derived symbolically from the equations of motion, and  
 1170 the solution is checked against a sphere of homogeneous concentric shells  
 1171 in closed form, by unitarity and reciprocity, and by an independent imple-  
 1172 mentation that agrees to  $2 \times 10^{-11}$  (`TakeuchiSaito.wl`, `ContinuumLimit_`  
 1173 `GradedSphere.wl`, `scripts/crosscheck_graded_sphere.py`). Each voxel  
 1174 carries the mean of the contrast over its cell. Table D.18 shows the error  
 1175 falling at second order from  $n_{\text{sub}} = 6$  to 24 at both frequencies (`scripts/`  
 1176 `pilot_graded_voxel_sphere.py`): the apparent order between successive  
 1177 grids is 2.02 to 2.06 at  $k_S a = 0.5$  and 2.00 at  $k_S a = 1$ , and the least-squares  
 1178 orders of Fig. 6(d) are  $2.045 \pm 0.004$  and  $1.994 \pm 0.004$ . With 24 voxels across  
 1179 the error is  $1.8 \times 10^{-4}$  and  $8.3 \times 10^{-4}$ . At  $n_{\text{sub}} = 4$  a voxel is as wide as the  
 1180 graded shell and the order is not yet asymptotic. This is the order the layer

1181 predicts for the mean-only voxel, and the first test of that prediction on a  
1182 body with genuine three-dimensional structure.

1183     What makes the sequence this clean is that the voxel carries a single error.  
1184 The collocation voxel of the sharp sphere applied to this body, each cell at  
1185 the profile's value at its centre and only the cells whose centres lie inside the  
1186 sphere kept, carries two: its own internal-field error and the error of sampling  
1187 the profile, both  $O(h^2)$ . Its error falls from  $1.0 \times 10^{-1}$  at  $n_{\text{sub}} = 4$  to  $4.7 \times 10^{-3}$ ,  
1188  $2.6 \times 10^{-3}$ ,  $6.4 \times 10^{-4}$  and  $2.3 \times 10^{-4}$  at 6, 8, 12 and 16, apparent orders 7.6,  
1189 2.0, 3.5 and 3.6 at  $k_S a = 0.5$ , and at  $k_S a = 1$  the apparent orders are 1.7, 2.5  
1190 and 2.4 from 6 to 16: on these grids its error is below the mean-only voxel's,  
1191 but its apparent order is not that of any single term and cannot be read from  
1192 them. The same collocation voxel with the cell mean of the contrast, and  
1193 every overlapping cell kept, gives  $3.4 \times 10^{-3}$ ,  $1.9 \times 10^{-3}$  and  $8.2 \times 10^{-4}$  at 6,  
1194 8 and 12, apparent orders 2.03 and 2.05: the sampling of the profile was the  
1195 second term, and with it removed the collocation and mean-only voxels differ  
1196 only by their constants, as in Table 7. The cell mean, with every overlapping  
1197 cell kept, represents the same body on every grid, and the order the layer  
1198 predicts is then the one observed.

1199     The first-moment voxel of §8, extended to three dimensions with a con-  
1200 trast linear in each cell, converges at fourth order on this same sphere (Ap-  
1201 pendix Appendix E): its apparent orders are 4.01 and 4.13 where the dis-  
1202 cretisation error is isolated at weak contrast, and at the full contrast its error  
1203 with 16 voxels across is  $4.9 \times 10^{-6}$  and  $5.9 \times 10^{-6}$ , 82 and 317 times below  
1204 the mean-only voxel's on the same grid.

1205 *The near field, an impedance march, and a sharp sphere.* Three further tests  
 1206 are in Appendix Appendix D. Read at finite distance, down to a quarter  
 1207 of a radius from the surface, the graded sphere converges at the orders of  
 1208 the far field, second for the mean-only voxel and 3.8 to 3.95 for the first-  
 1209 moment voxel, while the asymptotic readout would be in error there by 48%  
 1210 to 85% (Table D.17). Against the impedance march of Haines et al. [59],  
 1211 which needs far fewer unknowns, the first-moment voxel reaches the march's  
 1212 accuracy with six voxels across and continues at close to fourth order below  
 1213 it, where the march is limited by the scattering between the spheres of the  
 1214 periodic array it must solve (Table D.19). And a sphere with a sharp surface  
 1215 does not converge: its error tracks the volume of its staircase, which changes  
 1216 with the grid (Table D.20); the smooth sphere, like the layer, measures the  
 1217 scheme.

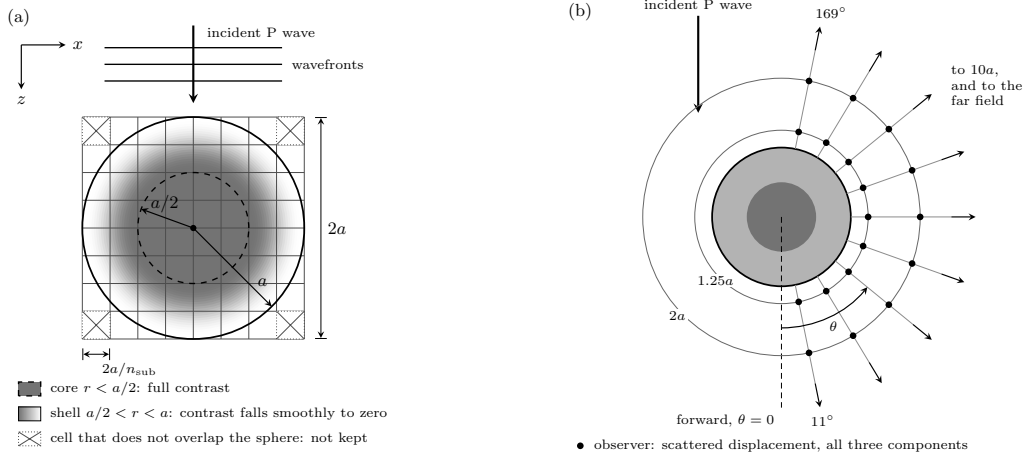


Figure 5: The sphere experiment, in the section through the centre that contains the direction of incidence. (a) The body and the grid. The contrast is full in the core  $r < a/2$  and falls smoothly to zero across the shell  $a/2 < r < a$ . The circumscribing cube of side  $2a$  is cut into  $n_{\text{sub}}^3$  cells, here  $n_{\text{sub}} = 8$ ; every cell that overlaps the sphere is kept and carries the projection of the contrast it contains. A P plane wave is incident along  $+z$ . (b) The observers. The scattered displacement is taken at nine scattering angles  $\theta$ , measured from the forward direction and equally spaced from  $11^\circ$  to  $169^\circ$ , on spheres of several radii about the centre, and compared with the exact displacement at the same points.

*Alt text:* Two panels. On the left, a disc on a square grid of eight by eight cells: the inner half of the disc is uniformly dark, and the outer half fades from dark to white at the rim, which is drawn as a solid circle; the four corner cells of the grid, which the disc does not reach, are crossed out. Horizontal lines above the grid with a downward arrow mark the incident wave. On the right, the same disc drawn smaller, with two concentric circles around it at one and a quarter and at two radii; nine dots on each circle lie on nine rays from the disc, fanning from just off the downward, forward direction to just off the upward, backward one, and each ray ends in an outward arrow.

## 1218 11. The gradient hierarchy as a voxel scheme

1219 The Legendre voxels of §8 represent the field of a voxel by its moments  
 1220 over the voxel. The hierarchy of §6 represents it by its gradients at the  
 1221 centre. Both are polynomial, and §6.4 shows that on the layer the hierarchy  
 1222 of degree  $\ell = 3$  is fourth order, as the first-moment voxel is. This section  
 1223 builds the hierarchy into a voxel scheme in three dimensions and compares  
 1224 the two.

1225 *The scheme.* Each voxel  $c$ , centred at  $\mathbf{x}_c$ , carries the coefficients  $U_{j;W}^c$  of  
 1226 Eq. (20). The equation and its derivatives to order  $\ell$  are imposed at ev-  
 1227 ery centre (with  $\ell = 1$  and no density contrast this is the static numerical  
 1228 equivalent inclusion method on cuboidal elements, with uniform eigenstrains,  
 1229 equivalence at the element centres and a fast Fourier transform, of Zhou et al.  
 1230 [61]):

$$U_{i;P}^c - \sum_{c'} \left\{ \omega^2 \Delta \rho \sum_W \frac{K_{ij}[P; W](\mathbf{x}_c - \mathbf{x}_{c'})}{|W|!} U_{j;W}^{c'} + \sum_W \frac{K_{in}[Pk; W](\mathbf{x}_c - \mathbf{x}_{c'})}{|W|!} \delta c_{nkrj} U_{j;rW}^{c'} \right\} = \partial_P u_i^0(\mathbf{x}_c), \quad (46)$$

$$K_{in}[D; W](\Delta) = \int_V (\partial_D G_{in})(\Delta - \xi) \xi_W d\xi. \quad (47)$$

1232 The two kinds of block are of different character. For  $c' = c$  the integrand is  
 1233 singular and the integral is a distribution: it is the moment of the hierarchy,  
 1234  $K_{in}[D; W](0) = (-1)^{|D|} \mathbf{g}_{in}[D; W]$ , in closed form on the cube. For  $c' \neq c$   
 1235 the field point is the centre of another voxel, at least half a voxel from the  
 1236 source voxel, so the integrand is smooth and a tensor Gauss rule integrates  
 1237 it; the derivatives of the Green's tensor, to any order, come from the family  
 1238  $x^a y^b z^c r^p e^{ikr}$ , which is closed under differentiation. All the singular work  
 1239 of the scheme is thus in the self block, where it is done once and in closed

1240 form; the Legendre voxel's self block is a double integral of the same singular  
 1241 kernel. The blocks depend on the offset alone, so the product of the system  
 1242 with a vector is a convolution over the lattice and is done exactly by the fast  
 1243 Fourier transform. Where the contrast varies within a voxel it is projected  
 1244 there on polynomials of degree  $r$ , as in §9, and the source is the product of  
 1245 that polynomial and the field's: the moments then carry weights to degree  
 1246  $\ell + r$ , and a block is a sum, over the monomials of the contrast, of blocks  
 1247 with the weight shifted.

1248 *Checks of the construction.* The derivatives of the Green's tensor to fourth  
 1249 order agree with the closed forms of its value, gradient and second derivative,  
 1250 and with central differences beyond them, to  $8 \times 10^{-7}$ . A lattice of one voxel  
 1251 reproduces the single site of §6.5 to  $10^{-14}$ . Only the offsets of one octant  
 1252 are computed, the others following from the parity of each index; a block  
 1253 obtained that way equals one computed directly to  $10^{-15}$ . The iterative  
 1254 solution equals a dense solution to  $7 \times 10^{-11}$  where both can be run.

1255 *A body with corners.* The cube of §6.5, at  $k_{sh} = 0.2$ , is cut into  $n^3$  voxels.  
 1256 Table 10 gives the far-field error against the converged reference, with the  
 1257 Legendre voxels on the same body. On this body every degree of the hierarchy  
 1258 converges at about the same rate, an order near two between the two finest  
 1259 grids, and the degree buys a constant: thirty times from  $\ell = 1$  to  $\ell = 3$   
 1260 on 64 voxels. That rate is the body's and not the scheme's, for the mean-  
 1261 only Legendre voxel shows it too: a homogeneous cube has edges and corners,  
 1262 whose fields no polynomial follows. The uniform-strain closure and the mean-  
 1263 only voxel agree on one cell, as they must, being the same approximation.  
 1264 And the first-moment Legendre voxel is far ahead: one cell of 36 unknowns is

Table 10: A homogeneous cube cut into  $n^3$  voxels,  $k_S h = 0.2$ : relative far-field error of the hierarchy of degree  $\ell$  and of the Legendre voxels, with the unknowns of each voxel.

|                        | unknowns a voxel | $n = 1$              | 2                    | 3                    | 4                    |
|------------------------|------------------|----------------------|----------------------|----------------------|----------------------|
| hierarchy, $\ell = 1$  | 12               | $5.4 \times 10^{-3}$ | $1.7 \times 10^{-3}$ | $8.3 \times 10^{-4}$ | $5.0 \times 10^{-4}$ |
| hierarchy, $\ell = 2$  | 30               | $1.6 \times 10^{-3}$ | $6.7 \times 10^{-4}$ | $3.7 \times 10^{-4}$ | $2.4 \times 10^{-4}$ |
| hierarchy, $\ell = 3$  | 60               | $3.4 \times 10^{-4}$ | $7.0 \times 10^{-5}$ | $3.0 \times 10^{-5}$ | $1.7 \times 10^{-5}$ |
| Legendre, mean only    | 9                | $5.4 \times 10^{-3}$ | $1.3 \times 10^{-3}$ | $5.8 \times 10^{-4}$ | $3.3 \times 10^{-4}$ |
| Legendre, first moment | 36               | $4.8 \times 10^{-5}$ | $6.0 \times 10^{-6}$ |                      |                      |

more accurate than eight voxels of the third-gradient hierarchy with 480. A projection, which the Legendre voxel is, follows a field that is not polynomial better than an expansion about a point: the advantage of a Galerkin over a Taylor discretisation of the Lippmann–Schwinger equation that Brisard et al. [47] established for the static equivalent inclusion method.

*A smooth medium.* The comparison that bears on voxel schemes is in a medium without corners. First in one dimension: the layer of §6.4 with a contrast that varies in depth, projected in each cell to the degree of the field, against the exact graded layer. With the profile  $\sin^2(\pi z/D)$  the orders from 16 to 32 cells are 1.99, 1.98 and 3.99 for  $\ell = 1, 2$  and 3: a varying contrast leaves the pairs of §6.4 as they are. With the smoothstep of §10 as the profile they are 1.97, 1.99 and 3.84, and on coarser grids they are erratic, the third-gradient scheme being worse than the second on four cells. The orders appear only when about four cells cross the graded part of the profile (`scripts/measure_layer_taylor_hierarchy_graded.py`). That fixes the conditions of the three-dimensional test. The graded sphere of §10 has a shell

1281 5 m wide, and a solver that reaches fourteen voxels across puts fewer than  
1282 four voxels across it; so the core is reduced here from  $a/2$  to  $a/10$ , which  
1283 widens the shell to 9 m, and a second profile,  $\sin^2(\pi x/2)$  in the fraction  $x$  of  
1284 the shell's width measured from the surface, is run beside the smoothstep.  
1285 The exact reference is the radial solution of §10 for each body. The contrast  
1286 is linear in each voxel for  $\ell = 3$ , which suffices for fourth order (§9), and of  
degree  $\ell$  for  $\ell = 1$  and 2.

Table 11: The graded sphere with a shell 9 m wide,  $k_{Sa} = 0.5$ : relative far-field error  
against the exact solution for  $n_{\text{sub}}$  voxels across, with the apparent order between the two  
finest grids of each row. The lower block is the profile  $\sin^2$ .

|                        | $n_{\text{sub}} = 4$ | 6                    | 8                    | 10                   | 12                   | 14                   | order |
|------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|-------|
| hierarchy, $\ell = 1$  | $1.5 \times 10^{-3}$ | $8.4 \times 10^{-4}$ | $4.9 \times 10^{-4}$ | $3.2 \times 10^{-4}$ |                      |                      | 1.94  |
| hierarchy, $\ell = 2$  | $3.3 \times 10^{-4}$ | $2.1 \times 10^{-4}$ | $1.5 \times 10^{-4}$ | $1.0 \times 10^{-4}$ |                      |                      | 1.56  |
| hierarchy, $\ell = 3$  | $2.6 \times 10^{-4}$ | $5.0 \times 10^{-5}$ | $1.7 \times 10^{-5}$ | $7.2 \times 10^{-6}$ | $3.6 \times 10^{-6}$ | $2.0 \times 10^{-6}$ | 3.80  |
| Legendre, mean only    | $7.3 \times 10^{-3}$ | $3.2 \times 10^{-3}$ | $1.8 \times 10^{-3}$ | $1.1 \times 10^{-3}$ |                      |                      | 2.02  |
| Legendre, first moment | $2.9 \times 10^{-4}$ | $6.4 \times 10^{-5}$ | $2.2 \times 10^{-5}$ | $9.4 \times 10^{-6}$ | $4.6 \times 10^{-6}$ | $2.5 \times 10^{-6}$ | 3.92  |
| hierarchy, $\ell = 3$  | $1.9 \times 10^{-4}$ | $3.8 \times 10^{-5}$ | $1.3 \times 10^{-5}$ | $5.2 \times 10^{-6}$ | $2.7 \times 10^{-6}$ | $1.5 \times 10^{-6}$ | 3.84  |
| Legendre, first moment | $2.2 \times 10^{-4}$ | $4.8 \times 10^{-5}$ | $1.6 \times 10^{-5}$ | $6.8 \times 10^{-6}$ | $3.4 \times 10^{-6}$ | $1.8 \times 10^{-6}$ | 4.02  |

1287  
1288 Table 11 gives the result. The first-gradient scheme converges at second  
1289 order, and is three to four times more accurate than the mean-only Legendre  
1290 voxel on every grid, for twelve unknowns a voxel against nine. The second-  
1291 gradient scheme is more accurate again but gains no order, as §6.4 predicts;  
1292 its apparent order is still rising towards two. The third-gradient scheme  
1293 converges at an apparent order between 3.7 and 4.1 from grid to grid, 3.8

1294 over the range from six to fourteen voxels across, for both profiles; the first-  
 1295 moment Legendre voxel does the same, between 3.7 and 4.0 and 3.85 over  
 1296 the range. Neither has settled on four at fourteen voxels across, and the  
 1297 second profile, gentler than the first, changes neither: the two fourth-order  
 1298 schemes behave alike. On every grid from six voxels across the hierarchy's  
 1299 error is the smaller, by a fifth to a quarter. It carries sixty unknowns a voxel  
 1300 against thirty-six, so that for a given number of unknowns the Legendre voxel  
 1301 remains somewhat the more accurate; the wide margin of Table 10 was the  
 1302 corners.

1303 *What the hierarchy is for.* The two schemes reach the same order by different  
 1304 closures of the same equation, and they differ in what they cost and in what  
 1305 they give besides. The Legendre voxel is the more economical in unknowns.  
 1306 The hierarchy's blocks are simpler: its self block is known in closed form on  
 1307 four constants (§6.2), and its coupling is a single integral of a smooth func-  
 1308 tion where the Legendre voxel's is a double integral. The iteration converged  
 1309 in five steps at every size; the time goes into the tables of blocks, which are  
 1310 computed once for a grid and a frequency. Formed on the same footing, one  
 1311 block for each orbit of offsets under the 48 symmetries of the cube, the others  
 1312 following exactly, and every point value of the Green's tensor or its deriva-  
 1313 tives from the same compiled radial ladder, the hierarchy's tables are the  
 1314 cheaper: 0.6, 1.9 and 4.5 s for 6, 10 and 14 voxels across, against 4.0, 9.3 and  
 1315 23.3 s for the first-moment Legendre voxel, with eight times fewer kernel eval-  
 1316 uations. Both have a one-off cost besides, the hierarchy's larger (13 s against  
 1317 3 s), which is the closed-form constants of its self block (`scripts/measure_`  
 1318 `hierarchy_table_cost.py`; one machine, five threads). And the hierarchy

1319 is, in addition, a single-site  $T$ -matrix in closed form, whose accuracy improves  
1320 block by block (§6.5), which no voxel defined by moments over the cell pro-  
1321 vides. The lattice results of this section come from one implementation,  
1322 checked against exact solutions and, where it can be, against a dense solve;  
1323 they are at one frequency and on one family of bodies (`scripts/gradient_`  
1324 `voxel_lattice.py`, `scripts/measure_lattice_gradient_hierarchy.py`,  
1325 `scripts/measure_graded_sphere_gradient_hierarchy.py`, `scripts/measure_`  
1326 `graded_sphere_gradient_hierarchy_fft.py`).

## 1327 12. Discussion

1328 Figure 6 collects the convergence of every test, with the order fitted to  
1329 each by least squares (`scripts/plot_convergence_orders.py`).

1330 *What each basis is for.* A voxel holds two functions, and the results give  
1331 each basis its part. The basis of the wavefield sets how well the voxel carries  
1332 the field across itself: in a uniform medium the scheme converges at order  
1333  $2p + 2$  with the constant  $c_p$  of Eq. (40), for P and S waves and at normal and  
1334 oblique incidence alike. The basis of the medium sets how well the voxel *is*  
1335 the medium: with degree  $r$  the order is capped at  $2r + 2$  (Eq. (42)). Neither  
1336 can make up for the other, and the second-order term shows why. The field  
1337 scattered once is a copy of the medium’s profile, so the wavefield basis must  
1338 carry the profile as well as the wave, and the error of that term is the relative  
1339 projection error of the profile at the lower of the two degrees (Eq. (44)). At  
1340 the frequencies of interest the wave varies little across a cell and the medium  
1341 may vary much, so it is the medium, not the wavelength, that decides the  
1342 degree and the size of the cells. Two cases bracket the choice. Where the

1343 medium is piecewise constant on the cells, as in a layered model with voxel  
 1344 faces on its interfaces, a constant is exact for the medium, the projection  
 1345 error vanishes, and the degree of the field is spent entirely on the wave.  
 1346 Where the medium varies smoothly within the cells, the two degrees should  
 1347 be equal: a higher degree for the contrast than for the field is wasted, and a  
 1348 higher degree for the field improves only the Born term. That the two degrees  
 1349 can be chosen independently is the practice of higher-order electromagnetic  
 1350 volume methods [1]; what the present results add is the rule for choosing  
 1351 them, and the error each choice leaves.

1352 *Where the error of a voxel scheme lives.* The results separate the two ap-  
 1353 proximations a voxel scheme makes. The *coupling* between voxels makes  
 1354 none: averaged over the source cell it is the continuum's cell integral exactly,  
 1355 between planes by the form factor's zeros and within a plane by the tiling  
 1356 identity, and the self term it omits is exactly the one the single site carries.  
 1357 Every error therefore comes from the *single site*: from the internal field it  
 1358 assumes and, where the medium varies within the cell, from the medium it  
 1359 is given. Its  $T$ -matrix sums the rescattering within the voxel exactly, but  
 1360 for the voxel as represented, and the split of the second-order term in §9  
 1361 measures the consequence: the error sits in the part whose two scatterings  
 1362 lie in one cell, and the part between cells carries at most 0.2% of it. For a  
 1363 uniform field that error is the midpoint rule's, known in closed form; for a  
 1364 linear field it is the next term of the same expansion. The converse is the  
 1365 practical warning. A kernel that is not the cell integral (the point Green's  
 1366 tensor, or a cell average that does not match the single site's closure) leaves  
 1367 a local term that does not vanish as the voxels shrink (§4): the evanescent

1368 lattice orders carry a static contribution of fixed size per voxel volume. A  
1369 scheme built that way converges to the wrong limit, and refinement cannot  
1370 reveal it, because the error is scale-invariant.

1371 *What the single site is.* The moment hierarchy of §6 turns the uniform-field  
1372 single site from a modelling choice into a derived object with a stated error  
1373 (§6.7 collects what that gives). Its static part is exact for a sphere, and for  
1374 a cube it differs from the sphere’s only by the cubic split of Eq. (32). Its  
1375 dynamic part departs from exact scattering by a sphere only through the  
1376 real  $(ka)^2$  term of Eq. (F.1); the radiation damping is exact. That term and  
1377 the collocation error of Eq. (36) have the same origin, the variation of the  
1378 field across the scatterer, and the same remedy, a richer internal field. In a  
1379 lattice the remedy is cheap: the field a voxel must represent is the smooth  
1380 continuum field, not the field of an isolated cube, with its concentrations at  
1381 the edges and corners.

1382 *Relation to other methods.* The partition of the error is the coupled-dipole  
1383 method’s choice of polarisability seen from the other side: for space-filling  
1384 cubes the tiling identity fixes the local field exactly, and the polarisability  
1385 is neither needed nor a free parameter. The voxel of degree  $p$  is close kin  
1386 to the spectral element of seismology; the two share a Galerkin condition  
1387 on polynomial bases, and differ as a differential formulation from an integral  
1388 one, and so suit different problems. Appendix Appendix H sets out both  
1389 relations, and where each method is the better choice.

1390 *The tension between the two error sources.* The coupled-dipole method has  
1391 two sources of error, the single site and the coupling between sites, and each

1392 has been corrected on its own: the polarisability by the Clausius–Mossotti  
 1393 relation, the lattice dispersion relation or the exact static limit, and the  
 1394 coupling by integrating or filtering the Green tensor. The two are not in-  
 1395 dependent. The point kernel’s evanescent orders carry a static local term  
 1396 whose size per voxel volume does not shrink with the voxels (§4), and the  
 1397 single site must cancel exactly that term. A polarisability tuned for one  
 1398 kernel is therefore wrong for another: the lattice dispersion relation corrects  
 1399 the point-dipole lattice, and changing the kernel to an integrated or a fil-  
 1400 tered one changes the term it must cancel. Tuning either alone moves the  
 1401 error between the two, and refinement cannot reveal a mismatch, because  
 1402 the surviving term is scale-invariant. For space-filling cubes the tension is  
 1403 resolved rather than balanced: the cell-averaged kernel omits exactly the self  
 1404 term that the closure carries (Eq. (35)), so the coupling error vanishes iden-  
 1405 tically and every remaining error is the single site’s, the variation of the field  
 1406 across the cell. That error has a second tension of its own. It is a midpoint-  
 1407 rule error whose constants have opposite signs in reflection and transmission  
 1408 (Eq. (36)), so any scalar adjustment of the single site that improves one wors-  
 1409 ens the other; the package’s far-field corrections reduce the reflection error  
 1410 by a factor 4.45 and increase the transmission error by 1.7. No polarisability,  
 1411 however chosen, can remove it. What removes it is a richer internal field:  
 1412 the first moment lowers the error by two orders in  $kd$  in both observables at  
 1413 once.

1414 *The cost of higher order, and a fast solver.* A first-moment voxel carries four  
 1415 times as many unknowns as a uniform one, but by the closed-form constants it  
 1416 reaches a reflection error of  $10^{-6}$  with voxels 58 times larger in each direction;

1417 the stratified model of §9 measures it directly, 347 planes of uniform-field  
 1418 voxels against 8 first-moment voxels. And because the discretisation error  
 1419 is known in advance and the coupling between cells is exact, a fast product  
 1420 need be no more accurate than that error, and can be budgeted separately  
 1421 from it. Appendix Appendix I sets out the cost, and what the error analysis  
 1422 offers a fast solver.

1423 *Limitations and outlook.* The results concern laterally uniform media, a ho-  
 1424 mogeneous layer and a randomly stratified one, for which the discrete sys-  
 1425 tem is exactly a chain of planes; a laterally heterogeneous medium needs  
 1426 the first-moment voxel in a full three-dimensional solver, where the gradi-  
 1427 ent channels couple through the evanescent lattice orders (§8) and the tiling  
 1428 identity holds only to  $O(k_{\parallel}h)$ . The smoothly graded sphere of §10 shows  
 1429 the mean-only voxel converging at second order on a laterally heterogeneous  
 1430 body against an exact solution, and the first-moment voxel converges at  
 1431 fourth order there (§11). A contrast that varies within a cell, which §9 shows  
 1432 is needed beyond second order in a smoothly varying medium, has a further  
 1433 consequence in three dimensions: its linear part is odd about the cell’s centre,  
 1434 so it couples the single site’s uniform-field and first-moment blocks, which  
 1435 parity keeps apart for a homogeneous cube (§6.3). A graded voxel therefore  
 1436 needs at least the first-moment block; that voxel is built, and tested on the  
 1437 graded sphere, in the companion paper. The numerical evidence is at mod-  
 1438 erate, physically typical contrasts (up to 15% in density and moduli) and  
 1439 at  $k_S h \leq 0.3$ , the range in which the single-site  $T$ -matrix is validated; the  
 1440 closed forms are first order in the contrast, and the departures from them  
 1441 at the full contrast are of second order. What is established is the *order* of

1442 convergence and its leading constant, not the behaviour at strong contrast,  
 1443 where the single site's own closure may limit the accuracy before the lattice  
 1444 coupling does. The single site's dynamic error is in closed form for a sphere  
 1445 only; for a cube, for which no exact solution exists, it is known through the  
 1446 layer (§§7–8) and, for the isolated cube, by comparison with a converged  
 1447 solution (§6.5). The hierarchy beyond the first gradient is established less  
 1448 widely than the rest: its orders and its error factor on the layer are derived  
 1449 symbolically and checked independently to the third degree, and its moments  
 1450 are confirmed by two routes, but its three-dimensional results, the single site  
 1451 against the sphere and the cube and the voxel scheme of §11, come from one  
 1452 implementation, at one frequency for the voxel scheme, checked against ex-  
 1453 act solutions. There the fourth order is approached and not attained on the  
 1454 grids reached (3.8 at fourteen voxels across, as for the first-moment voxel), the  
 1455 fifth-gradient system, whose moments are in hand, has not been assembled in  
 1456 three dimensions, and the cost of forming the blocks of the two fourth-order  
 1457 schemes has not been compared. Two extensions follow directly. The first  
 1458 is the three-dimensional solver itself. The second is a stratified *background*,  
 1459 a layered reference medium, as distinct from the stratified contrast of §9.  
 1460 For a laterally uniform layer of voxels the orders, the first-order factor and  
 1461 the projection-error law all survive it, at normal and at oblique incidence,  
 1462 for P, SV and SH waves, in a background of two half-spaces and in one of  
 1463 three layers (§9). What remains is a body of finite lateral extent in such a  
 1464 background, where the image of a voxel in an interface it touches is singular.

### 13. Conclusions

A voxel of a volume-integral scheme must represent the medium within it and the wavefield within it. This paper has determined what each basis must be and what error a given choice leaves, by an experiment in which nothing else is approximated: a layer of cubes is the layer itself, so its exact solution isolates the discretisation error.

With the wavefield expanded to Legendre degree  $p$  and the contrast to degree  $r$  in each voxel, the scheme converges at order  $\min(2p+2, 2r+2)$ , with the leading error  $c_p(kd)^{2p+2}$  in closed form (Eq. (40)) for P, SV and SH waves at normal and oblique incidence, in a whole space and in a plane-stratified background. A contrast held constant in each cell caps every voxel at second order. The error of the term of second order in the contrast is the relative projection error of the medium at degree  $\min(p, r)$ : it is the single site's, whose  $T$ -matrix sums the rescattering within a voxel exactly for the voxel as represented, so the two bases must be matched, and where the medium is piecewise constant on the cells it vanishes. A sphere whose contrast falls smoothly to zero at its surface converges at the orders the layer predicts.

The single site is derived rather than assumed. Three properties of the cube make the experiment exact: the source-cell-averaged coupling is the continuum's cell integral exactly, the moments of the Green's tensor over the cube give the single site with no polarisability to choose, and its self term cancels the kernel's, so the simplest scheme is a collocation of the continuum equation. The uniform-strain closure is the first member of a moment hierarchy, the elastodynamic extension of the Taylor form of the equivalent inclusion method [46] to bodies that are not ellipsoids, whose moments are

1490 in closed form on four constants. Its blocks gain accuracy in pairs, with the  
1491 error of each truncation in closed form: as a single site the third-gradient  
1492 system is three hundred times more accurate than the uniform-strain closure  
1493 against the exact sphere, and as a voxel scheme it is fourth order, as the  
1494 first-moment voxel is, on a smoothly graded sphere, where in a lattice the  
1495 Legendre voxel is the more economical. The results and the evidence for each  
1496 are collected in Appendix Appendix J.

1497 *The error is known in advance.* For every discretisation studied here the  
1498 error is not something to be discovered by refining until the answer stops  
1499 changing. Its order and its leading constant are known before anything is  
1500 computed:  $+(kd)^2/8$  in reflection and  $-(kd)^2/24$  in transmission for the  
1501 collocation voxel (Eq. (36)),  $(kd)^2/12$ ,  $(kd)^4/720$  and  $(kd)^6/100800$  for the  
1502 voxels of degree 0, 1 and 2 (Eq. (40)), and the order  $\min(2p + 2, 2r + 2)$   
1503 where the medium varies within a cell (Eq. (42)), with the size of the error  
1504 there given by the relative projection error of the medium (Eq. (44)), which  
1505 is computed from the model and the grid alone. That is worth more than  
1506 the order alone, for three reasons. First, the cell size and the degree of the  
1507 voxel that a required accuracy needs follow from the wavenumber and the  
1508 medium, before the computation and without a refinement study. Secondly,  
1509 it turns a refinement study from a hope into a test. A scheme of this kind can  
1510 converge smoothly to the wrong limit (§4), and smooth convergence alone  
1511 cannot reveal it; but the difference between two grids must have the known  
1512 order and the known constant, and a computation whose differences do not  
1513 is in error from some cause other than the discretisation. Thirdly, it shows  
1514 where effort pays. The error is that of representing the field and the medium

1515 within a cell, so raising the degree of the cell does what multiplying the cells  
1516 cannot: in the stratified model one first-moment voxel per layer reaches  $10^{-6}$   
1517 where the uniform-field voxel needs about forty. Whether the error of a body  
1518 with three-dimensional structure can be known in advance in the same way  
1519 is the question taken up in a following paper [65], where it decides where the  
1520 cells of an adaptive grid are placed.

## 1521 **Appendix A. The moment engine: gates and a second route**

1522 Two checks, whose answers do not depend on the engine, gate every grade.  
1523 Because  $\nabla^2 r^m = m(m+1)r^{m-2}$  and  $\nabla^2(1/r) = -4\pi\delta$ ,

$$\sum_p E[m; ppR; W] = m(m+1) E[m-2; R; W], \quad \sum_p E[-1; ppR; W] = -4\pi (-1)^{|R|} (\partial_R x_W)(0), \quad (\text{A.1})$$

1524 for any further derivatives  $R$  and any weight  $W$ . The second is the sum rule  
1525 (27) at every grade: it tests the  $\delta$  function of the inclusion problem and its  
1526 derivatives, which is where a quadrature fails. Both hold for every grade of  
1527 Table 1 with two or more derivatives, 910 traces of each kind; the grades with  
1528 fewer, for which the integrand is integrable, agree with numerical integration  
1529 over the cube.

1530 The closed forms are also confirmed by a route that shares nothing with  
1531 the faces of the cube. The cube is its inscribed ball plus the remainder.  
1532 On the ball the derivatives are moved onto the sphere, where the surface  
1533 integrals of monomials are Gamma functions, and all the distributional con-  
1534 tent is there; on the remainder the integrand is an ordinary function, in-  
1535 tegrated radially in closed form and over directions through the six faces  
1536 by Gauss–Legendre quadrature. This evaluation reproduces all 4224 stored

1537 values, the largest difference being  $1.1 \times 10^{-12}$ , and satisfies Eq. (A.1) by  
1538 itself to  $5 \times 10^{-10}$  or better up to eight derivatives (`CubeMomentCore.wl`,  
1539 `CubeMomentHigherGrades.wl`, `CubeMomentStore.wl`, `CubeMomentCanonical.`  
1540 `wl`; `scripts/crosscheck_cube_moments_ball_shell.py`).

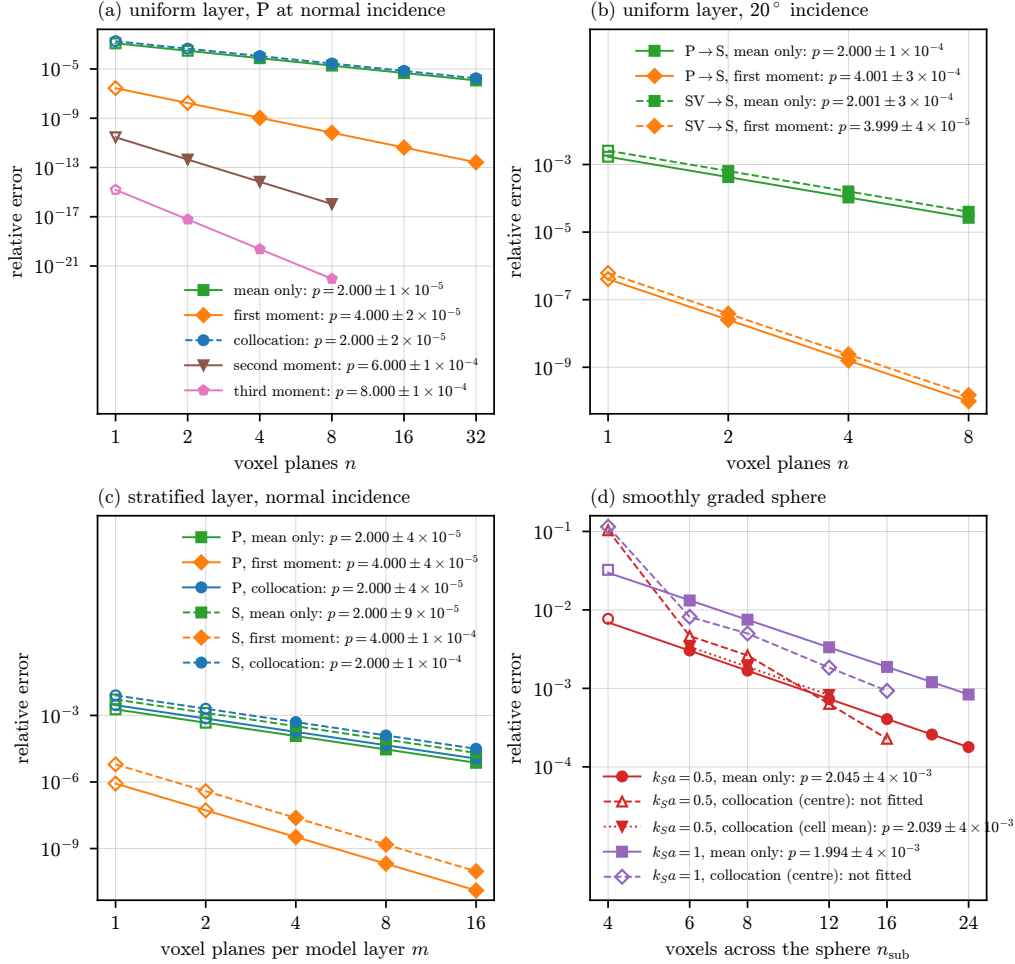


Figure 6: Convergence of every test, with least-squares regression lines. Each line fits  $\log e = c - p \log n$  to the filled points; open points are pre-asymptotic and are not fitted. The legend gives the order  $p$  and its standard error. The marker, and in colour the hue, marks the scheme, so that the figure reads in grey scale: collocation (circles; §6), the mean-only voxel (squares) and the voxels carrying Legendre moments to degree  $p = 1$  (diamonds), 2 (triangles) and 3 (pentagons) (Eqs. (38) and (B.1)). (a) The uniform layer, P reflection at normal incidence ( $D = 2$  m,  $\omega = 300$  rad s $^{-1}$ , the contrast of §7), fitted from  $n = 4$ ; the collocation line is dashed and lies on the mean-only line, from which its constant differs only by the factor 3/2 of Table 7; the second- and third-moment voxels, computed in 40-digit arithmetic, fitted from  $n = 2$ . (b) The uniform layer at  $20^\circ$ , fitted from  $n = 2$ : the S wave converted from incident P (solid) and reflected from incident SV (dashed). (c) The stratified layer of §9, reflection at normal incidence for P (solid) and S (dashed), against the number  $m$  of voxel planes per model layer, fitted from  $m = 4$ . (d) The smoothly graded sphere of §10 against its exact solution, for  $n_{\text{sub}}$  voxels across: the mean-only voxel, fitted from  $n_{\text{sub}} = 6$  (at 4 a voxel is as wide as the graded shell); the collocation voxel with the contrast sampled at the cell centre, joined by dashed lines and

1541 **Appendix B. Further tests of the Legendre voxel**

1542 *Oblique incidence.* The scheme is built spectrally. By Poisson summation  
 1543 the Bloch coupling between planes  $i$  and  $j$  at lateral wavenumber  $\mathbf{k}_{\parallel}$  is

$$C_{ab}(i, j) = \frac{1}{d^2} \sum_{\mathbf{g}} L_a(\boldsymbol{\kappa}) L_b(-\boldsymbol{\kappa}) \iint \phi_a(z) \hat{\mathbf{G}}(\boldsymbol{\kappa}; z-z') \phi_b(z') dz dz', \quad \boldsymbol{\kappa} = \mathbf{k}_{\parallel} + \mathbf{g}, \quad (\text{B.1})$$

1544 with  $L_a$  the lateral Legendre factors. The whole-space tensor decays as  
 1545  $1/k_z^2$  and the nine-component state adds at most two powers of  $k_z$ , so each  
 1546 entry of  $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z)$  is a  $\delta(z)$  of constant weight plus, per mode,  $[\mathrm{i}\alpha/2\gamma +$   
 1547  $\mathrm{i}\beta \operatorname{sgn}(z)/2] \mathrm{e}^{\mathrm{i}\gamma|z|}$ ,  $\gamma^2 = k_W^2 - \kappa^2$ : every  $z$  integral is closed-form. Double  
 1548 averaging makes the lattice sum converge without Ewald splitting, to three  
 1549 digits at  $|p|, |q| \leq 8$ . Against the exact layer, for a P wave incident at  $20^\circ$   
 and  $40^\circ$  (`ContinuumLimit_Oblique.wl`; Table B.12). The mean-only voxel

Table B.12: Relative error of the reflected P ( $R_{PP}$ ) and converted S ( $R_{PS}$ ) waves for a P wave incident at  $20^\circ$ , with the mean-only and first-moment voxels of Eq. (B.1). At  $40^\circ$  the orders are again 2 and 4.

| $n$   | mean only            |                      | mean and first moments |                       |
|-------|----------------------|----------------------|------------------------|-----------------------|
|       | $R_{PP}$             | $R_{PS}$             | $R_{PP}$               | $R_{PS}$              |
| 1     | $9.3 \times 10^{-4}$ | $1.7 \times 10^{-3}$ | $7.2 \times 10^{-8}$   | $4.1 \times 10^{-7}$  |
| 2     | $2.3 \times 10^{-4}$ | $4.2 \times 10^{-4}$ | $4.5 \times 10^{-9}$   | $2.5 \times 10^{-8}$  |
| 8     | $1.5 \times 10^{-5}$ | $2.7 \times 10^{-5}$ | $1.7 \times 10^{-11}$  | $9.9 \times 10^{-11}$ |
| order | 2                    | 2                    | 4                      | 4                     |

1550

1551 is second order and the first-moment voxel fourth order in both the reflected  
 1552 P and the converted S wave: the breakdown of the tiling identity at  $O(k_{\parallel}h)$

1553 and the evanescent coupling of the gradient channels do not cost an order.  
1554 At  $\mathbf{k}_{\parallel} = 0$  the construction reproduces Eq. (38) to all printed digits.

1555 *Incident S.* Nothing in the construction is specific to P. For a normally in-  
1556 cident S wave the layer is the one-dimensional problem of Eq. (38) with  
1557  $M_P \rightarrow \mu$ ,  $k_P \rightarrow k_S$  and  $\Delta M \rightarrow \Delta\mu$ ; the collocation and mean-only schemes  
1558 converge at order 2.00 and the first-moment voxel at 4.00, in reflection and  
1559 transmission, and the full  $9 \times 9$  construction with an incident SH wave re-  
1560 produces that one-dimensional result to all printed digits. At oblique inci-  
1561 dence, against the exact layer (`ContinuumLimit_IncidentS.wl`), the orders  
are those of Table B.13. The orders, and with  $k = k_S$  the closed-form con-

Table B.13: Observed order of convergence ( $n = 2 \rightarrow 8$ ,  $\omega = 300 \text{ rad s}^{-1}$ ) for incident SV and SH waves, with the mean-only and first-moment voxels. The SV angles lie below the critical angle  $\arcsin(\beta/\alpha) = 36.9^\circ$ .

|                                            | mean only   | mean and first moments |
|--------------------------------------------|-------------|------------------------|
| SV at $20^\circ$ : $R_{SS} / R_{SP}$       | 2.00 / 2.00 | 4.00 / 3.99            |
| SV at $30^\circ$ : $R_{SS} / R_{SP}$       | 1.98 / 2.00 | 3.97 / 4.00            |
| SH at $20^\circ$ and $40^\circ$ : $R_{SH}$ | 2.00        | 4.00                   |

1562

1563 stants of Eqs. (36) and (39), are those of the incident P wave.

1564 *Degree two at oblique incidence.* In three dimensions the voxel of degree  
1565 2 carries, on each of the nine components of its state, the ten Legendre  
1566 products of total degree at most two in  $(z, x, y)$ . In the construction of  
1567 Eq. (B.1) it converges at sixth order against the exact layer for incident  
1568 P, SV and SH waves (Table B.14; `scripts/measure_oblique_degree.py`).

1569 The test is run at  $\omega = 3000 \text{ rad s}^{-1}$  ( $k_P D = 1.2$ ,  $k_S D = 2$ ), ten times the  
1570 frequency of the other tables, because at  $300 \text{ rad s}^{-1}$  a sixth-order error on  
1571 these grids lies below the round-off of double precision. The exact reflection  
1572 coefficient is that of Kennett's recursion, an independent algorithm, and the  
1573 same computation with the bases of degree 0 and 1 returns orders 2 and 4  
1574 in all five cases. At normal incidence the error on four planes,  $6.95 \times 10^{-9}$ , is  
1575 the  $c_2(kd)^6 = 7.2 \times 10^{-9}$  of Eq. (40) to 4%. Doubling the number of lattice  
1576 orders retained in Eq. (B.1) changes the errors by less than 1%. Unlike the  
1577 degrees 0 and 1, this degree is implemented once at oblique incidence; its  
check is the exact solution.

Table B.14: The voxel of degree 2 (ten Legendre products per state component) at  $\omega = 3000 \text{ rad s}^{-1}$ : relative error of the same-type reflection coefficient against the exact layer on  $n$  planes, and the order over the last halving, with the orders of the degrees 0 and 1 from the same computation.

| incident wave               | error, $p = 2$       |                      |                      | order   | orders          |
|-----------------------------|----------------------|----------------------|----------------------|---------|-----------------|
|                             | $n = 1$              | $n = 2$              | $n = 4$              | $p = 2$ | $p = 0 / p = 1$ |
| P, normal: $R_{PP}$         | $3.4 \times 10^{-5}$ | $4.6 \times 10^{-7}$ | $7.0 \times 10^{-9}$ | 6.05    | 2.04 / 4.05     |
| P at $20^\circ$ : $R_{PP}$  | $7.3 \times 10^{-6}$ | $1.0 \times 10^{-7}$ | $1.5 \times 10^{-9}$ | 6.04    | 2.04 / 4.06     |
| SV at $20^\circ$ : $R_{SS}$ | $2.2 \times 10^{-4}$ | $2.3 \times 10^{-6}$ | $3.4 \times 10^{-8}$ | 6.11    | 2.10 / 4.14     |
| SH at $20^\circ$ : $R_{SH}$ | $2.3 \times 10^{-4}$ | $2.5 \times 10^{-6}$ | $3.6 \times 10^{-8}$ | 6.11    | 2.10 / 4.14     |
| SH at $40^\circ$ : $R_{SH}$ | $3.0 \times 10^{-4}$ | $3.6 \times 10^{-6}$ | $5.2 \times 10^{-8}$ | 6.09    | 2.12 / 4.05     |

1578

1579 The spectral construction is implemented twice. The second implementa-  
1580 tion (`scripts/crosscheck_first_moment_voxel.py`) shares only the defi-  
1581 nition of the scheme: it takes the depth dependence of the transformed kernel  
1582 from an independent residue construction, obtains the local weight as a nu-

merical large- $k_z$  limit of the inverted Christoffel matrix, evaluates every double integral by quadrature and separates P from S by sampling the upgoing field at two depths. The raw reflected displacements of the two agree to  $9 \times 10^{-16}$  in all 36 cases compared: incident P, SV and SH, normal incidence and  $20^\circ$ ,  $n = 1, 2, 4$ , with and without the first moments. The one-dimensional scheme of any degree is also implemented twice. The second implementation (`scripts/crosscheck_second_moment_voxel.py`) evaluates every integral by Gauss quadrature, split at the receiver within a voxel, and the exact layer from its own interface system, in double precision. Its errors agree with the 40-digit ones within its round-off floor ( $3 \times 10^{-14}$  in reflection,  $1.6 \times 10^{-13}$  in transmission); at  $\omega = 1200 \text{ rad s}^{-1}$  it measures orders 4.02 / 3.99 and 6.02 / 6.00; for one plane as the layer thins its differences from the exact layer fall as  $D^{3.00}$ ,  $D^{5.00}$  and  $D^{7.00}$  for  $p = 0, 1$  and  $2$ ; and it rederives  $c_p$  for  $p \leq 5$ . Figure 6(a,b) shows the convergence, with the orders fitted to it.

## Appendix C. The stratified-background construction

The experiment is built in four steps.

*The kernel.* At each lateral wavenumber  $\boldsymbol{\kappa}$  the whole-space kernel between two depths factorises over the plane-wave modes, three going down and three going up (P, SV, SH):  $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z - z') = \mathbf{D}_\downarrow \mathbf{E}(z - z') \mathbf{S}_\downarrow$  for  $z > z'$ , with  $\mathbf{S}_\downarrow$  the  $3 \times 9$  matrix of mode amplitudes radiated downwards by a nine-component source,  $\mathbf{E}(L) = \text{diag}(e^{i\gamma_m L})$  the propagation over a distance  $L$  with  $\gamma_m$  the vertical wavenumber of mode  $m$ , and  $\mathbf{D}_\downarrow$  the  $9 \times 3$  matrix of the modes' states; and likewise upwards. With the interfaces at  $z_A$  above and  $z_B$  below,  $H = z_B - z_A$ , and  $\mathbf{R}_A, \mathbf{R}_B$  the  $3 \times 3$  reflection matrices of the two interfaces

1607 seen from inside the layer, a source at  $z'$  sends  $\mathbf{s}_\downarrow = \mathbf{S}_\downarrow q$  down and  $\mathbf{s}_\uparrow = \mathbf{S}_\uparrow q$   
 1608 up, and the reverberation it sets up is

$$\begin{aligned} \mathbf{d}_A &= (\mathbf{I} - \mathbf{R}_A \mathbf{E}(H) \mathbf{R}_B \mathbf{E}(H))^{-1} \mathbf{R}_A [\mathbf{E}(z' - z_A) \mathbf{s}_\uparrow + \mathbf{E}(H) \mathbf{R}_B \mathbf{E}(z_B - z') \mathbf{s}_\downarrow], \\ \mathbf{u}_B &= \mathbf{R}_B [\mathbf{E}(z_B - z') \mathbf{s}_\downarrow + \mathbf{E}(H) \mathbf{d}_A], \\ \hat{\mathbf{G}}_{\text{rev}}(\boldsymbol{\kappa}; z, z') q &= \mathbf{D}_\downarrow \mathbf{E}(z - z_A) \mathbf{d}_A + \mathbf{D}_\uparrow \mathbf{E}(z_B - z) \mathbf{u}_B, \end{aligned} \tag{C.1}$$

1609 with  $\mathbf{d}_A$  the downgoing amplitudes leaving the upper interface and  $\mathbf{u}_B$  the  
 1610 upgoing ones leaving the lower. It is added to the whole-space kernel in  
 1611 every block of Eq. (B.1), at every order of the reciprocal lattice. It is smooth  
 1612 and separable in  $z$  and  $z'$ , so its integrals over the cells are one-dimensional  
 1613 moments of exponentials. For two half-spaces  $\mathbf{R}_A = 0$  and only the second  
 1614 line survives.

1615 *The interfaces.*  $\mathbf{R}_A$ ,  $\mathbf{R}_B$  and the transmission matrices into and out of the  
 1616 layer follow from continuity of displacement and traction at each interface:  
 1617 six equations in the states of the six modes on either side, solved for each  
 1618 lateral wavenumber, evanescent orders included.

1619 *The incident field and the output.* A plane wave of unit amplitude arrives  
 1620 from the upper half-space. Its field in the layer, a downgoing part and an  
 1621 upgoing part with the reverberation summed as in Eq. (C.1), is the right-  
 1622 hand side of the scheme. The output is the upgoing wave of the same type  
 1623 as the incident one, arriving at  $z_A$  from the voxels and transmitted into the  
 1624 upper half-space: the change that the layer of voxels makes to the same-type  
 1625 reflection coefficient ( $R_{PP}$ ,  $R_{SS}$  or  $R_{SH}$ ), a quantity that does not depend on  
 1626 how the waves are normalised.

1627 *The reference, and a check of the construction.* The exact change is the

1628 same-type reflection coefficient of the stack with the layer, less that of the  
 1629 background, both from Kennett’s recursion [58], an independent algorithm.  
 1630 Before any voxel is introduced, the background’s own reflection coefficient  
 1631 computed from the modes and the interface matrices above agrees with Ken-  
 1632 nett’s to  $5 \times 10^{-15}$  in all twelve cases.

1633 It is computed by two independent algorithms. The first is invariant  
 1634 imbedding, the impedance march of Haines et al. [59]: the six-component  
 1635 state of displacement and traction obeys  $dy/dz = A(z)y$ , and with the four  
 1636  $3 \times 3$  blocks  $A_{ij}$  of  $A$  the impedance  $Y$ , which gives the traction from the  
 1637 displacement, obeys the Riccati equation

$$\frac{dY}{dz} = A_{21} + A_{22}Y - YA_{11} - YA_{12}Y. \quad (C.2)$$

1638 It is started below the stack with the impedance of the downgoing waves  
 1639 of the lower half-space, which is the radiation condition, marched upwards,  
 1640 exactly through the homogeneous gaps and by fourth-order Runge–Kutta  
 1641 with the medium sampled at the stages through the graded layer, and at  
 1642 the top the reflection matrix follows from continuity of displacement and  
 1643 traction with the modes of the upper half-space.  $Y$  stays bounded, so no  
 1644 growing exponential is formed. The second is the propagator: the same  
 1645 state is integrated across the graded layer as a  $6 \times 6$  matrix, with  $A(z) =$   
 1646  $F(z)\text{diag}(i\gamma_m)F(z)^{-1}$  built at each depth from the plane-wave modes of the  
 1647 medium there,  $F$  the matrix of the six modes’ displacements and tractions  
 1648 and  $\gamma_m$  their vertical wavenumbers, and the radiation conditions of the two  
 1649 half-spaces are imposed on the result. For the graded layer the two agree  
 1650 to  $10^{-12}$  in all twelve cases with 400 steps of the march, whose error falls  
 1651 sixteenfold for each doubling of the steps; for a uniform layer the propagator

1652 equals Kennett's recursion to  $3 \times 10^{-14}$ .

1653 *Oblique incidence, and a three-layer background.* At oblique incidence the  
1654 test is the three-dimensional scheme of Eq. (B.1), with P, SV and SH waves,  
1655 in two backgrounds: the two half-spaces above, and a three-layer background  
1656 in which the voxels lie in a layer between a slower half-space above ( $\alpha = 4200$ ,  
1657  $\beta = 2400 \text{ m s}^{-1}$ ,  $\rho = 2300 \text{ kg m}^{-3}$ ) and the faster one below (6500, 3700,  
1658 2800), so that the layer reverberates between two interfaces. The construc-  
1659 tion is the one set out above: the reverberation added to the whole-space  
1660 kernel at every order of the reciprocal lattice, Eq. (C.1), the interface ma-  
1661 trices, the incident field, and the reference, Kennett's recursion [58], against  
1662 which the background's own reflection coefficient is reproduced to  $5 \times 10^{-15}$   
1663 in all twelve cases before any voxel is introduced.

1664 The results are in Table C.15, at  $\omega = 3000 \text{ rad s}^{-1}$  ( $k_S D = 2$ ; at  $300 \text{ rad s}^{-1}$   
1665 the sixth-order errors lie below round-off, as in Table B.14) and  $20^\circ$  in the  
1666 upper half-space. In all thirty-six cases the order is that of the whole space,  
1667 with the interfaces 3m from the layer and with the voxels against them  
1668 (`scripts/measure_oblique_stratified_background.py`). The errors are  
1669 close to those of the whole space at the same frequency (Table B.14): the  
1670 background changes the field the voxels must represent, not how well they  
1671 represent it.

1672 *The two terms of the error at oblique incidence.* The orders say that the  
1673 scheme converges; the two-term analysis says what its error is. Both terms  
1674 are tested at  $20^\circ$ , in the whole space and in the stratified backgrounds, for  
1675 P, SV and SH waves.

Table C.15: The layer of voxels in a stratified background at  $20^\circ$ ,  $\omega = 3000 \text{ rad s}^{-1}$ : relative error of the change of the same-type reflection coefficient on  $n = 4$  planes, and the order over the last halving ( $n = 2 \rightarrow 4$ ), for field degrees  $p = 0, 1$  and  $2$ .

| background                                    | wave | $p = 0$                     | $p = 1$                     | $p = 2$                     |
|-----------------------------------------------|------|-----------------------------|-----------------------------|-----------------------------|
| two half-spaces, interface 3 m below          | P    | $6.1 \times 10^{-3}$ (2.03) | $2.7 \times 10^{-6}$ (4.02) | $2.3 \times 10^{-9}$ (6.04) |
|                                               | SV   | $1.5 \times 10^{-2}$ (2.11) | $2.6 \times 10^{-5}$ (4.13) | $3.7 \times 10^{-8}$ (6.09) |
|                                               | SH   | $1.6 \times 10^{-2}$ (2.11) | $2.7 \times 10^{-5}$ (4.13) | $4.1 \times 10^{-8}$ (6.09) |
| two half-spaces, interface against the voxels | P    | $5.1 \times 10^{-3}$ (2.04) | $3.8 \times 10^{-6}$ (4.04) | $2.1 \times 10^{-9}$ (6.00) |
|                                               | SV   | $1.6 \times 10^{-2}$ (2.10) | $2.5 \times 10^{-5}$ (4.14) | $3.3 \times 10^{-8}$ (6.11) |
|                                               | SH   | $1.7 \times 10^{-2}$ (2.10) | $2.4 \times 10^{-5}$ (4.15) | $3.4 \times 10^{-8}$ (6.11) |
| three layers, interfaces 3 m away             | P    | $6.3 \times 10^{-3}$ (2.03) | $8.7 \times 10^{-7}$ (4.04) | $2.4 \times 10^{-9}$ (6.06) |
|                                               | SV   | $1.0 \times 10^{-2}$ (2.10) | $5.8 \times 10^{-6}$ (4.08) | $5.2 \times 10^{-8}$ (6.11) |
|                                               | SH   | $1.3 \times 10^{-2}$ (2.10) | $3.0 \times 10^{-6}$ (3.95) | $9.3 \times 10^{-8}$ (6.10) |
| three layers, interfaces against the voxels   | P    | $4.8 \times 10^{-3}$ (2.04) | $2.4 \times 10^{-6}$ (4.04) | $4.1 \times 10^{-9}$ (6.03) |
|                                               | SV   | $8.5 \times 10^{-3}$ (2.09) | $7.1 \times 10^{-6}$ (4.01) | $5.0 \times 10^{-8}$ (6.09) |
|                                               | SH   | $1.4 \times 10^{-2}$ (2.10) | $5.8 \times 10^{-6}$ (3.92) | $8.5 \times 10^{-8}$ (6.12) |

1676 *The first-order term.* The scheme is written as  $(\mathbf{M} - \mathbf{K})\mathbf{c} = \mathbf{b}$  with the  
 1677 output  $u = \mathbf{r} \mathbf{c}$ , as in Eq. (G.8), so that  $T_1^{\text{scheme}} = \mathbf{r} \mathbf{M}^{-1} \mathbf{b}$  with no differencing.  
 1678 The prediction is built from plane waves. The background field in the layer  
 1679 is a sum of six, P, SV and SH going down and going up, and the voxels  
 1680 radiate into six. For a pair made of an incident wave of wavevector  $\mathbf{k}_{\text{in}}$  and  
 1681 a radiated wave of wavevector  $\mathbf{k}_{\text{out}}$ , a cubic voxel of half-side  $h$  whose field is  
 1682 held to total degree  $p$  gives the exact first-order term of that pair times

$$1 + E_p(\mathbf{k}_{\text{in}}, \mathbf{k}_{\text{out}}) = \frac{\sum_{|a| \leq p} \prod_{i=1}^3 (2a_i + 1) j_{a_i}(k_{\text{in},i}h) j_{a_i}(k_{\text{out},i}h)}{\prod_{i=1}^3 j_0((k_{\text{in},i} - k_{\text{out},i})h)}, \quad (\text{C.3})$$

1683 the sum being over the Legendre products  $P_{a_1} P_{a_2} P_{a_3}$  of the voxel's basis and  
 1684  $j_n$  the spherical Bessel functions: the numerator is the projection of the inci-  
 1685 dent wave onto the basis paired with that of the outgoing wave, the denomi-  
 1686 nator the exact pairing. Its leading departure from one is  $-d^2 \mathbf{k}_{\text{in}} \cdot \mathbf{k}_{\text{out}}/12$  for  
 1687  $p = 0$ , with  $d = 2h$ , and along one axis it is the factor of Eq. (39). The pre-  
 1688 dicted first-order term is the sum over the thirty-six pairs of the exact term  
 1689 of each pair, an integral of two exponentials over the voxel, times its factor,  
 1690 the radiated waves then being carried to the observer by the reverberation  
 1691 of Eq. (C.1) exactly. On one and two planes of voxels of degree 0, 1 and  
 1692 2, at  $\omega = 3000 \text{ rad s}^{-1}$ , the scheme's first-order term equals this prediction  
 1693 to  $2 \times 10^{-15}$  in all twelve combinations of background and wave type. The  
 1694 first-order error is therefore known in closed form at oblique incidence, in a  
 1695 stratified background, for every wave type.

1696 *The second-order term.* For the law of Eq. (44) the contrast must vary

1697 within the voxels. Each plane of voxels is given the smooth profile of §9  
 1698 in depth, projected to Legendre degree  $r = p$ , and the coupling is that of  
 1699 Eq. (41) with the reverberation added. The second-order term of the scheme  
 1700 is  $T_2^{\text{scheme}} = \mathbf{r} \mathbf{M}^{-1} \mathbf{K} \mathbf{M}^{-1} \mathbf{b}$ . The exact  $T_2$  needs the exact stack with a layer  
 1701 whose properties vary continuously in depth, at oblique incidence, for which  
 1702 a recursion over homogeneous layers is not exact. It is computed by two  
 1703 independent algorithms, invariant imbedding [59] and the propagator of the  
 1704 six-component state, which agree to  $10^{-12}$  in all twelve cases (above).  $T_2$  then  
 1705 follows from Eq. (G.9) at  $\epsilon = 0.1$ , where halving  $\epsilon$  changes it by  $2 \times 10^{-8}$ .  
 1706 The ratio of the relative error of  $T_2$  to the relative projection error  $\mathcal{E}_p$  is in  
 1707 Table C.16. On four planes the law holds to 3% for  $p = 0$  and 1 and to 6%  
 1708 for  $p = 2$  at  $k_S D = 0.2$ , and to 0.6% for all three degrees at  $k_S D = 0.067$ , in  
 1709 every background and for every wave type: the departure falls about ninefold  
 1710 as the frequency falls threefold, as  $(kD)^2$ . On two planes the ratios for  $p = 0$   
 1711 and 1 lie within 0.03 of those in the table; for  $p = 2$  they are lower, 0.89 to 0.99  
 1712 at  $k_S D = 0.2$  and 0.988 to 0.999 at  $k_S D = 0.067$ . The law is loosest for the  
 1713 highest degree at the highest frequency, because the projection error of degree  
 1714 2 is so small ( $1.6 \times 10^{-5}$  on four planes) that the departure from the long-wave  
 1715 limit is a larger share of it. Both terms of the error are thus known in advance  
 1716 at oblique incidence and in a stratified background, as they are at normal  
 1717 incidence in the whole space (`scripts/measure_oblique_two_term.py`).

1718 Two limits stand. The oblique construction is implemented once, and  
 1719 its checks are the exact solutions. And the voxels here fill their plane, so  
 1720 that the image of a voxel in an interface it touches is the image of a whole  
 1721 plane; a body of finite lateral extent against an interface, where that image

Table C.16: The relative error of the second-order term over the relative projection error of the medium, at  $20^\circ$ , on  $n = 4$  planes with  $r = p$ ; the law of Eq. (44) predicts one in the long-wave limit.

|                                             |      | $k_S D = 0.2$ |         |         | $k_S D = 0.067$ |         |         |
|---------------------------------------------|------|---------------|---------|---------|-----------------|---------|---------|
| background                                  | wave | $p = 0$       | $p = 1$ | $p = 2$ | $p = 0$         | $p = 1$ | $p = 2$ |
| whole space                                 | P    | 1.000         | 0.998   | 1.011   | 1.000           | 1.000   | 1.000   |
|                                             | SV   | 0.998         | 0.992   | 1.032   | 1.000           | 0.999   | 1.004   |
|                                             | SH   | 0.999         | 0.992   | 1.031   | 1.000           | 0.999   | 1.003   |
| two half-spaces, interface 3 m below        | P    | 1.010         | 1.009   | 1.019   | 1.001           | 1.001   | 1.001   |
|                                             | SV   | 1.012         | 1.008   | 1.046   | 1.001           | 1.001   | 1.005   |
|                                             | SH   | 1.027         | 1.025   | 1.059   | 1.003           | 1.003   | 1.006   |
| three layers, interfaces 3 m away           | P    | 1.004         | 1.003   | 1.013   | 1.000           | 1.000   | 1.000   |
|                                             | SV   | 0.979         | 0.973   | 1.002   | 0.997           | 0.996   | 1.000   |
|                                             | SH   | 1.007         | 1.004   | 1.035   | 1.001           | 1.001   | 1.004   |
| three layers, interfaces against the voxels | P    | 1.002         | 1.001   | 1.011   | 1.000           | 1.000   | 1.000   |
|                                             | SV   | 0.995         | 0.988   | 1.018   | 0.999           | 0.999   | 1.002   |
|                                             | SH   | 1.004         | 1.002   | 1.032   | 1.000           | 1.000   | 1.003   |

1722 is singular, has not been run.

## 1723 **Appendix D. The sphere: near field, an impedance march and a** 1724 **sharp surface**

1725 *The near field.* A receiver is seldom at infinity, and the scheme does not  
1726 require it to be. The scattered field at a point  $\mathbf{x}$  outside the voxels is the  
1727 propagator applied to the source that each voxel carries, the contrast times  
1728 the field found in it:

$$\psi^{\text{sc}}(\mathbf{x}) = \sum_j \int_{V_j} \mathbf{K}(\mathbf{x} - \mathbf{x}') \boldsymbol{\Delta}(\mathbf{x}') e_j(\mathbf{x}') d\mathbf{x}', \quad (\text{D.1})$$

1729 with  $\mathbf{K}$  the nine-component Green's tensor of §2, the same one that couples  
1730 the voxels in the solve,  $\boldsymbol{\Delta}$  the contrast operator of Eq. (5) and  $e_j$  the poly-  
1731 nomial internal state of voxel  $j$ . Its first three rows are the displacement and  
1732 its last six the strain. Nothing is dropped: the terms in  $1/r^2$  and  $1/r^3$  are in  
1733  $\mathbf{K}$ . The integrand's source is a polynomial in each voxel and is integrated by  
1734 a tensor Gauss rule with six nodes a side; the rule's own error is measured  
1735 by repeating the readout with four nodes, and from 8 voxels across the two  
1736 differ by less than  $4 \times 10^{-6}$  at  $1.25a$  and by less than  $10^{-10}$  at  $2a$ . The ob-  
1737 servers are the nine angles of Fig. 5(b) on spheres of radius  $1.25a$ ,  $2a$  and  
1738  $10a$ , where  $k_S r$  runs from 0.6 to 10; the reference is the exact partial-wave  
1739 displacement at the same points; and the error is the measure of the far-field  
1740 test. Table D.17 gives it for the mean-only voxel and for the first-moment  
1741 voxel (Appendix Appendix E). The orders of the far field hold in the near  
1742 field: second for the mean-only voxel at every distance and both frequencies,  
1743 and 3.8 to 3.95 for the first-moment voxel, still rising towards four as in the

Table D.17: The graded sphere read in the near field: relative error of the scattered displacement at 1.25, 2 and 10 radii from the centre, against the exact displacement at the same points, and the apparent order between the two finest grids.

| $n_{\text{sub}}$           | mean only            |                      |                      | first moment         |                      |                      |
|----------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                            | 1.25a                | 2a                   | 10a                  | 1.25a                | 2a                   | 10a                  |
| $k_S a = 0.5$              |                      |                      |                      |                      |                      |                      |
| 4                          | $3.3 \times 10^{-2}$ | $1.2 \times 10^{-2}$ | $5.3 \times 10^{-3}$ | $1.4 \times 10^{-2}$ | $1.6 \times 10^{-3}$ | $6.3 \times 10^{-4}$ |
| 6                          | $1.4 \times 10^{-2}$ | $5.2 \times 10^{-3}$ | $1.9 \times 10^{-3}$ | $1.2 \times 10^{-3}$ | $3.8 \times 10^{-4}$ | $1.9 \times 10^{-4}$ |
| 8                          | $8.0 \times 10^{-3}$ | $2.9 \times 10^{-3}$ | $1.1 \times 10^{-3}$ | $4.7 \times 10^{-4}$ | $1.4 \times 10^{-4}$ | $7.3 \times 10^{-5}$ |
| 12                         | $3.5 \times 10^{-3}$ | $1.2 \times 10^{-3}$ | $4.7 \times 10^{-4}$ | $7.8 \times 10^{-5}$ | $3.1 \times 10^{-5}$ | $1.7 \times 10^{-5}$ |
| 16                         | $2.0 \times 10^{-3}$ | $6.7 \times 10^{-4}$ | $2.6 \times 10^{-4}$ | $2.5 \times 10^{-5}$ | $1.0 \times 10^{-5}$ | $5.7 \times 10^{-6}$ |
| order, 12 $\rightarrow$ 16 | 2.03                 | 2.06                 | 2.02                 | 3.95                 | 3.85                 | 3.77                 |
| $k_S a = 1.0$              |                      |                      |                      |                      |                      |                      |
| 4                          | $4.8 \times 10^{-2}$ | $3.5 \times 10^{-2}$ | $3.3 \times 10^{-2}$ | $9.9 \times 10^{-3}$ | $1.3 \times 10^{-3}$ | $6.0 \times 10^{-4}$ |
| 6                          | $2.0 \times 10^{-2}$ | $1.4 \times 10^{-2}$ | $1.4 \times 10^{-2}$ | $9.3 \times 10^{-4}$ | $3.0 \times 10^{-4}$ | $2.0 \times 10^{-4}$ |
| 8                          | $1.2 \times 10^{-2}$ | $8.1 \times 10^{-3}$ | $7.8 \times 10^{-3}$ | $3.5 \times 10^{-4}$ | $1.1 \times 10^{-4}$ | $7.4 \times 10^{-5}$ |
| 12                         | $5.2 \times 10^{-3}$ | $3.6 \times 10^{-3}$ | $3.5 \times 10^{-3}$ | $6.0 \times 10^{-5}$ | $2.5 \times 10^{-5}$ | $1.7 \times 10^{-5}$ |
| 16                         | $2.9 \times 10^{-3}$ | $2.0 \times 10^{-3}$ | $1.9 \times 10^{-3}$ | $1.9 \times 10^{-5}$ | $8.3 \times 10^{-6}$ | $5.7 \times 10^{-6}$ |
| order, 12 $\rightarrow$ 16 | 2.01                 | 2.01                 | 2.00                 | 3.95                 | 3.84                 | 3.78                 |

1744 far field at this contrast. A quarter of a radius from the surface, with 16 vox-  
 1745 els across, the first-moment voxel's error is  $2.5 \times 10^{-5}$  and  $1.9 \times 10^{-5}$ , 78 and  
 1746 153 times below the mean-only voxel's. The error grows towards the body,  
 1747 by a factor of 1.5 to 7 between  $10a$  and  $1.25a$ , because the near field carries  
 1748 the higher multipoles of the source, which a given grid resolves less well; the  
 1749 order does not change. On the coarsest grid a voxel is as wide as the graded  
 1750 shell and twice as wide as the gap to the nearest observer, and the sequence  
 1751 is not yet asymptotic. Read at these distances with the asymptotic formula,  
 1752 the same solutions would be in error by 48% to 85%: the far-field readout  
 1753 is for the far field (`scripts/measure_graded_sphere_near_field.py`; the  
 1754 readout is `graded_field` of the package, tested against the propagator of a  
 1755 point source, against the gradient of its own displacement, and against the  
 1756 far-field readout at great distance).

1757 *The voxels against an impedance march.* A second route to the same body  
 1758 is invariant imbedding. The impedance of §9, Eq. (C.2), is marched up-  
 1759 wards through the depth range of the sphere with the lateral variation of the  
 1760 medium carried inside the march, on a laterally periodic grid, and a second  
 1761 sweep carries the field down [59]; the march returns the reflected plane waves  
 1762 on a plane above the body and the transmitted ones on a plane below. It is  
 1763 run here on the graded sphere and set beside the voxels, to ask how many  
 1764 voxels a given accuracy costs against an established alternative. Four things  
 1765 are fixed so that the comparison is of like with like.

- 1766 1. *The planes.* Both methods are read on the planes  $z = -1.25a$  and  
 1767  $z = +1.25a$ . On the tangent planes  $z = \pm a$  themselves the observers lie  
 1768 on the faces of the voxels at the poles, where the Gauss rule of the voxel

Table D.18: The smoothly graded sphere, voxelised with the mean-only voxel on grids of  $n_{\text{sub}}$  voxels across, against its exact solution: relative far-field error, and the apparent order between successive grids. Nine unknowns per cell, every cell overlapping the sphere kept.

| $n_{\text{sub}}$ | unknowns | $k_{\text{sa}} = 0.5$ |       | $k_{\text{sa}} = 1.0$ |       |
|------------------|----------|-----------------------|-------|-----------------------|-------|
|                  |          | error                 | order | error                 | order |
| 4                | 576      | $7.70 \times 10^{-3}$ |       | $3.24 \times 10^{-2}$ |       |
| 6                | 1 872    | $3.04 \times 10^{-3}$ | 2.30  | $1.32 \times 10^{-2}$ | 2.22  |
| 8                | 3 672    | $1.68 \times 10^{-3}$ | 2.05  | $7.52 \times 10^{-3}$ | 1.95  |
| 12               | 11 520   | $7.30 \times 10^{-4}$ | 2.06  | $3.34 \times 10^{-3}$ | 2.00  |
| 16               | 24 984   | $4.06 \times 10^{-4}$ | 2.04  | $1.88 \times 10^{-3}$ | 2.00  |
| 20               | 47 304   | $2.58 \times 10^{-4}$ | 2.03  | $1.20 \times 10^{-3}$ | 2.00  |
| 24               | 79 344   | $1.79 \times 10^{-4}$ | 2.02  | $8.34 \times 10^{-4}$ | 2.00  |

1769 readout (Eq. (D.1)) cannot integrate the singular kernel: its change  
1770 from four to six nodes is 2% to 10% there, more than the scheme's error,  
1771 and below  $10^{-4}$  a quarter of a radius away. The march's amplitudes are  
1772 carried across the same homogeneous gap by each order's own phase,  
1773 which is exact.

1774 2. *The periodic array.* The march needs a laterally periodic grid, so it  
1775 solves a square array of spheres. Its answer is scored against the exact  
1776 solution of one sphere summed over the array, by Poisson summation,  
1777 which is exact but for scattering between the spheres. That part, and  
1778 the lateral resolution, are errors of the method as used on a single body,  
1779 and are counted.

1780 3. *The error.* For the march, the norm of the difference over the plane-

1781 wave orders of the grid, relative to the norm of the exact scattered  
1782 orders; for the voxels, the norm of the difference of the scattered dis-  
1783 placement over a patch of side  $5a$  on each plane, relative to the norm  
1784 of the exact one. The two are the same quantity in wavenumber and  
1785 in space, for the array and for the single body.

1786 4. *The refinement.* For the march, the lateral grid, with the depth step  
1787 refined until the error no longer changes; for the voxels, the grid and  
1788 the degree.

Table D.19: The graded sphere at  $k_S a = 0.5$  read on the planes  $z = \mp 1.25a$ : relative error in reflection and in transmission. The march's unknowns are those of one plane, on which it carries a dense matrix; the period is  $5a$  unless stated.

| method              | setting                        | unknowns | reflection           | transmission         |
|---------------------|--------------------------------|----------|----------------------|----------------------|
| impedance march     | $12 \times 12$ , 16 steps      | 432      | $1.2 \times 10^{-2}$ | $9.9 \times 10^{-3}$ |
|                     | $16 \times 16$ , 32 steps      | 768      | $4.3 \times 10^{-3}$ | $3.6 \times 10^{-3}$ |
|                     | $20 \times 20$ , 32 steps      | 1200     | $7.6 \times 10^{-4}$ | $7.3 \times 10^{-4}$ |
|                     | $28 \times 28$ , 48 steps      | 2352     | $2.5 \times 10^{-4}$ | $5.6 \times 10^{-4}$ |
|                     | $30 \times 30$ , period $7.5a$ | 2700     | $7.2 \times 10^{-4}$ | $4.7 \times 10^{-4}$ |
| mean-only voxel     | 8 across                       | 3672     | $5.0 \times 10^{-3}$ | $7.9 \times 10^{-3}$ |
|                     | 16 across                      | 24 984   | $1.2 \times 10^{-3}$ | $2.0 \times 10^{-3}$ |
| first-moment voxel  | 6 across                       | 7488     | $3.3 \times 10^{-4}$ | $5.2 \times 10^{-4}$ |
|                     | 8 across                       | 14 688   | $1.4 \times 10^{-4}$ | $2.2 \times 10^{-4}$ |
|                     | 16 across                      | 99 936   | $1.1 \times 10^{-5}$ | $1.8 \times 10^{-5}$ |
| second-moment voxel | 8 across                       | 36 720   | $1.4 \times 10^{-5}$ | $2.3 \times 10^{-5}$ |
|                     | 10 across                      | 69 120   | $2.6 \times 10^{-6}$ | $4.2 \times 10^{-6}$ |

1789 Table D.19 gives the result. The march converges quickly in the lateral grid,

1790 because the body is smooth, and with far fewer unknowns than any voxel  
 1791 scheme: at  $20 \times 20$  it is at  $7 \times 10^{-4}$  with 1200 unknowns on a plane. The  
 1792 mean-only voxel does not reach that with sixteen voxels across and twenty  
 1793 times the unknowns; on a sharp sphere, where it also carries a staircase, it  
 1794 fares worse. The first-moment voxel reaches the march's accuracy with six  
 1795 voxels across, and below about  $2 \times 10^{-4}$  it continues at close to fourth order  
 1796 where the march does not: in transmission the march's error at a period  
 1797 of  $5a$  stops near  $5 \times 10^{-4}$ , which is the scattering between the spheres of  
 1798 its array, since lengthening the period lowers it and refining the grid does  
 1799 not. The costs are of different kinds. The march's unknowns are few but its  
 1800 operator is a dense matrix on the plane, marched through the depth, and a  
 1801 finer lateral grid requires a smaller depth step (with eight steps the  $20 \times 20$   
 1802 march diverged); the voxel schemes have many unknowns and a product done  
 1803 by fast Fourier transform. In the timings of these runs, taken on one machine  
 1804 with other work in progress and therefore only indicative, the first-moment  
 1805 voxel at six across was solved in about a sixth of the time of the  $20 \times 20$  march.  
 1806 The march has what the comparison does not use: it returns the reflection  
 1807 and transmission for every incident plane wave of the grid at once, and a  
 1808 laterally periodic or laterally extended medium is its natural problem and  
 1809 not a cost. The comparison is at one frequency (`scripts/measure_graded_`  
 1810 `sphere_march.py`, `scripts/measure_graded_sphere_planes.py`).

1811 *A sphere with a sharp surface.* The homogeneous sphere is the familiar test,  
 1812 and its voxelisation changes with every grid. Table D.20 shows its error for  
 1813 grids of 4, 6 and 8 voxels across: 4.6%, then 19%, then 4.2% at  $k_S a = 0.5$ .  
 1814 The error tracks the staircase's volume, which is 5%, 17% and 4% too large

1815 on these grids. Rescaled by the ratio of the sphere’s volume to the staircase’s,  
1816 it falls below 1% at  $k_S a = 0.5$  and to at most 3% at  $k_S a = 1.0$ , but it still  
1817 does not converge: what remains is the staircase’s shape. A refinement study  
1818 of a sharp sphere measures its staircase; the smooth sphere, like the layer,  
1819 measures the scheme.

Table D.20: A homogeneous sphere voxelised on grids of  $n_{\text{sub}}$  voxels across, against exact elastic scattering: relative far-field error, raw and rescaled by the ratio of the sphere’s volume to the staircase’s. It does not converge with refinement, because the staircase changes with the grid.

| $n_{\text{sub}}$ | voxels | $k_S a = 0.5$         |                      | $k_S a = 1.0$         |                      |
|------------------|--------|-----------------------|----------------------|-----------------------|----------------------|
|                  |        | raw                   | volume-rescaled      | raw                   | volume-rescaled      |
| 4                | 32     | $4.6 \times 10^{-2}$  | $1.4 \times 10^{-3}$ | $5.0 \times 10^{-2}$  | $5.4 \times 10^{-3}$ |
| 6                | 136    | $1.94 \times 10^{-1}$ | $7.9 \times 10^{-3}$ | $1.68 \times 10^{-1}$ | $3.0 \times 10^{-2}$ |
| 8                | 280    | $4.2 \times 10^{-2}$  | $2.3 \times 10^{-3}$ | $3.5 \times 10^{-2}$  | $8.9 \times 10^{-3}$ |

## 1820 Appendix E. The three-dimensional voxel solver

1821 The three-dimensional results of §§10 and 11 use the Legendre voxels  
1822 of §8 on a cubic grid, with the following construction (`cubic_scattering/`  
1823 `graded_voxel`).

1824 *The voxel.* Each cell carries the nine-component state of §2, displacement  
1825 and engineering strain, with the strain an *independent* unknown and not the  
1826 gradient of the displacement: a strain derived from a cell’s projected dis-  
1827 placement jumps across the faces between cells and leaves a spurious source  
1828 there (§8). The state is expanded in the Legendre products of degree at

1829 most  $p$  in the cell's local coordinates  $\boldsymbol{\xi} \in [-1, 1]^3$ , and the contrast opera-  
 1830 tor  $\boldsymbol{\Delta}$  in those of degree at most  $r$ , its coefficients the  $L^2$  projection of the  
 1831 medium;  $p = r = 1$  gives the first-moment voxel of 36 weights. The source  
 1832  $\boldsymbol{\Delta}\psi$  is then a polynomial of degree  $p + r$ , expanded in monomials  $m_c$ . The  
 1833 Lippmann–Schwinger equation is tested with each basis function over each  
 1834 cell (Galerkin), and cells  $m$  and  $n$  couple through the blocks

$$\mathbf{K}_{ac}(\mathbf{R}) = \int_{V_m} \int_{V_n} L_a(\boldsymbol{\xi}) \mathbf{P}(\mathbf{x} - \mathbf{x}') m_c(\boldsymbol{\xi}') d^3x d^3x', \quad \mathbf{R} = \mathbf{x}_m - \mathbf{x}_n, \quad (\text{E.1})$$

1835 with  $\mathbf{P}$  the  $9 \times 9$  point propagator of §2.

1836 *One integral in place of two.* With  $\mathbf{s} = \mathbf{u} - \mathbf{u}'$  the difference of the two  
 1837 local positions, Eq. (E.1) becomes  $\mathbf{K}_{ac}(\mathbf{R}) = \int W_{ac}(\mathbf{s}) \mathbf{P}(\mathbf{R} + \mathbf{s}) d^3s$  over  
 1838  $\mathbf{s} \in [-2h, 2h]^3$ , where  $W_{ac}$  is a product over the three axes of one-dimensional  
 1839 cross-correlations of the test and source polynomials: continuous, polynomial  
 1840 on  $[-2h, 0]$  and  $[0, 2h]$ , with kinks at 0 and  $\pm 2h$ . For cells that do not touch,  
 1841 the kernel is smooth on each of the eight pieces, and a three-dimensional  
 1842 Gauss rule integrates it, with 14, 10 or 8 points per axis at a Chebyshev  
 1843 distance of 2, 3 or 4, and 5 or more cells; this agrees with six-dimensional  
 1844 quadrature of Eq. (E.1) to  $10^{-9}$  one cell beyond contact.

1845 *The self cell and its 26 neighbours.* The static part of  $\mathbf{P}$  is a sum of derivatives  
 1846 of  $r^{-1}$  and  $r$ , distributions at the origin. Derivatives are moved onto  $W$  until  
 1847 the kernel is at most as singular as  $1/r$ , and the kinks of  $W$  then carry  
 1848 plane  $\delta$  functions; every piece is a box or a rectangle with its corners on the  
 1849 breakpoints, so the singular point is either a vertex of a piece, integrated  
 1850 with Duffy's pyramids, or at least  $2h$  away. The  $\delta$  function of the inclusion  
 1851 problem is carried by those plane  $\delta$  functions and is not added by hand.

1852 The construction reproduces the sum rule  $\sum_p \int_V \int_V \partial_p \partial_p |\mathbf{x} - \mathbf{x}'|^{-1} d^3x d^3x' =$   
1853  $-4\pi V$ , the biharmonic rule for  $r$  and the cube's Coulomb self-energy to  $10^{-11}$ ,  
1854 and every block equals the sum of the 64 blocks between the two cells' half-  
1855 size sub-cells, with the polynomials re-expressed on them, to  $2 \times 10^{-15}$ . The  
1856 dynamic remainder of  $\mathbf{P}$  is at most weakly singular and is integrated by the  
1857 Gauss rule of the pieces.

1858 *Symmetry, and the solve.* A block depends only on the offset  $\mathbf{o}$  between the  
1859 two cells. The background is isotropic and the cube is mapped to itself  
1860 by its 48 signed permutations  $\mathbf{Q}$ , under which  $\mathbf{P}(\mathbf{Q}\mathbf{r}) = \mathbf{S}\mathbf{P}(\mathbf{r})\mathbf{S}^\top$ , with  
1861  $\mathbf{S} = \text{diag}(\mathbf{Q}, \mathbf{R}_6(\mathbf{Q}))$  and  $\mathbf{R}_6$  the representation of  $\mathbf{e} \mapsto \mathbf{Q}\mathbf{e}\mathbf{Q}^\top$  on Voigt  
1862 vectors. With the test and source polynomials transforming by matrices  $\mathbf{T}$   
1863 and  $\mathbf{U}$ ,  $\mathbf{K}_{ac}(\mathbf{Q}\mathbf{o}) = \sum_{b,d} T_{ab} U_{cd} \mathbf{S} \mathbf{K}_{bd}(\mathbf{o}) \mathbf{S}^\top$ , so one block is computed for each  
1864 orbit of offsets and the rest follow exactly. The coupling is then a convolution  
1865 over the grid, applied by the fast Fourier transform on a  $(2n-1)^3$  circular  
1866 embedding, and the system is solved by GMRES preconditioned with each  
1867 cell's own block. Where both can be run, the solution equals that of a dense  
1868 solve to  $10^{-10}$  relative (`tests/test_graded_voxel_fft.py`). Every cell that  
1869 overlaps the body is kept, each with the  $L^2$  projection of its contrast by a  
1870 six-point Gauss rule per axis; keeping only the cells whose centre lies inside  
1871 the body drops the contrast of the partly filled cells, and the mean-only  
1872 voxel then converges at order one instead of two. The far field is read at  
1873  $5 \times 10^8$  radii, where the near-field terms that the asymptotic form omits are  
1874 negligible.

1875 *The first-moment voxel on the graded sphere.* The voxel with  $p = r = 1$  is  
1876 applied to the graded sphere of §10. To isolate the discretisation error from

1877 the rescattering the grid cannot yet resolve, the contrast is first scaled by  
 1878  $10^{-6}$ , and the exact sphere is solved at the same scaled contrast, so that no  
 1879 Born approximation enters the reference: the relative error is then that of  
 1880 the scheme's first-order term, in which the field is the incident one carried  
 1881 across each cell. With 4, 6 and 8 voxels across, 2304 to 14688 unknowns, the  
 1882 error is  $7.5 \times 10^{-5}$ ,  $5.0 \times 10^{-6}$  and  $1.6 \times 10^{-6}$  at  $k_S a = 0.5$ , and  $2.3 \times 10^{-4}$ ,  
 1883  $9.1 \times 10^{-5}$  and  $2.8 \times 10^{-5}$  at  $k_S a = 1$ . The apparent orders between 6 and  
 1884 8 voxels are 4.01 and 4.13; between 4 and 6 they are not yet asymptotic, a  
 1885 voxel at  $n = 4$  being as wide as the graded shell (`scripts/pilot_graded_`  
 1886 `voxel_sphere.py--weak`). At the full contrast the fast solve is carried to 16  
 1887 voxels across, where the error is  $4.9 \times 10^{-6}$  at  $k_S a = 0.5$  and  $5.9 \times 10^{-6}$  at  
 1888  $k_S a = 1$ , and the apparent order approaches four at a rate set by the number  
 1889 of voxels across the graded shell.

## 1890 **Appendix F. The uniform-strain closure against the exact sphere**

1891 *Against the exact sphere.* These statements are tested against the exact elas-  
 1892 tic Mie solution for a sphere, with no Eshelby formula in between. The  
 1893 concentration of Mie order  $n$  is  $E_n = a_n/a_n^{\text{Born}}$  (orders 0, 1, 2 are the dilata-  
 1894 tional, density and shear channels), extrapolated to  $ka \rightarrow 0$ . The formulation  
 1895 is applied to a ball of radius  $a$ , with its moment  $\mathbf{M}$  evaluated by the surface  
 1896 route of Eq. (26) on  $|\mathbf{r}| = a$ . Its dilatational and shear channels equal Mie's  
 1897 to  $2 \times 10^{-10}$ , and the density channel is 1. The cube's closed forms meet  
 1898 the same numbers. Its dilatational channel is Mie's  $E_0$  to  $2 \times 10^{-10}$ , and its  
 1899 isotropic shear average, Eq. (32), is the depolarisation read from Mie's  $E_2$   
 1900 to  $10^{-8}$ . The same moment taken as a principal value, over the ball with

1901 a small sphere excluded around the origin, has zero depolarisation in every  
 1902 channel. The whole static response of the sphere is the  $\delta$  term.

1903 At finite frequency the closure uses the dynamic ball moments, which are  
 1904 in closed form. It departs from Mie at second order in  $ka$  and first order in  
 1905 the contrast, and the departure is closed-form too: the ball's moments are  
 1906 elementary, and Mie's low-frequency series with the contrast kept exact gives  
 1907  $E_n$  to relative order  $(ka)^3$ . To first order in the contrast,

$$\frac{E_n^{\text{closure}}}{E_n^{\text{Mie}}} - 1 = \begin{cases} -\frac{(k_P a)^2}{10} \frac{\Delta\lambda}{\lambda + 2\mu}, & n = 0, \\ \frac{2(k_S a)^2 + (k_P a)^2}{30} \frac{\Delta\rho}{\rho}, & n = 1, \\ -\frac{3(k_S a)^2 + 2(k_P a)^2 \beta^2 / \alpha^2}{75} \frac{\Delta\mu}{\mu}, & n = 2, \end{cases} \quad (\text{F.1})$$

1908 with no  $O(ka)$  term and no  $O((ka)^3)$  term. The imaginary  $(ka)^3$  terms,  
 1909 the radiation damping, agree exactly in all three channels. So the closure's  
 1910 only error at this order is the real  $(ka)^2$  term, the uniform-field ansatz's  
 1911 own. That is the order the hierarchy assigns to the omitted second gradi-  
 1912 ents (§6.3), and it has the same origin, the variation of the field across the  
 1913 scatterer, as the collocation error of §7. Eq. (F.1) is derived symbolically  
 1914 (`SphereClosureDynamic.wl`). It agrees with the coefficient extracted from  
 1915 the numerical Mie solve and an independent implementation of the closure  
 1916 to  $5 \times 10^{-4}$ , which is the size of the next order in the contrast.

## 1917 **Appendix G. The scattering series of the layer, and the projection-** 1918 **error law**

1919 *The scattering series.* Multiply the whole contrast,  $\Delta\lambda$ ,  $\Delta\mu$  and  $\Delta\rho$   
 1920 together, by a number  $s$ , and let  $u(s)$  be the scattered displacement at an

1921 observer at depth  $z_o$  outside the layer. It vanishes at  $s = 0$  and is analytic  
 1922 in  $s$  there, and its expansion is the Born series,

$$u(s) = s T_1 + s^2 T_2 + s^3 T_3 + \dots, \quad (\text{G.1})$$

1923 so that  $T_n$  is the part of the field at the full contrast,  $s = 1$ , that has been  
 1924 scattered  $n$  times.

1925 *The first two terms of the exact layer.* They are integrals over the layer.  
 1926 With the incident state  $w^0 = (u^0, \varepsilon^0)$ , the contrast  $\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$   
 1927 with profile  $f(z)$ , and the kernel  $\mathbf{K}$  of Eq. (38), whose first row  $(g, g')(z_o - z)$   
 1928 gives the displacement at the observer,

$$T_1 = \int_0^D (g, g')(z_o - z) \Delta_1 f(z) w^0(z) dz, \quad (\text{G.2})$$

1929

$$w^1(z) = \int_0^D \mathbf{K}(z - z') \Delta_1 f(z') w^0(z') dz', \quad T_2 = \int_0^D (g, g')(z_o - z) \Delta_1 f(z) w^1(z) dz. \quad (\text{G.3})$$

1930 Here  $w^1$  is the state scattered once, inside the layer; its strain includes the  
 1931 local term of the kernel,  $-f(z) \Delta M \varepsilon^0(z)/M_P$ . Each further term repeats  
 1932 the step,  $w^n(z) = \int_0^D \mathbf{K}(z - z') \Delta_1 f(z') w^{n-1}(z') dz'$ . For a plane force at  $z_s$   
 1933 above the layer,  $w^0(z) = c e^{ik(z-z_s)}(1, ik)$  with  $c = i/(2M_P k)$ , every integrand  
 1934 is an exponential times the profile. Writing  $a = \omega^2 \Delta \rho$  and  $b = \Delta M$ , and  
 1935 carrying out the products of Eqs. (G.2) and (G.3), the terms seen in reflection  
 1936 ( $z_o < 0$ ) and in transmission ( $z_o > D$ ) are, for any profile,

$$T_1^R = c^2 e^{-ik(z_o + z_s)} (a + bk^2) \int_0^D f(z) e^{2ikz} dz, \quad T_1^T = c^2 e^{ik(z_o - z_s)} (a - bk^2) \int_0^D f(z) dz, \quad (\text{G.4})$$

1937

$$\begin{aligned}
T_2^R &= c^2 e^{-ik(z_o+z_s)} \left[ c(a^2 - b^2 k^4) \int_0^D \int_0^D f(z) f(z') e^{ik(z+z'+|z-z'|)} dz' dz - \frac{b^2 k^2}{M_P} \int_0^D f(z)^2 e^{2ikz} dz \right], \\
T_2^T &= c^2 e^{ik(z_o-z_s)} \left[ c(a - bk^2)^2 \int_0^D \int_0^z f(z) f(z') dz' dz + c(a + bk^2)^2 \int_0^D \int_z^D f(z) f(z') e^{2ik(z'-z)} dz' dz \right. \\
&\quad \left. + \frac{b^2 k^2}{M_P} \int_0^D f(z)^2 dz \right].
\end{aligned} \tag{G.5}$$

1938 The first-order reflection is the Fourier transform of the profile at the wavenum-  
1939 ber  $2k$ , and the first-order transmission its integral. In each second-order  
1940 term the last integral is the local term of the kernel, in which both scat-  
1941 terings are at one point and the profile appears squared; the others are the  
1942 two scatterings at different depths. In transmission the wave scattered twice  
1943 forwards carries the contrast of velocity twice,  $(a - bk^2)^2$ , and the wave scat-  
1944 tered backwards and then forwards again the contrast of impedance twice,  
1945  $(a + bk^2)^2$ . For the uniform layer ( $f = 1$ ) the integrals give

$$T_1^R = c^2 e^{-ik(z_o+z_s)} (a + bk^2) \frac{e^{2ikD} - 1}{2ik}, \quad T_1^T = c^2 e^{ik(z_o-z_s)} (a - bk^2) D, \tag{G.6}$$

1946

$$\begin{aligned}
T_2^R &= \frac{c^2 e^{-ik(z_o+z_s)}}{4k^3 M_P} \left[ -i(a^2 + b^2 k^4) + e^{2ikD} (a^2(i + 2kD) + b^2 k^4(i - 2kD)) \right], \\
T_2^T &= \frac{i c^2 e^{ik(z_o-z_s)}}{8k^3 M_P} \left[ a^2(1 - e^{2ikD} + 2kD(i + kD)) - 2abk^2(e^{2ikD} - 1 + 2kD(kD - i)) \right. \\
&\quad \left. + b^2 k^4(1 - e^{2ikD} + 2kD(kD - 3i)) \right].
\end{aligned} \tag{G.7}$$

1947 The reflection sees the contrast of impedance,  $a + bk^2 = k^2 M_P(\Delta\rho/\rho +$   
1948  $\Delta M/M_P)$ , and the transmission that of velocity,  $a - bk^2$ . As  $kD \rightarrow 0$ ,  $T_2^R \rightarrow$   
1949  $-c^2 e^{-ik(z_o+z_s)} b^2 k^2 D/M_P$ : the local term alone, which is the starting point of  
1950 the argument at the end of this appendix. For the smooth profile the integrals

1951 of Eqs. (G.4) and (G.5) are evaluated in closed form symbolically, and by  
 1952 quadrature they return the same values to  $10^{-16}$ . Equations (G.6) and (G.7)  
 1953 equal, identically, the first two Taylor coefficients in  $s$  of the exact reflection  
 1954 and transmission of the layer of §3 (`ContinuumLimit_BornTerms.wl`). At  
 1955  $\omega = 300 \text{ rad s}^{-1}$ ,  $|T_2|/|T_1|$  is 0.039 in reflection and 0.17 in transmission for  
 1956 the uniform layer.

1957 *The first two terms of the scheme.* Write Eq. (38) as  $(\mathbf{M} - \mathbf{K}) \mathbf{c} = \mathbf{b}$ , with  
 1958  $\mathbf{M}$  the diagonal matrix of the  $\int_i \phi_a^2 dz$ ,  $\mathbf{K}$  the matrix of the coupling blocks  
 1959 times  $\Delta_1$ , and  $\mathbf{b}$  the moments of the incident field, and let  $\mathbf{r}$  be the row that  
 1960 gives the scattered displacement at the observer from the coefficients,  $u = \mathbf{r} \mathbf{c}$ .  
 1961 Scaling the contrast by  $s$  scales  $\mathbf{K}$  and  $\mathbf{r}$  by  $s$ , so  $u(s) = s \mathbf{r} (\mathbf{M} - s \mathbf{K})^{-1} \mathbf{b}$  and

$$T_1^{\text{scheme}} = \mathbf{r} \mathbf{M}^{-1} \mathbf{b}, \quad T_2^{\text{scheme}} = \mathbf{r} \mathbf{M}^{-1} \mathbf{K} \mathbf{M}^{-1} \mathbf{b}. \quad (\text{G.8})$$

1962  $\mathbf{M}^{-1} \mathbf{b}$  is the projection of the incident field onto each cell's functions: the first  
 1963 term needs no solve. Splitting  $\mathbf{K}$  into its blocks within a cell and its blocks  
 1964 between cells splits  $T_2^{\text{scheme}}$  into the part scattered twice in one cell and the  
 1965 part scattered in two; the same split of the inner integral of Eq. (G.3) splits  
 1966 the exact  $T_2$ .

1967 *A check by differences.* The terms can also be read from solutions at four  
 1968 scaled contrasts,  $s = \pm\epsilon$  and  $\pm 2\epsilon$ , with no knowledge of the formulas above.  
 1969 The odd and even parts of  $u$  in  $s$  separate the odd orders from the even,  
 1970 and the two step sizes remove the next term of each (the five-point central  
 1971 differences for  $u'(0)$  and  $\frac{1}{2}u''(0)$ ):

$$\begin{aligned} T_1 &= \frac{8[u(\epsilon) - u(-\epsilon)] - [u(2\epsilon) - u(-2\epsilon)]}{12\epsilon} + 4\epsilon^4 T_5 + \dots, \\ T_2 &= \frac{16[u(\epsilon) + u(-\epsilon)] - [u(2\epsilon) + u(-2\epsilon)]}{24\epsilon^2} + 4\epsilon^4 T_6 + \dots \end{aligned} \quad (\text{G.9})$$

1972 A negative  $s$  is a medium slower and lighter than the background by the  
 1973 same fractions, as legitimate a medium as the other. With  $\epsilon = 0.01$  the  
 1974 remainders  $4\epsilon^4 T_5/T_1$  and  $4\epsilon^4 T_6/T_2$  are below  $2 \times 10^{-12}$  for this layer, and the  
 1975 formulas applied to the exact layer return Eqs. (G.6) and (G.7) less exactly  
 1976 those remainders, to  $10^{-17}$ . The measurements of §9 use the differences for  
 1977 the scheme and the exact layer alike, each in its own arithmetic, and they  
 1978 agree with Eqs. (G.2), (G.3) and (G.8).

1979 *Why the terms must be separated.* The law of Eq. (44) is a statement  
 1980 about  $T_2$ . At the full contrast the terms of third and higher order are 7% of  
 1981 the second and carry errors of their own, and if every order above the first is  
 1982 taken together the ratio of Eq. (44) reads 0.90 to 0.94 where the second-order  
 1983 term alone gives 1.00 to 1.01.

1984 Equation (44) is exact in the long-wave limit, for a reason that shows what  
 1985 each basis is for. The strain-strain entry of the kernel,  $g'' = -k^2 g - \delta(z)/M_P$ ,  
 1986 is local as  $kD \rightarrow 0$ , and the incident strain  $\varepsilon_0$  is then uniform across the layer,  
 1987 so the strain scattered once is  $-(\Delta M/M_P) f(z) \varepsilon_0$ : it is the profile itself. A  
 1988 cell holds the contrast as  $\Pi_r f$  and the field to degree  $p$ , so it holds the once-  
 1989 scattered strain as  $\Pi_p(\Pi_r f) = \Pi_q f$  with  $q = \min(p, r)$ . The second-order  
 1990 term pairs the contrast with that strain:

$$T_2 \propto \int f \cdot f \, dz = \|f\|^2, \quad T_2^{\text{scheme}} \propto \int (\Pi_r f)(\Pi_q f) \, dz = \|\Pi_q f\|^2, \quad (\text{G.10})$$

1991 the last because  $\Pi_q f$  lies in the range of  $\Pi_r$ . The relative error is  $1 -$   
 1992  $\|\Pi_q f\|^2 / \|f\|^2 = \mathcal{E}_q$ . Because the kernel is local in this limit, the second  
 1993 scattering takes place in the cell of the first:  $T_2$  is the second-order term of  
 1994 the single-site  $T$ -matrix, and Eq. (44) is a statement about the single site.

1995 The  $T$ -matrix sums the rescattering within a cell exactly for the cell as it  
 1996 is represented, a contrast  $\Pi_r f$  and a field of degree  $p$ ; its error is that the  
 1997 cell is not the medium. The split is measured directly (`scripts/measure_`  
 1998 `layer_bases.py`): at  $\omega = 300 \text{ rad s}^{-1}$  on 4, 8 and 16 cells,  $T_2$  of the scheme  
 1999 and of the exact layer are each divided into the part whose two scatterings  
 2000 lie in one cell and the part between different cells. For all nine pairs the  
 2001 error of the same-cell part is 0.998 to 1.013 times  $|T_2| \mathcal{E}_{\min(p,r)}$ , and that of  
 2002 the part between cells at most 0.002 times it. The coupling between cells  
 2003 contributes only the departure from the limit. Measured, the ratio of the  
 2004 two sides of Eq. (44) for  $p = r = 1$  on 8 cells departs from one by  $1.7 \times 10^{-2}$ ,  
 2005  $4.3 \times 10^{-3}$ ,  $4.8 \times 10^{-4}$  and  $4 \times 10^{-5}$  at  $kD = 0.24, 0.12, 0.04$  and  $0.012$ :  
 2006 as  $(kD)^2$  (`scripts/measure_layer_bases.py`). The pairing in Eq. (G.10)  
 2007 is verified symbolically for all nine pairs, and the projection errors are com-  
 2008 puted exactly (`ContinuumLimit_LayerDefect.wl`); they fall by  $4^{q+1}$  for each  
 2009 halving of the cells.

## 2010 **Appendix H. Relation to other methods**

2011 *Relation to the coupled-dipole method.* The same partition appears in the  
 2012 coupled-dipole method as the choice of polarisability. The Clausius–Mossotti  
 2013 relation, the lattice dispersion relation [11], the polarisability taken from the  
 2014 exact static solution [12] and the integrated Green tensor [13] all adjust the  
 2015 single site so that the lattice reproduces the continuum. For space-filling  
 2016 cubes the adjustment is neither needed nor a free parameter: the tiling iden-  
 2017 tity fixes the local field exactly, the self term of the collocation closure is the  
 2018 one that makes it cancel, and what is left is a closed-form error rather than

2019 an empirical one. That the method is a collocation of the volume integral  
 2020 equation was shown for the electromagnetic case by Lakhtakia [9]; Eq. (35)  
 2021 is the same statement for the elastic lattice with the self term evaluated over  
 2022 the finite cube, and it is what makes the error a midpoint-rule error. The  
 2023 filtered coupled dipoles of Piller and Martin [14] and Yurkin et al. [15] band-  
 2024 limit the source to the grid; the source-cell average does the complementary  
 2025 thing exactly, its form factor vanishing on the reciprocal lattice, and the  
 2026 spectral construction of Eq. (B.1) is the periodic coupled-dipole method of  
 2027 Chaumet et al. [16] with the Galerkin double average in place of the point  
 2028 dipole. The second-order convergence reported for the DDA by Yurkin et al.  
 2029 [3] corresponds to the uniform-field voxel's, and the opposite-signed con-  
 2030 stants of Eq. (36) explain why a polarisability correction can improve one  
 2031 observable only at the expense of another, as the package's own form-factor  
 2032 corrections do in reflection and transmission. Going beyond second order  
 2033 requires a richer internal field, which is the point of higher-order bases for  
 2034 volume integral equations [66, 2]; the first-moment voxel is the lowest such  
 2035 basis, and in its spectral form the Galerkin double average makes the lattice  
 2036 sum converge without Ewald splitting.

2037 *Relation to the spectral element method.* The voxel of degree  $p$  is close kin to  
 2038 the spectral element [67], which was taken up for seismic wave propagation in  
 2039 the 1990s, is set out for three-dimensional models by Komatitsch and Tromp  
 2040 [68], and is now a standard tool for it. Both divide the medium into hexahe-  
 2041 dra, represent the field in each by polynomials of high degree, and determine  
 2042 the coefficients by a Galerkin condition on a weak form. The spectral element  
 2043 uses Lagrange interpolants through the Gauss–Lobatto–Legendre points of

2044 the element; the voxel uses the Legendre polynomials themselves. Four of  
 2045 the present results have their counterparts there.

2046 1. *A diagonal mass matrix.* The spectral element obtains one by integrat-  
 2047 ing at the Gauss–Lobatto–Legendre points, at which each interpolant  
 2048 vanishes but one [68]. That rule is not exact for the product of two  
 2049 interpolants, so that the diagonal form is reached, in their words, by a  
 2050 process of subintegration, as in the finite-element methods that lump  
 2051 the mass. Here the Gram matrix  $\mathbf{M}$  of Eq. (G.8) is diagonal with no  
 2052 approximation, because the Legendre functions are orthogonal on the  
 2053 cell; this is why the first-order term needs no solve. The two routes  
 2054 serve different ends: theirs an explicit march in time, ours the terms of  
 2055 the scattering series.

2056 2. *The doubled order.* A scattered amplitude is a functional of the field,  
 2057 and a Galerkin method gives a functional to twice the order of the field  
 2058 itself. The order  $2p+2$  is this doubling: the leading error of Eq. (39) is  
 2059 the product of the part of the incident wave that the cell’s polynomials  
 2060 miss and the part of the outgoing wave that they miss, each of order  
 2061  $h^{p+1}$ .

2062 3. *The medium to the degree of the field.* In a spectral element the den-  
 2063 sity and the elastic moduli are evaluated at the element’s own grid  
 2064 points and may vary from one to the next, so that a fully heterogeneous  
 2065 medium is held to the degree of the field [68]. The rule  $\min(2p+2, 2r+2)$   
 2066 says why that is the right choice, and what a coarser representation of  
 2067 the medium would cost.

2068 4. *Elements that honour the interfaces.* A spectral-element mesh is de-

signed to honour the free surface and the main geological boundaries of the model [68]. The stratified layer of §9 is the same requirement seen from the integral side: with voxel faces on the interfaces the projection error vanishes, and a jump inside a voxel costs the order.

The two methods differ as a differential formulation differs from an integral one, and they suit different problems. The spectral element discretises the weak form of the equations of motion in the time domain. Its coupling is local and its matrices are sparse; it treats heterogeneity, anisotropy and topography of any form throughout a three-dimensional model; and the price is that the whole model is meshed, that the outgoing waves are absorbed at an artificial boundary, and that over a long simulation the accuracy is governed by the time scheme [68]. The voxel scheme discretises the volume integral equation in the frequency domain. Only the departure from a stratified background is meshed, the field of each voxel is independent of its neighbours', and propagation through the background, the radiation condition included, is carried exactly by the background's Green's tensor; the price is a coupling between every pair of voxels, and a background restricted to one whose Green's tensor is known. It differs also from the boundary integral methods, which are restricted to a finite number of homogeneous regions [68]: a volume integral equation admits a medium that varies within the scatterer. The voxel scheme is therefore a candidate where the heterogeneity is localised in a layered medium, and not a replacement for the spectral element in a model that is heterogeneous throughout. A spectral-element mesh is designed by a rule of resolution, about five grid points to the shortest wavelength at polynomial degrees between five and ten [68]. The error analysis of this paper is

2094 the counterpart of that rule for the integral formulation: it gives the size and  
 2095 the degree of the voxels for a required accuracy in closed form, and it adds a  
 2096 term, the projection error of the medium.

## 2097 **Appendix I. The cost of higher order, and what the error analysis** 2098 **offers a fast solver**

2099 *The cost of fourth order.* A first-moment voxel carries four times as many  
 2100 unknowns as a uniform one (36 against 9 in three dimensions). The closed  
 2101 forms show what that buys. To reach an error  $\varepsilon$  in reflection, the uniform  
 2102 voxel needs  $kd = \sqrt{8\varepsilon}$  and the first-moment voxel  $kd = (720\varepsilon)^{1/4}$ ; at  $\varepsilon = 10^{-6}$   
 2103 that is  $kd = 2.8 \times 10^{-3}$  against 0.16, voxels 58 times larger in each direction.  
 2104 In a three-dimensional model the voxel count falls by  $58^3 \approx 2 \times 10^5$  and the  
 2105 number of unknowns by about  $5 \times 10^4$  (these are the Born-order constants;  
 2106 at the contrast studied the first-moment voxel's measured errors lie within  
 2107 5% of them, while the uniform voxel's transmission constant grows with the  
 2108 contrast, so the comparison is if anything conservative). The stratified model  
 2109 of §9 measures it directly: 347 planes of uniform-field voxels against 8 first-  
 2110 moment voxels for the same accuracy. Higher moments continue the trend  
 2111 in one dimension: by Eq. (40) the second-moment voxel reaches  $\varepsilon = 10^{-6}$  at  
 2112  $kd = 0.68$  and the third-moment voxel at  $kd = 1.5$ , where a voxel is already  
 2113 a quarter of a wavelength. In three dimensions, with the Legendre products  
 2114 of total degree at most  $p$ , a voxel carries 10 ( $p = 2$ ) or 20 ( $p = 3$ ) weights per  
 2115 state component, against 4 for the first-moment voxel. The voxel of degree  
 2116 2 converges at sixth order in the layer at oblique incidence (Table B.14);  
 2117 that of degree 3 has not been built in three dimensions. The gain is largest

2118 precisely where it matters: when the model varies on scales that are coarse  
2119 compared with the accuracy the wavefield must have.

2120 *What the error analysis offers a fast solver.* A voxel scheme is dense, and  
2121 for a large medium it is solved iteratively with a fast product. The adaptive  
2122 integral method [62] and the precorrected fast Fourier transform [63] are  
2123 the standard forms: the sources are projected onto a regular grid, in the  
2124 first by matching their multipole moments and in the second by matching  
2125 the potentials they produce at a set of test points, the interactions between  
2126 distant cells are evaluated on that grid by fast Fourier transforms, and those  
2127 between near cells are computed directly and corrected. Such a method has  
2128 three accuracies to set, that of the discretisation, that of the projection, and  
2129 the radius of the near zone, and they are usually set by trial. The results of  
2130 this paper set them beforehand, in five ways.

- 2131 1. *The target is known.* The discretisation error is known before the solve,  
2132 in order and in size: the first-order factor of a cell, Eq. (39) and its  
2133 form for any pair of wavenumbers (§9), and the projection error of the  
2134 medium, Eq. (44). A fast product need be no more accurate than that,  
2135 and it is known when it is accurate enough.
- 2136 2. *The two errors do not mix.* The error of the scheme is the single site's:  
2137 the coupling between cells, averaged over the cells, is exact (§5). An  
2138 approximation of the far coupling to a set tolerance therefore adds its  
2139 own error to the discretisation's and does not alter it, so the two can  
2140 be budgeted separately.
- 2141 3. *For cubes on a lattice there is nothing to project.* The cells are the reg-  
2142 ular grid. The coupling depends on the separation of the cells alone, so

2143 the product is a convolution, block by block in the polynomial degrees,  
 2144 and the fast Fourier transform applies to it exactly, as in Georgakis  
 2145 et al. [2] for the linear voxel. What the projection onto the grid ap-  
 2146 proximates in general is here an identity.

2147 4. *For cells of several sizes the moments are already in hand.* Where the  
 2148 cells differ in size or degree and are mapped to a uniform auxiliary  
 2149 grid, the quantities to be matched are the polynomial moments of a  
 2150 cell, which are the unknowns of the scheme; and the first-order factor  
 2151 says how many of them matter at a given  $kd$ .

2152 5. *A stratified background splits as the method requires.* The Green's ten-  
 2153 sor of a plane-stratified background is the whole-space tensor, which is  
 2154 invariant under translation in three dimensions and carries all the sin-  
 2155 gularity, plus a reverberation, Eq. (C.1), which is smooth, is invariant  
 2156 under lateral translation, and separates in depth into one-dimensional  
 2157 moments. The first is the convolution above; the second is a fast Fourier  
 2158 transform laterally and a product in depth, with no near-zone correc-  
 2159 tion except where a voxel touches an interface. A splitting of the kernel  
 2160 into a term that carries the singularity and a nonsingular remainder is  
 2161 used to a related end by Sharma and Triverio [64], for surface integral  
 2162 equations in homogeneous and in layered backgrounds: there the sin-  
 2163 gular term is independent of frequency, so that the direct near-zone  
 2164 integrals are computed once for a whole sweep in frequency. And be-  
 2165 cause the orders and the error law are unchanged by the background  
 2166 (§9), the error budget of the whole space carries over to it.

2167 No such solver is built here. These are consequences of the analysis, and they

2168 are to be tested.

## 2169 **Appendix J. The evidence for each result, and where to reproduce** 2170 **it**

2171 Table J.21 lists each result with the evidence for it. Every numeri-  
2172 cal value is reproducible from the Mathematica notebooks `Mathematica/`  
2173 `ContinuumLimit_X.wl`, for X in Reference, ThinLayer, SpecularSums, Stati-  
2174 cLocal, AveragedSums, CubeT, Chain, ErrorConstant, FourthOrder, FourthOrder-  
2175 Error, Reduction3D, Oblique, IncidentS, Heterogeneous, SecondMoment, Grad-  
2176 edContrast, LayerDefect and Figure (data for the figure), and their ref-  
2177 erence dumps `scripts/continuum_limit_*.py`; the moment integrals are  
2178 `Mathematica/CubeMomentCore.wl` (gate `CubeMomentCoreTest.wl`), `CubeA22Block.`  
2179 `wl`, `CubeMomentArchiveCheck.wl` and `CubeShearSplit.wl`, with the Python  
2180 half of the last in `scripts/gate_cube_shear_split.py`; the stratified refer-  
2181 ence against Kennett's recursion is `scripts/gate_heterogeneous_reference_`  
2182 `vs_kennett.py`; the voxelised spheres are `scripts/pilot_sphere_voxel_`  
2183 `vs_mie.py` (sharp) and `scripts/pilot_graded_voxel_sphere.py` (graded,  
2184 with the mean-only voxel of `cubic_scattering/graded_voxel`), the graded  
2185 sphere's exact solution `Mathematica/ContinuumLimit_GradedSphere.wl` with  
2186 the library `Mathematica/TakeuchiSaito.wl` and `scripts/crosscheck_graded_`  
2187 `sphere.py`; the test against exact Mie is `scripts/gate_sphere_closure_`  
2188 `vs_mie.py`, with the closed forms in `Mathematica/SphereClosureDynamic.`  
2189 `wl`; the exact kernel is `cubic_scattering/cell_averaged_lattice.py`; the  
2190 independent implementation of the first-moment voxel is `scripts/crosscheck_`  
2191 `first_moment_voxel.py`, and of the voxel of any degree `scripts/crosscheck_`

2192 `second_moment_voxel.py`, and of the graded cell `scripts/crosscheck_`  
2193 `graded_contrast.py`; the projection-error law and its split are measured by  
2194 `scripts/measure_layer_bases.py`, in a stratified background by `scripts/`  
2195 `measure_layer_stratified_background.py`, `scripts/measure_oblique_`  
2196 `stratified_background.py` and `scripts/measure_oblique_two_term.py`,  
2197 and the voxel of degree 2 at oblique incidence by `scripts/measure_oblique_`  
2198 `degree.py`; the convergence figure is drawn by `scripts/plot_convergence_`  
2199 `orders.py`; the near field of the sphere is measured by `scripts/measure_`  
2200 `graded_sphere_near_field.py`; the moments of every grade are in Mathematica/  
2201 `CubeMomentHigherGrades.wl`, Mathematica/`CubeMomentStore.wl` and Mathematica/  
2202 `CubeMomentCanonical.wl`, with their independent evaluation in `scripts/`  
2203 `crosscheck_cube_moments_ball_shell.py`; the hierarchy on the layer is  
2204 Mathematica/`ContinuumLimit_GradientHierarchy.wl` and `scripts/measure_`  
2205 `layer_taylor_hierarchy.py`, and in three dimensions `scripts/measure_`  
2206 `ball_gradient_hierarchy.py`, `scripts/measure_cube_gradient_hierarchy.`  
2207 `py` and `scripts/measure_graded_sphere_gradient_hierarchy_fft.py`. All  
2208 are available at <https://github.com/tmnestor/MultipleScatteringCalculations>.

Table J.21: The results of the paper, and the evidence for each.

| Result                                                     | Evidence                                                          |
|------------------------------------------------------------|-------------------------------------------------------------------|
| Continuous layer, exact and to $O(D^k)$                    | reference to $2.5 \times 10^{-8}$ ; series order $k$              |
| Thin-layer series, Eq. (7)                                 | each coefficient symbolically, to $D^4$                           |
| Point-kernel local term survives $d \rightarrow 0$         | $11 \times$ the continuum term at $m = 1$                         |
| Averaged coupling between planes = continuum $\times$ sinc | $2.8 \times 10^{-16}$ , 81 entries                                |
| Tiling identity, Eq. (13)                                  | 21/21 components to $4 \times 10^{-9}$                            |
| Exact kernel, Eq. (14)                                     | replaces a $9.4 \times 10^{-4}$ static error                      |
| Moments as distributions, Eq. (26)                         | sum rule (27) exact; M, Q equal an independent archive, 81/81     |
| Channels (29)–(31)                                         | a production code by another route, $2 \times 10^{-15}$           |
| Cube vs sphere, Eq. (32)                                   | symbolic; and numerically at five $\nu$ , $4 \times 10^{-16}$     |
| Ball closure = exact Mie, static                           | $2 \times 10^{-10}$ ; cube's bulk and shear average $10^{-8}$     |
| Ball closure vs Mie, dynamic, Eq. (F.1)                    | symbolic; numerical Mie $5 \times 10^{-4}$ ; $(ka)^3$ terms exact |
| Self term cancels, Eq. (35)                                | $4 \times 10^{-16}$                                               |
| Package $T$ = closure statically                           | $2 \times 10^{-5}$ at $k_S h = 0.01$                              |
| Second-order convergence                                   | order 2.000, three frequencies                                    |
| Closed-form error, Eq. (36)                                | $n = 1$ –16, to $10^{-4}$ of the error                            |
| First-moment voxel, Eq. (38)                               | order 4.00, three frequencies                                     |
| First-moment voxel, oblique, Eq. (B.1)                     | order 4, P and S, $20^\circ$ and $40^\circ$                       |

## 2209 **Acknowledgments**

2210 This work began thirty years ago in the author’s doctoral research at  
2211 the Research School of Earth Sciences of the Australian National University,  
2212 Canberra [21].

## 2213 **Declaration of competing interest**

2214 The author declares no known competing financial interests or personal  
2215 relationships that could have appeared to influence the work reported in this  
2216 paper.

## 2217 **Declaration of generative AI and AI-assisted technologies in the** 2218 **manuscript preparation process**

2219 During the preparation of this work, the author used Claude (Anthropic)  
2220 in order to check the existing Mathematica notebooks, Python and Fortran  
2221 code, to build tests to ensure their correctness, and to write the TikZ code  
2222 of the diagrams. After using this tool, the author reviewed and edited the  
2223 content as needed and takes full responsibility for the content of the published  
2224 article.

## 2225 **Data availability**

2226 The Mathematica notebooks and Python scripts that derive and check  
2227 every result, and the package they test, are available at [https://github.](https://github.com/tmnestor/MultipleScatteringCalculations)  
2228 [com/tmnestor/MultipleScatteringCalculations](https://github.com/tmnestor/MultipleScatteringCalculations).

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